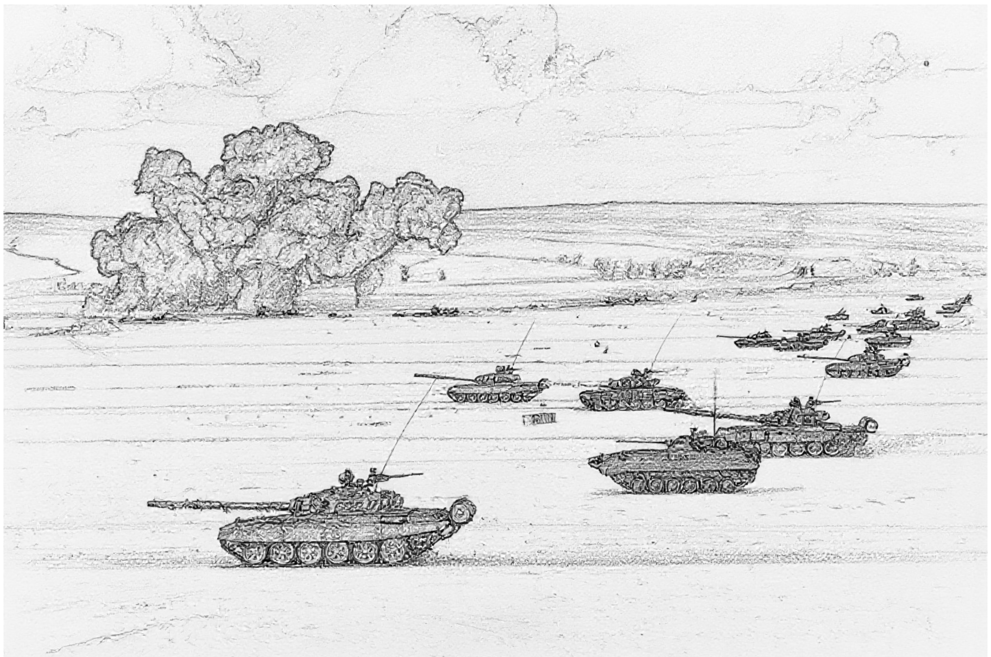
Air Power

Ground Unit Briefing

Alan Watson Forster

23 June 2026



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PREFACE

For many, the principal focus of the *Air Power* game is air-to-air combat. That said, no less an authority on air-to-air combat than Robin Olds stated:[1]

> "You're not going to win a war by shooting down a damn MiG, you're going to win a war by bombing the bejesus out of whatever you were sent to bomb."

This briefing document deals with the ground units and ground targets that are subject to air-to-ground attacks. It first presents ground-combat and transport units, then air-defense units (AAA, FCR, SAM, EWR, and CCU units), and finally ground targets. An appendix gives some examples of battalion-sized units.

Much of the data here is adapted from J.D. Webster's *Air Strike*, *Eagles of the Gulf*, and *The Speed of Heat*, from Malcolm Pipe's extensive compendium of AAA and SAM units, and to a lesser degree from *Air Power Journal.* However, the data have been carefully revised for accuracy and internal consistency, a wider variety of ground-combat units are covered to give a richer experience, and the descriptions of the units and their weapon systems have been expanded. Extensive notes follow the main text, providing provenance of data, sources of information, and justifications for decisions.

I am grateful to Carl Smeaton for many discussions on the data presented here and to Malcolm Pipes for leading the way with his compilations of AAA and SAM units.

Alan Watson-Forster
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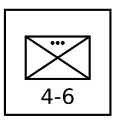
GROUND-COMBAT AND TRANSPORT UNITS

Ground-combat units are formations of infantry, tanks, IFVs, APCs, other AFVs, or artillery whose primary role is combat with other ground units. Transport units provide operational and strategic mobility to infantry and towed artillery and are the lifeblood of logistic networks. These units are presented below with their counters. The Ground-Combat and Transport Unit Table summarizes their properties. Some representative examples of vehicles are given in the descriptions, and the Vehicle Types Table gives a more extensive list. All ground-combat and transport platoons and batteries are also available as sections and squads.[2]

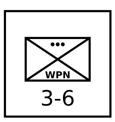
INFANTRY UNITS

Infantry are soldiers trained to fight on foot and, in some cases, from IFVs. Their primary roles in modern warfare are seizing, clearing, and holding ground, often as part of a combined-arms formation with tanks and artillery, highly mobile offensive and defensive operations in combination with helicopters, and combat in difficult terrain, including urban areas, mountainous zones, and dense jungle or forest.

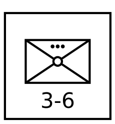
An important secondary characteristic of infantry is any associated transport. Motorized infantry units are transported by trucks, giving them operational mobility. Mechanized infantry are transported by APCs or IFVs, giving them both tactical and operational mobility. Light infantry units have no or fewer supporting vehicles, although modern light infantry is often airborne.



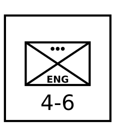
Infantry Platoon: An infantry platoon is armed with small arms, light machine guns, and sometimes with light grenade launchers and portable anti-tank weapons such as bazookas, LAWs, MAWs, and portable ATGMs. The platoon is the primary combat unit in larger infantry formations.



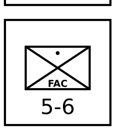
Infantry Weapons Platoon: An infantry weapons platoon is equipped with heavier weapons, such as medium or heavy machine guns, recoilless rifles, portable ATGMs, and automatic grenade launchers. It is often a company- or battalion-level asset, and its role is to provide direct-fire support to the infantry platoons in its formation.[3]



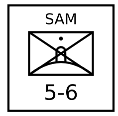
Infantry Mortar Platoon: An infantry mortar platoon is equipped with 60-mm, 81-mm, 82-mm, or similar mortars. It is often a battalion-level asset, and its role is to provide indirect fire support and smokescreens to the infantry platoons in its formation. Due to the weight of ammunition required for sustained fire, it will often be transported by a light truck platoon even in light infantry formations.



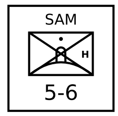
Infantry Engineers Platoon: An infantry engineers platoon is additionally equipped and trained for combat engineering tasks such as mine-laying and mine-clearing. It is often battalion-level asset.



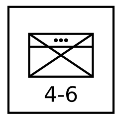
Infantry FAQ Squad: An infantry FAC squad has a forward air controller with other soldiers providing communication and protection. An infantry FAC squad will often form a composite platoon with an infantry or infantry weapons platoon. If operating independently, it will often be transported by a light truck squad.



Infantry SAM Squad: Infantry SAM squads are equipped with shoulder-launched SAMs, with one launcher per squad. Their SAM capabilities and VP values are given elsewhere. An infantry SAM squad will often form a composite platoon with an infantry or infantry weapons platoon. If operating independently, it will often be transported by a light truck squad.



Infantry Heavy SAM Squad: Infantry heavy SAM squads are equipped with pedestal-mounted SAMs, with one launcher per squad. Their SAM capabilities and VP values are given in a subsequent section. An infantry heavy SAM squad will often form a composite platoon with an infantry or infantry weapons platoon. If operating independently, it will often be transported by a light truck squad.

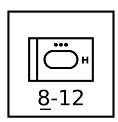


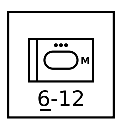
Infantry HQ Platoon: An infantry HQ platoon is the HQ unit of an infantry battalion or regiment. It consists of the formation commander, their staff, and their protection detail.[4]

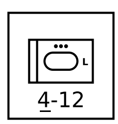
ARMORED VEHICLE UNITS

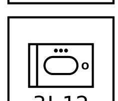
Armored ground-combat vehicle units combine armored protection for their crew and weapons, mobility, and either a firepower or transport capability.

The primary factors that determine an armored-vehicle unit's type are its weapons and protection level. The means of traction is a secondary consideration. Armored cars and armored half-tracks are considered to be light tanks, light IFVs, or light or open APCs, light anti-tank missile vehicles, or light reconnaissance vehicles according to their role and armament.

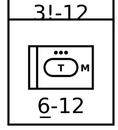


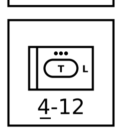




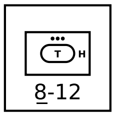


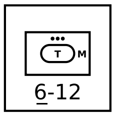
Tank Platoon: A tank platoon has armored vehicles principally armed with a direct-fire gun. The gun is most commonly mounted in a turret, but AFVs with casemate guns such as the SU-100, Kanonen JPz, and Strv 103 are also considered to be tanks. The role of a tank is the support of infantry and the destruction of other tanks and armored vehicles. A tank platoon may have heavy, medium, light, or open armor.[5]
Heavy examples: M1 Abrams, T-64, and T-72/80.
Medium examples: M60, T-34, T-54/55, and T-62.
Light examples: M114A2, M551 Sheridan, and PT-76.
Open example: ASU-57.

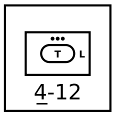


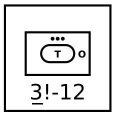


IFV Platoon: An infantry fighting vehicle (IFV) platoon is equipped with armored vehicles that provide infantry with tactical and operational mobility, protection, and fire support, being armed with a 20 mm or larger cannon and sometimes an anti-tank missile launcher. In most cases, the infantry must dismount to fight effectively. An IFV platoon may have medium or light armor.[6]
Medium example: M2A2 Bradley.
Light examples: M2/M2A1 Bradley and BMP-1/2.

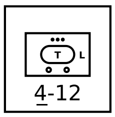


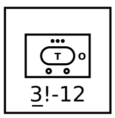




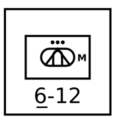


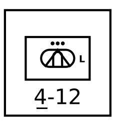
APC Platoon: An APC platoon is equipped with armored vehicles that provide infantry with tactical and operational transport and protection, but unlike IFVs do not provide fire support and at most are armed with a heavy machine gun. An APC platoon may have heavy, medium, light, or open armor. Heavy and medium APCs are typically converted tanks.[7]
Heavy example: Nakpadon.
Medium example: Nagmachon.
Light examples: M113 and BTR-50PK/60PB/70.
Open examples: M3 half-track and BTR-50P/60P.



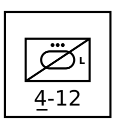


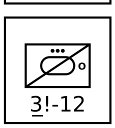
Armored Truck Platoon: An armored truck platoon has armored wheeled vehicles that can transport medium or heavy loads of infantry, towed weapons, and supplies, including fuel and ammunition. Unlike a truck platoon, it has armored protection, but unlike an APC platoon, it has more limited off-road mobility. An armored truck platoon may have light or open armor.[8]
Light example: BTR-152K.
Open examples: BTR-40 and other BTR-152 versions.



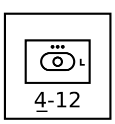


Armored Anti-Tank Missile Platoon: An armored anti-tank missile platoon has armored vehicles equipped with anti-tank guided missiles. Its mission is the destruction of other armored vehicles. Conventional gun-armed tank destroyers such as the Kanonen JPz are considered to be tanks. An armored anti-tank missile platoon is often a battalion- or regiment-level asset. It may have medium or light armor.[9]
Medium examples: Raketen JPz and Jaguar.
Light examples: M150, M901, and BRDM.

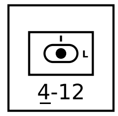


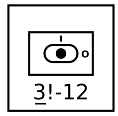


Armored Reconnaissance Platoon: An armored reconnaissance platoon has armored vehicles specialized in scouting and observation and with no offensive weapons more powerful than machine guns. (Vehicles used for reconnaissance that do have significant offensive weapons are typically classified as light tanks or IFVs.) An armored reconnaissance platoon may have light or open armor.[10]
Light examples: M114 and BRDM-2.
Open example: M20.

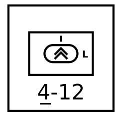


Armored Mortar Platoon: An armored mortar platoon has mortars mounted in armored vehicles (most commonly converted APCs or IFVs). It is typically a battalion-level asset in mechanized-infantry or tank battalions, and its role is indirect fire support. An armored mortar platoon has light armor.[11]
Examples: M106 and M125.

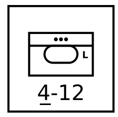




Armored Artillery Battery: An armored artillery battery has guns or howitzers for indirect-fire support in armored vehicles providing some protection, in particular from counter-battery fire, and mobility. It is typically a regiment-, brigade-, or division-level asset. An armored artillery battery may have light or open armor.[12]
Light examples: M109, 2S3, and 2S1.
Open example: M7 Priest.



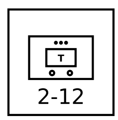
Armored Rocket Artillery Battery: An armored rocket artillery battery is the rocket-armed equivalent of an armored artillery battery. An armored rocket artillery battery has light armor. [13]
Example: M270.



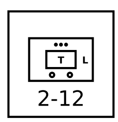
Armored HQ Platoon: An armored HQ platoon is the HQ unit of a tank, mechanized-infantry, or armored-artillery battalion, regiment, or brigade. It consists of the formation commander, their staff, and their protection detail. It is typically equipped with modified APCs or IFVs. An armored HQ platoon has light armor.

UNARMORED VEHICLE UNITS

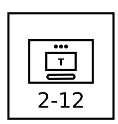
Unarmored or soft vehicles provide mobility, but do not provide (or provide only incomplete) armor protection for their crew and weapons. They include armored vehicles whose crew or weapons are exposed. They may be wheeled or tracked.



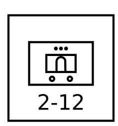
Truck Platoon: A truck platoon has unarmored wheeled vehicles that can transport medium or heavy loads of infantry, towed weapons, and supplies, including fuel and ammunition. It can provide operational mobility to infantry and towed artillery, but due to its extreme vulnerability to direct- and indirect-fire, is not typically used in a tactical role.



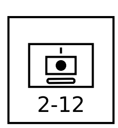
Light Truck Platoon: A light truck platoon has unarmored wheeled vehicles that can transport light loads of infantry, towed weapons, and supplies. It can provide transport to independent infantry FAC or SAM units, but in these roles it can avoid close contact with enemy combat units, mitigating its vulnerability.
Examples: M151 Jeep, HMMWV, and GAZ-69.



Tracked Carrier Platoon: A tracked carrier platoon is equipped with unarmored tracked vehicles that can transport medium loads of infantry, towed weapons, and supplies, including fuel and ammunition. It serves the same roles as trucks or light trucks, but in terrain that requires their greater mobility, in particular arctic regions.
Example: Bv 202.

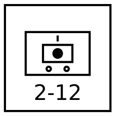


Mobile Anti-Tank Missile Platoon: A mobile anti-tank missile platoon is equipped with anti-tank missiles mounted on unarmored light trucks. It is extremely vulnerable to direct- and indirect-fire, and must rely on concealment and mobility.
Examples: M151 TOW and M966 HMMWV TOW.

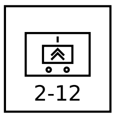


Tracked Artillery Battery: A tracked artillery battery has guns or howitzers mounted on tracked vehicles that provide mobility, but give only limited protection from small arms or artillery shrapnel. It is typically a regiment-, brigade-, or division-level asset, and its

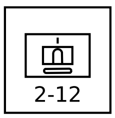
role is indirect fire support.[14]
Examples: M107, M110, 2S4, and 2S7.



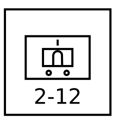
Mobile Artillery Battery: A mobile artillery battery has howitzers mounted on wheeled vehicles that provide mobility, but give only limited protection from small arms or artillery shrapnel. Lacking armor, it must avoid counter-battery fire by using their mobility to rapidly change firing positions. It is typically a regiment-, brigade-, or division-level asset, and their role is indirect fire support.[15]
Examples: CAESAR and Archer.



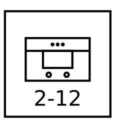
Mobile Rocket Artillery Battery: A mobile rocket artillery battery is the rocket-armed equivalent of a mobile artillery battery. Again, it relies on mobility to avoid counter-battery fire.[16]
Examples: M142 HIMARS and BM-21 Grad.



Tracked Missile Artillery Battery: A tracked rocket artillery battery is the rocket-armed equivalent of a tracked artillery battery. The tracked chassis provides mobility, but the missiles are exposed. Nevertheless, the long range of the missile permits deployment far behind the front line, mitigating its vulnerability.[17]
Example: M752 Lance.



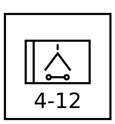
Mobile Missile Artillery Battery: A mobile missile artillery battery is the missile-armed equivalent of a mobile artillery battery. Again, the long range of the missile permits deployment far behind the front line, mitigating its vulnerability.[18]
Examples: M1003 Pershing II and OTR-21 Tocha (SS-21).



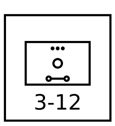
Mobile HQ Platoon: A mobile HQ platoon is typically the HQ unit of a battalion, regiment, or brigade other than those of infantry, tank, mechanized infantry, or armored artillery. It is equipped with trucks, and its personnel are not specialized in fighting as infantry.

TOWED UNITS

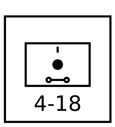
Towed weapons provides neither inherent mobility nor protection. However, they are light and cheap compared to similar vehicle-mounted weapons. Towed weapons are typically provided with appropriate ground or helicopter transport.



Towed Anti-Tank Gun Battery: A towed anti-tank gun battery has anti-tank guns on towed mounts with at most only very limited protection. It is often a regimental-, brigade-, or divisional-level asset.
Example: 2A18 (T-12) and 2A45 (Sprut).



Towed Mortar Platoon: A towed mortar platoon has 120 to 160 mm mortars on unprotected, towed mounts. It is often a regimental- or brigade- asset. Smaller infantry mortars are found in infantry mortar platoons.
Example: M1943.



Towed Artillery Battery: A towed artillery battery has guns or howitzers on towed mounts with at most only very limited protection. It is often a regimental-, brigade-, or divisional-level asset.
Examples: M119, M198, and 2A18 (D-30).

GROUND-COMBAT AND TRANSPORT UNIT TABLE

Type	Size[a]	Defense Strength	Sighting Range	Mobility	VPs 3D/2D/D	AAA Class[b]
Infantry	Platoon	4	6	U	5/3/2	B2
Infantry Weapons	Platoon	3	6	U	6/4/2	B2
Infantry Mortar	Platoon	3	6	U	6/4/2	B2
Infantry FAC	Squad	5	6	U	–/–/8	—
Infantry SAM	Squad	5	6	U		—
Infantry Heavy SAM	Squad	5	6	U		—
Infantry HQ	Platoon	4	6	U	10/7/3	B2
Heavy Tank	Platoon	8	12	U	12/8/4	B3
Medium Tank	Platoon	6	12	U	10/7/3	B3
Light Tank	Platoon	4	12	U	6/4/2	B3
Open Tank	Platoon	3!	12	U	5/3/2	B3
Medium IFV	Platoon	6	12	U	10/7/3	B3
Light IFV	Platoon	4	12	U	6/4/2	B3
Heavy APC	Platoon	8	12	U	10/7/3	B3
Medium APC	Platoon	6	12	U	8/5/3	B3
Light APC	Platoon	4	12	U	6/4/2	B3
Open APC	Platoon	3!	12	U	5/3/2	B3
Light Armored Truck	Platoon	4	12	G/P	6/4/2	B3
Open Armored Truck	Platoon	3!	12	G/P	5/3/2	B3
Medium Armored Anti-Tank Missile	Platoon	6	12	U	12/8/4	B3
Light Armored Anti-Tank Missile	Platoon	4	12	U	10/7/3	B3
Light Armored Reconnaissance	Platoon	4	12	U	6/4/2	B3
Open Armored Reconnaissance	Platoon	3!	12	U	4/3/1	B3
Light Armored Mortar	Platoon	4	12	U	10/7/3	B3
Light Armored Artillery	Battery	4	12	U	12/8/4	B3
Open Armored Artillery	Battery	3!	12	U	12/8/4	B3
Light Armored Rocket Artillery	Battery	4	12	U	12/8/4	B3
Light Armored HQ	Platoon	4	12	U	12/8/4	B3
Truck	Platoon	2	12	G/P	4/3/1	—
Light Truck	Platoon	2	12	G/P	4/3/1	—
Tracked Carrier	Platoon	2	12	U	4/3/1	—
Mobile Anti-Tank Missile	Platoon	2	12	G/P	8/5/3	—
Mobile Artillery	Battery	2	12	G/P	9/6/3	—
Tracked Artillery	Battery	2	12	U	10/7/3	—
Mobile Rocket Artillery	Battery	2	12	G/P	9/6/3	—
Tracked Missile Artillery	Battery	2	12	U	10/7/3	—
Mobile Missile Artillery	Battery	2	12	G/P	9/6/3	—
Mobile HQ	Platoon	2	12	G/P	10/7/3	—
Towed Mortar	Platoon	3	12	T	7/5/2	—
Towed Anti-Tank Gun	Battery	4	12	T	8/5/3	—
Towed Artillery	Battery	4	18	T	8/5/3	—

[a] Platoons and batteries are available as sections and squads with corresponding modifications to their damage resiliences and VPs. [b] Platoons with AAA class of B2 and B3 can employ barrage fire to altitudes of 2 and 3 levels, respectively.

VEHICLE TYPES TABLE

Heavy Tanks		
Al-Khalid/VT-1A	PK/CN	2001
Ariete	IT	1995
Arjun	IN	2004
Challenger 1	GB	1983
Challenger 2	GB	1998
Chieftain	GB	1967
Conqueror	GB	1955
IS-2	SU	1944
IS-3	SU	1945
K1	KR	1987
K2 Black Panther	KR	2014
Leclerc	FR	1992
Leopard 2	DE	1979
M-84	YU	1985
M1 Abrams	US	1980
Merkava 1	IL	1979
Merkava 2	IL	1983
Merkava 3	IL	1989
Merkava 4	IL	2004
PT-91	PL	1995
Stridsvagn 103	SE	1967
T-10	SU	1954
T-14 Armata	RU	2024
T-64	SU	1966
T-72	SU	1973
T-80	SU	1976
T-84	UA	1999
T-90	RU	1992
Type 10	JP	2012
Type 90	JP	1990
Type 99	CN	2001

Medium Tanks		
AMX-30	FR	1966
ASU-85	SU	1959
Centurion	GB	1945
Comet	GB	1944
Cromwell	GB	1944
ISU-122	SU	1944
ISU-152	SU	1943
Kanonen JPz	DE	1965
Leopard 1	DE	1965
M-50 Sherman	IL/FR	1956
M-51 Sherman	IL/FR	1961
M103	US	1957
M26 Pershing	US	1944
M4 Sherman	US	1942
M46 Patton	US	1949
M47 Patton	US	1951
M48 Patton	US	1952
M60	US	1959
Panzer 61	CH	1965
Panzer 68	CH	1971
SU-100	SU	1944
T-34	SU	1940
T-54	SU	1955
T-55	SU	1958
T-62	SU	1961
TAM	AR	1983
Type 59	CN	1959
Type 61	JP	1961
Type 69	CN	1972
Type 74	JP	1975
Type 79	CN	1984
Type 85	CN	1988
Type 88	CN	1988
Type 96	CN	1997
Type 15	CN	2018
Vickers MBT	GB	1965
Vijayanta	IN/GB	1965

Light Tanks		
AMX-10 RC	FR	1981
AMX-13	FR	1953
Fox	GB	1975
Commando (90 mm)	US	1964
LAV I Cougar	CA	1976
LAV II Coyote	CA	1990
M8/Greyhound	US	1943
M24 Chaffee	US	1944
M41 Walker Bulldog	US	1953
M113 C&V with 25 mm	US	1974
M114A2	US	1969
M551 Sheridan	US	1969
Panhard AML	FR	1961
Panhard EBR	FR	1951
PT-76	SU	1951
Saladin	GB	1958
Scimitar	GB	1974
Scorpion	GB	1973
SpPz Luchs	DE	1975
Stryker MGS	US/CA	2006
Strv 74	SE	1958
Type 62	CN	1961
Type 63	CN	1963
Type 05 AAV	CN	2005
Type 08 ARV	CN	2008

Open Tanks		
ASU-57	SU	1951
SU-76	SU	1943

Medium IFVs		
BMP-3	SU	1987
Boxer IFV	NL/DE	2011
Strf 9040/CV90	SE	1994
LAV III (add-on-armor)	CA	2009
M2A2/3/4 Bradley	US	1988
M3A2/3/4 Bradley	US	1988
SPz Marder	DE	1971
SPz Puma	DE	2015

Light IFVs		
AIFV	US	1979
AMX-10P	FR	1973
BMP-1	SU	1966
BMP-2	SU	1980
Commando (20 mm)	US	1963
FV432 RARDEN	GB	1975
LAV III	CA	1999
LAV-25	CA	1983
M2 and M2A1 Bradley	US	1981
M3 and M3A1 Bradley	US	1981
M113A1 FSC	AU	1972
Piranha IFV	CH	1974
Pbv 301	SE	1961
Pbv 302	SE	1966
SPz Kurz	FR	1959
SPz Lang HS.30	CH	1960
Stryker ICVD	US	2018
Type 86	CN	1992
Type 92	CN	1992
Type 05 IFV	CN	2005
Type 08 IFV	CN	2008
VCBI	FR	2008
Warrior	GB	1987
YPR-765	NL	1977

Heavy APCs		
Achzarit	IL	1988
Nakpadon	IL	1998
Namer	IL	2008
Puma	IL	1993

Medium APCs		
Boxer APC	NL/DE	2011
Nagmachon	IL	2000
Nagmashot	IL	1990

Light APCs		
Bronco/Warthog	SG	2001
BvS10/Viking	SE	2005
BTR-50PK	SU	1958
BTR-60PA/PAI/PB	SU	1965
BTR-70	SU	1972
BTR-80	SU	1986
Commando	US	1963
FV432	GB	1963
LAV I Grizzly	CA	1976
LAV II Bison	CA	1990
LVTP-5	US	1956
LVTP-7 (AAV)	US	1972
M59	US	1954
M75	US	1952
M113	US	1960
OT-64 SKOT	CS/PL	1963
OT-810 half-track	CS	1958
Piranha APC	CH	1974
Saracen	GB	1952
Spartan	GB	1978
Stryker ICV	US/CA	2002
TPz Fuchs	DE	1979
Type 63	CN	1978
Type 77	CN	1978
Type 89	CN	1978
Type 08 APC	CN	2008
VAB	FR	1979
YP-408	NL	1964

Open APCs		
AT-P	SU	1954
BTR-50P	SU	1952
BTR-60P	SU	1959
M3 half-track	US	1941

Light Armored Trucks		
BTR-152K	SU	1959
Saxon	GB	1983

Open Armored Trucks		
BTR-40	SU	1950
BTR-152A	SU	1950
SKP/VKP	SE	1944
SKPF/VKPF	SE	1956

Medium Armored Anti-Tank		
Raketen JPz 1	DE	1961
Raketen JPz 2	DE	1967
Jaguar 1	DE	1978
Jaguar 2	DE	1983

Light Armored Anti-Tank		
BRDM-2 ATGM	SU	1973
BRDM-3 ATGM	SU	1973
Fennek MRAT	DE/NL	2003
FV438 Swingfire	GB	1971
M150	US	1973
M901 ITV	US	1979
Striker	GB	1976
Stryker ATGM	US/CA	2003
VAB HOT	FR	1978
YP-408 PRAT	NL	1976
YPR-765 PRAT	NL	1980

Light Armored Reconnaissance		
BRDM-2	SU	1973
Fennek	DE/NL	2003
Ferret	GB	1952
M113 C&V/Lynx	US	1966
M114	US	1962

Open Armored Reconnaissance		
M20	US	1943

Light Armored Mortar		
FV432 with 81 mm	GB	1963
M106	US	1964
M125	US	1964
M1064	US	1990
Stryker MCV	US/CA	2005

Light Armored Artillery		
2S1 Gvozdika	SU	1972
2S3 Akatsiya	SU	1971
2S19 Msta-S	SU	1989
AMX-30 AuF1	FR	1977
AS-90	GB	1992
Bkan 1	SE	1967
FV433 Abbot	GB	1965
L-33 Ro'Em	IL	1973
M52	US	1955
M108	US	1962
M109	US	1963
Panzerhaubitze 2000	DE	1998

Open Armored Artillery		
AMX-105A	FR	1958
M-50	IL	1962
M7 Priest	US	1942
M40	US	1945
M44	US	1954

Light Armored Rocket Artillery		
M270 MLRS	US	1983
BM-1	RU	1988

Light Truck		
GAZ-69	SU	1953
HMMWV	US	1985
Ilitis	GE	1978
Landrover	UK	1949
M38 1/4-ton truck	US	1949
M151 1/4-ton truck	US	1960
UAZ-469	SU	1971
Willys MB 1/4-ton truck	US	1941

Tracked Carrier		
AT-T/S/L	SU	1950
Bv 202	SE	1964
Bv 206	SE	1980
BvS10/Beowulf	SE	2024
M548	US	1960
Sisu Nasu	FI	1986

Mobile Anti-Tank Missile		
M151 TOW	US	1970
M966 HMMWV TOW	US	1985

Mobile Artillery		
Archer	SE	2016
CAESAR	FR	2004

Tracked Artillery		
2S4 Tyulpan	SU	1975
2S7 Pion	SU	1976
M41	US	1942
M107	US	1962
M110	US	1963
AMX-13 F3	FR	1962

Mobile Rocket Artillery		
BM-8	SU	1941
BM-13	SU	1941
BM-31	SU	1944
BM-24	SU	1947
BM-14	SU	1952
BM-21 Grad	SU	1963
LARS	GE	1970
BM-27 Uragan	SU	1975
Valkyr	ZA	1982
BM-30 Smerch	SU	1989
M142 HIMARS	US	2005

Tracked Missile Artillery		
M752 Lance	US	1972
AMX-30 Pluton	FR	1974

Mobile Missile Artillery		
9K52 Luna-M (Frog-7)	SU	1962
Corporal	US	1954
Sergeant	US	1962
M34/M50 Honest John	US	1953
M74 Redstone	US	1958
M790 Pershing 1a	US	1962
M1003 Pershing II	US	1983
OTR-21 Tochka (SS-21)	SU	1975
RSD-10 Pioneer (SS-20)	SU	1976

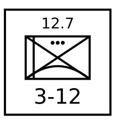
AAA UNITS

AAA units use guns to destroy attacking aircraft, but also have a secondary ground-combat capability. Some also have SAMs, giving a layered capacity. AAA units are presented below with their counters, ordered first by caliber and then by year of introduction.

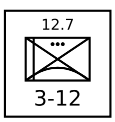
The AAA Unit Table summarizes the properties of AAA units. As well as the basic properties common to all ground units, the table gives: the AAA class (light, medium, or heavy); the maximum altitude; the limits of short, medium, and long range in hexes; the hit rolls at short, medium, and long range; the damage rating; whether the unit has an all-weather or ranging-only fire-control radar and its frequency band or a laser rangefinder; whether it has night infrared sights; and whether it is also equipped with a SAM system. If the unit has a SAM, this will be described in detail in the section on SAM launcher units.[19]

All AAA platoons and batteries are also available as sections (but not squads), with damage resiliences of 2, hit rolls reduced by 1, and VPs corresponding to a platoon or battery that has taken 1D damage.

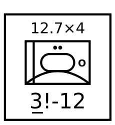
LIGHT AAA UNITS



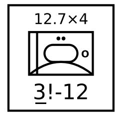
M2 Platoon: The Browning M2 .50 cal (12.7 mm) heavy machine gun is normally found on a low tripod for ground combat or mounted on armored fighting vehicles, but here is used on a tall dual-purpose mount suitable for air defense. A platoon has six guns. The M2 gun was introduced in 1933 and has been used by United States forces and their allies in numerous conflicts since then.[20]



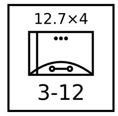
DShK Platoon: The DShK M1938 12.7 mm heavy machine gun is most commonly found on a low, wheeled mount for ground combat or mounted on armored fighting vehicles, but here is used on a tall tripod suitable for air defense. It is nicknamed the Dushka (darling) or Dochka (daughter). A platoon has six guns. The DShK entered service with the Soviet Army in 1938 and saw action in WW2 and numerous conflicts since then, including the Korean and Vietnam Wars, in which the Soviet Union participated or provided arms. In particular, it was the main air-defense gun for Chinese and North Korean field units in the early part of the Korean War.[21]



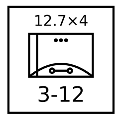
M16 Section: The M16 Multiple Gun Motor Carriage has an M45 turret with four M2 .50 cal (12.7 mm) heavy machine guns mounted on a modified M3 half-track. A section has two vehicles. The M16 is a development on the M13 and M14 Multiple Gun Motor Carriages, which have two M2 heavy machine guns in a similar turret. As a single M2 has an insufficient rate of fire to be effective even against WW2 aircraft, first two and then four were used on a single mount. The M16 was used by United States forces during WW2. It was also used by US, Commonwealth, and South Korean forces in the Korean War, primarily in an anti-infantry role. It was largely withdrawn form US services in late 1951 and officially retired in 1958. Israeli forces employed it during the 1956 Arab-Israeli War. Other post-WW2 users include Belgium, France, West Germany, Japan, Netherlands, Philippines, Portugal, and Thailand[22]



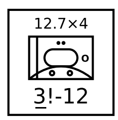
M17 Section: The M17 Multiple Gun Motor Carriage is the equivalent of the M16 Multiple Gun Motor Carriage, but based on the slightly different M5 half-track. A section has two vehicles. The M17 was supplied to the Soviet Union and entered service in 1944. It was presumably replaced by the BTR-40A.[23]



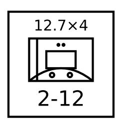
M55 Platoon: The M55 has an M45 turret with four M2 .50 cal (12.7 mm) heavy machine guns (the same turret as used in the M16) mounted on a towed carriage. A platoon has four gun mounts. The M55 was used by United States forces during the Korean War and by Pakistani forces during the 1965 India-Pakistan War.[24]



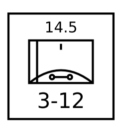
Vz.53 Platoon: The Czechoslovak Vz.53 has four Vz.38/46 12.7 mm machine guns (copies of the Soviet DShK) on a light, wheeled mount. It is sometimes referred to as the M53. A platoon has six guns. It entered service with the Czechoslovak Army in 1953, and was exported to Cuba, Egypt (from 1955), Iraq, Syria, and Vietnam. Cuba subsequently exported some to Grenada. It saw action with the Egyptian Army in the 1956 War, with Cuba in the Bay of Pigs Invasion in 1961, in the Vietnam War, in the Nicaraguan Revolution, with the Grenadian Army during the invasion by US force in 1983[25]

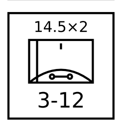


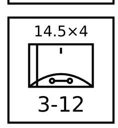
BTR-152A Vz.53 Section: The BTR-152A Vz.53 has a Czechoslovak Vz.53 gun mount with four 12.7 mm Vz.38/46 machine guns in the open, six-wheeled Soviet BTR-152 APC. The gun mount is also referred to as the M53. A section has two vehicles. It served only in the Egyptian Army, and saw action in the 1956 War.[26]



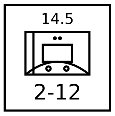
Mobile M45 Section: The mounted M45 has the M45 turret with four M2 .50 cal (12.7 mm) heavy machine guns (the same turret as used in the M16) removed from an M55 trailer and mounted on a truck. During the Vietnam War, in 1967, US forces improvised these *gun trucks* for convoy security[27]

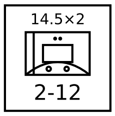


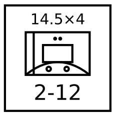




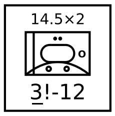
ZPU-1/2/4 Battery: The ZPU-1/2/4 have one, two, or four 14.5 mm heavy machine guns in open mounts on towed carriages. A battery has six gun mounts. The ZPU-1/2/4 was introduced in 1949 and saw extensive service with Soviet or Soviet-supported forces in many subsequent conflicts, including with communist forces in the Korean War and Vietnam War and with Arab forces in the various Arab-Israeli Wars from 1956 onwards.[28]



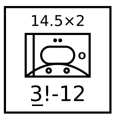


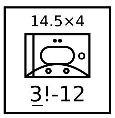


Mobile ZPU-1/2/4 Section: The mobile ZPU-1/2/4 is a light truck with a ZPU-1/2/4 gun mount. A section has two vehicles. Many users of the ZPU-1/2/4 adapt it in this way for increased mobility, and like the towed versions, these mounted versions have seen combat in many conflicts. Formal conversions tend to be used for air-defense, but informal conversions or *technicals* are mainly used for ground combat, serving as light cavalry or infantry support.[29]

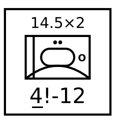


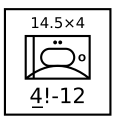
BTR-40A Section: The BTR-40A has a twin 14.5 mm ZPU-2 turret in the open, four-wheeled BTR-40 APC. A section has two vehicles. The BTR-40A entered service with the Soviet Army in 1950, presumably replacing the M17. It was also used by East Germany and Cuba.[30]



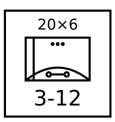


BTR-152A ZPU-2/4 Section: The BTR-152A has a twin 14.5 mm ZPU-2 or quad 14.5 mm ZPU-4 turret in the open, six-wheeled BTR-152 APC. A section has two vehicles. The BTR-152A entered service with the Soviet Army in 1951, presumably replacing the BTR-40A. The BTR-152D/E are similar, but based on the improved BTR-152V/V1 variants. Other users include Ethiopia in the Ogaden War and Egypt.[31]

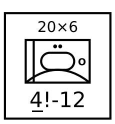




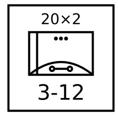
M113 ZPU-2/4 Section: The M113 ZPU-2/4 has a twin 14.5 mm ZPU-2 or quad 14.5 mm ZPU-4 turret mounted on an M113 APC. A section has two vehicles. They were created by Lebanese militias during the Lebanese Civil War and were used mainly for ground combat.[32]



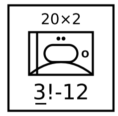
M167 Platoon: The M167 Vulcan Air Defense System (VADS) has a 20 mm six-barreled Vulcan gun in an open mount on a towed carriage. A platoon has four gun mounts. The fire-control system has optical sights and a range-only radar. The M167 was introduced into service with the US Army in 1967, principally for airfield defense, and subsequently served with other US allies.[33]



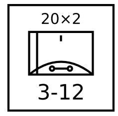
M163 Section: The M163 Vulcan Air Defense System (VADS) has a 20 mm six-barreled Vulcan gun in a lightly armored but open-topped turret on an adapted M113 APC. A section has two vehicles. The fire-control system has optical sights and a range-only radar. The M163 VADS was introduced into service with the US Army in 1968 and has since been adopted by other US allies. The M163 was used by US forces in the Vietnam War, the 1989 Invasion of Panama, and the 1991 Gulf War. It was also used by the Israel Army in the 1982 Lebanon War and in 2002 during the Second Intifada. From 1988 in the US Army, each M163 was typically issued with a FIM-92 Stinger launcher with two rounds, so each M163 section effectively carried a pair of FIM-92 squads.[34]



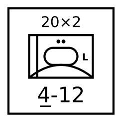
TCM-20 Platoon: The TCM-20 is an Israeli adaptation of the M55, with two Hispano-Suiza 20mm guns from obsolete aircraft replacing the four .50 cal machine guns. As with the M55, the fire-control system is entirely optical. A platoon has four gun mounts. The TCM-20 was employed by Israeli forces from 1970 and subsequently saw combat in the 1973 Arab-Israeli War and the 1982 Lebanon War.[35]



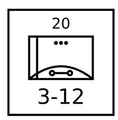
M3 TCM-20 Section: The M3 TCM-20 has a TCM-20 turret Hispano-Suiza 20mm guns mounted in an adapted M3 half-track, like the earlier M16. A section has two vehicles. The M3 TCM-20 was used by Israeli forces in the 1973 Arab-Israeli War and the 1982 Lebanon War.[36]



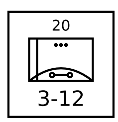
Rh-202 Battery: The Rheinmetall Rh-202 has a twin 20mm gun in an open mount on a towed carriage. The fire-control system is entirely optical. A battery has six gun mounts. It entered service with West German forces in 1972 and was used for airfield defense. Subsequently, it was exported and notably used by the Argentine Air Force in the 1982 South Atlantic War in the defense of BAM Malvinas (Port Stanley Airport) and BAM Condor (Goose Green Airfield).[37]



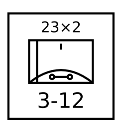
Panhard M3 VDA Section: The Panhard M3 VDA vehicles have an armored turret for twin HS.820 20 mm guns mounted on the Panhard M3 APC. A section has two vehicles. The guns have optical sights, but typically one vehicle in each section or platoon is equipped with a search-and-ranging radar and can share this information automatically with the others. The Panhard M3 VDA served with the armed forces of Côte d'Ivoire, Niger, and the UAE.[38]



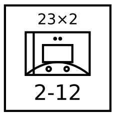
53 T1 Platoon: The 53 T1 has a single 20 mm F1 cannon in an open mount on a towed carriage. It is an adaptation of the much earlier 1950s 53 T1 with the MG 151 cannon replaced with a F1 cannon. The F1 cannon is an early version of the M693 and is derived from the HS 820. A platoon has four guns. In small numbers, the 53 T1 entered service with the French Army in 1976.[39]



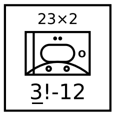
Tarasque Platoon: The 53 T2 Tarasque has a single 20 mm F1 cannon in an open mount on a towed carriage. It uses the same M693 cannon as the 53 T1, but mounted in a more ergonomic mount. It is named for a mythological beast from Provence with a poisonous breath. A platoon has four guns. The Tarasque entered service with the French Army in 1976.[40]



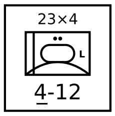
ZU-23 Battery: The ZU-23 has twin 23 mm cannons in an open mount on a towed carriage. It was designed as a more powerful replacement for the ZPU-1/2/4. A battery has six gun mounts. The fire-control system is entirely optical. The ZU-23 was introduced into service by the Soviet Army in 1960. It later served with the armed forces of many Soviet allies, including with North Vietnam in the Vietnam War, with Egypt and Syria in the 1967 and 1973 Wars, with Soviet forces in the Soviet-Afghan War, with Iraq in the Iraq-Iran War, with Syria in the 1982 Lebanon War, and with both sides in the Afghan Civil War.[41]



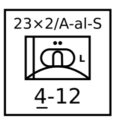
Mobile ZU-23 Section: This is a section of two light trucks mounting ZU-23 guns. As with the earlier ZPU-1/2/4, many users of the ZU-23 mount it on vehicles, including trucks, tracked carriers, and APCs, for better mobility. Informal conversions are typical used as *technicals* in ground combat.[42]

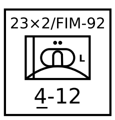


BTR-152A ZU-23 Section: The BTR-152A ZU-23 has a twin 23 mm ZU-23 turret in the open, six-wheeled BTR-152 APC. A section has two vehicles. It entered service with the PLO prior to the 1982 War in Lebanon; the in-service data is a guess.[43]

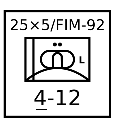


ZSU-23-4 Section: The ZSU-23-4 has four radar-controlled 23 mm guns, similar to those in the ZU-23, mounted in an armored turret on a hull adapted from the PT-76 light tank. A section has two vehicles. The ZSU-23-4 entered service in the Soviet Army in 1965, replacing the ZSU-57-2 and outclassing all NATO anti-aircraft guns at that time. A platoon of four was assigned to the air-defense battery of tank and motor-rifle regiments and later complemented by a platoon of four SA-9 vehicles. It was exported to Soviet allies and saw extensive combat with Arab forces in the 1973 Arab-Israeli War and with Iraqi forces in the Iran-Iraq War and the 1991 Gulf War.[44]

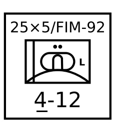




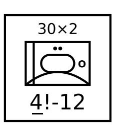
Sinai 23 Section: The Sinai 23 has a Dassault TA20 turret with two 23 mm guns and six Ayn-al-Saqr (similar to the Strela-2M) or Stinger missiles on an M113 chassis. A section has two vehicles. Each battery of two sections operates with one vehicle equipped with an RA 20S search and ranging radar. The Sinai 23 entered service with the Egyptian Army in 1992.[45]



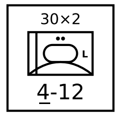
M2 Blazer: The M2 Blazer has a 25 mm GAU-12 rotary cannon mounted in a turret on the M2 Bradley IFV. It was proposed in 1985 for the "Line-of-Sight-Forward (Heavy)" component of the US Army's Forward Area Air Defense System, but did not progress beyond a prototype. It has a search radar, but fire-control is based optical tracking with a passive night IR sight and laser rangefinder. In addition to the gun, it is armed with four FIM-92D Stinger missiles.[46]



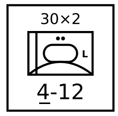
LAV-AD Section: The LAV-AD has a 25 mm GAU-12 rotary cannon and eight FIM-92D Stinger missiles mounted in a turret on the LAV-25 vehicle. A section has two vehicles. The LAV-AD does not have radar, but is equipped with a laser rangefinder and passive night IR sight. It entered service with the USMC in 1997, but was retired after only a few years.[47]



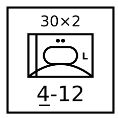
M53/59 Section: The M53/59 has a twin M53 30 mm gun mount on a partially armored truck. The fire-control system is entirely optical. A section has two vehicles. It entered service in the Czech Army in 1959 and was used instead of the ZSU-57-2. It was also exported to Iraq and saw combat in the Iran-Iraq War and the 1991 Gulf War.[48]



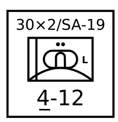
AMX-13 DCA Section: The AMX-13 DCA is equipped with a pair of 30 mm Hispano-Suiza HS.831A guns mounted in an armored turret on the hull of an AMX-13 tank. The Thomson-CSF Œil Noir radar provides search and ranging, but tracking is manual using an optical sight. Using such a heavy turret on the hull of a light tank gave relatively poor mobility and led directly to the development of the AMX-30 DCA. A section has two vehicles. The AMX-13 DCA served only in the French Army, entering service in 1969 and being retired in the 1990s.[49]



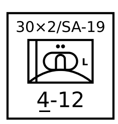
AMX-30 DCA Section: The AMX-30 DCA is equipped with a pair of 30 mm Hispano-Suiza HS.831A guns mounted in an armored turret on the hull of an AMX-30 tank. The turret, guns, radar, and fire-control system are very similar to those of the AMX-13 DCA, but the more powerful hull gives superior mobility and the ability to carry more ammunition. A section has two vehicles. The AMX-30 DCA was developed in the late 1960s and per se did not advance beyond the prototype state, but it did form the basis of the AMX-30SA.[50]



AMX-30SA Section: The AMX-30SA is equipped with a pair of radar-controlled 30 mm Hispano-Suiza HS.831A guns mounted in an armored turret on the hull of an AMX-30 tank. It is a development of the AMX-30 DCA, with adaptations for desert conditions and an improved Œil Vert radar. As in the AMX-13 DCA, the radar provides search and ranging, but tracking is manual using an optical sight. A section has two vehicles. AMX-30SA served only in the Saudi Army from 1979.[51]

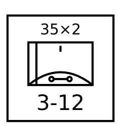


Tunguska Section: The Tunguska has twin radar-controlled 30mm guns and four SA-19 missiles in an armored turret on an armored and tracked hull. A section has two vehicles. The Tunguska was designed to counter the A-10 and AH-64, against which the earlier ZSU-23-4 had weaknesses. It entered service in the Soviet Army in 1984. It has been used by Russia, Ukraine, and Belarus after the break-up of the Soviet Union and has been exported. The Tunguska is referred to as the "ZSU-30-2" in *Air Strike*.[52]

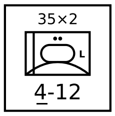


Tunguska-M Section: The Tunguska-M is an improved version of the Tunguska with eight missiles instead of four. A section has two vehicles. It entered service in the Soviet Army in 1990. It has been used by Russia, Ukraine, and Belarus after the break-up of the Soviet Union and has been exported.[53]

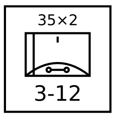
MEDIUM AAA UNITS



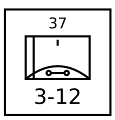
Oerlikon GDF-001 Battery: The Oerlikon GDF-001 has two 35 mm guns in an open mount on a towed carriage. The GDF-002/3/4 are similar. A battery has six gun mounts. The basic fire-control system is optical, but a section is often coupled with the Super Fledermaus or Skyguard fire-control radars. (These radars cannot control a whole battery, only a section.) The Oerlikon GDF entered service in 1963 and has served widely, including with the armed forces of Argentina, Brazil, Cameroon, China, Ecuador, Greece, Iran, Indonesia, Japan, South Korea, Nigeria, Romania, Singapore, South Africa, Switzerland, Taiwan, Turkey, and the UK. Early-model GDFs saw combat in the South Atlantic War with the Argentine forces in the defense of BAM Malvinas (Port Stanley Airport) and BAM Condor (Goose Green Airfield) and in the 2025 India-Pakistan conflict with Pakistan forces.[54]



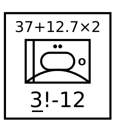
Gepard Section: The Gepard has two radar-controlled 35 mm guns in an armored turret on an adapted Leopard 1 tank hull. A section has two vehicles. The Gepard entered service in 1976 with the West German Army and also served with the Belgian and Dutch armies. After the end of the Cold War, they were retired by these armies, and many were sold on to other countries.[55]



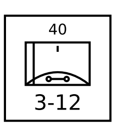
Oerlikon GDF-005 Battery: The Oerlikon GDF-005 has two 35 mm guns in an open mount on a towed carriage. A battery has six gun mounts. The Oerlikon GDF-005 is an upgraded version of the earlier GDF-004, featuring a laser rangefinder and a digital fire-control computer. Like earlier models, it is often used with the Skyguard FCR. The Oerlikon GDF-005 entered service in 1985 and has served widely, including with the armed forces of Austria, Bahrain, Bangladesh, Canada, Chile, Colombia, Cyprus, Kuwait, Malaysia, Oman, Pakistan, Saudi Arabia, South Africa, Spain, Switzerland, Thailand, and UAE.[56]



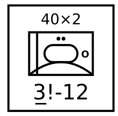
61-K Battery: The 61-K (M1939) has a single 37 mm gun in an open mount on a towed carriage. A battery has six gun mounts. The fire-control system is entirely optical, and (as an exception to the normal rule allowing medium AAA units to employ external FCRs) the 61-K cannot be coupled to an external fire-control radar. After entering service in 1939 with the Soviet Army, it served in WW2 and after until replaced by the S-60 57 mm gun. It also served with many Soviet allies, and in particular with communist forces in the Korean and Vietnam Wars. The 61-K is referred to as the "M-38" in *Air Strike*, *Eagles of the Gulf*, and *The Speed of Heat*.[57]



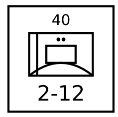
M15 Section: The M15 Combination Gun Motor Carriage has a M1 37-mm cannon and two Browning M2 .50 cal (12.7 mm) machine guns in an M42 turret mounted on a modified M3 half-track. A section has two vehicles. The M15 entered service with the US Army in 1942, saw combat in WW2 and the Korean War, in the latter mainly in a ground-support role, and was retired in 1953. It also served with the armed forces of Japan and Yugoslavia.[58]



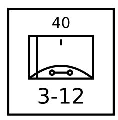
Bofors L/60 Battery: The Bofors L/60 has a single 40 mm gun in an open mount on a towed carriage. A battery has six gun mounts. The fire-control system is entirely optical. The Bofors L/60 entered service in the 1940s and was widely used by Allied forces during WW2, both on land and sea. After the war, many continued to see service, although it was increasingly inadequate against faster jet aircraft and in many cases was replaced by the new L/70 model.[59]



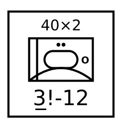
M19 Section: The M19 Multiple Gun Motor Carriage has two Bofors 40 mm L/60 guns in a partially armored mount on a tracked chassis based on that of the M24 light tank. A section has two vehicles. The fire-control system is entirely optical. The M19 entered service in 1944 with the US Army, but did not see combat in WW2. It was used in the Korean War, mostly in against ground targets, but was retired in 1953 and replaced by the M42. It served in the Netherlands Army from 1951 to 1978.[60]



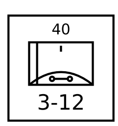
M34 Section: The M34 Gun Motor Carriage has a Bofors 40 mm L/60 cannon in an open mount on a modified M15 half-track. Many M15s were converted into M34s as supplies of 37-mm shells for the M1 cannon became scarce. Unlike the M15, the gun crew have no armored protection. A section has two vehicles. The M34 entered service with the US Army in 1951, saw combat in the Korean War, mainly in a ground-support role, and was likely retired in 1953.[61]



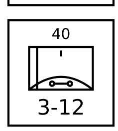
Bofors L/70 Battery: The Bofors L/70 has a single 40 mm gun in an open mount on a towed carriage. A battery has six gun mounts. Between the 1930s and the end of WW2, aircraft speed increased substantially, and so the engagement time of the Bofors L/60 became inadequate. The new L/70 fired a lighter shell at a higher velocity and achieved a significant improvement in range and engagement time. The basic fire-control system is optical, but various external fire-control radars can be used with it, including the Super Fledermaus and Flycatcher systems. NATO forces and others adopted it widely starting in 1952. Israel used batteries with the Super Fledermaus FCR for point-defense in the 1967 war.[62]



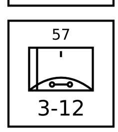
M42 Section: The M42 Duster has two Bofors 40 mm L/60 guns in a partially armored mount on a tracked chassis based on that of the M41 light tank. A section has two vehicles. The fire-control system is entirely optical. The M42 entered service in 1953 with the US Army and US Marine Corps, replacing the M19, saw combat in the Vietnam War against ground targets, and served at least with reserve units until 1988. It was also used by Austria, Greece, Japan, Jordan, Lebanon, Pakistan, Taiwan, Thailand, Tunisia, Turkey, Venezuela, Vietnam (South and North), and West Germany.[63]



Bofors L/70 BOFI: The BOFI version of the Bofors L/70 adds modern sights, a laser rangefinder, and proximity-fuzed shells. A battery has six gun mounts. The BOFI is used by the armed forces of Brazil.[64]

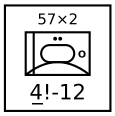


Bofors L/70 BOFI-R Battery: The BOFI-R version of the Bofors L/70 also adds an integrated fire-control radar to the BOFI version. A battery has six gun mounts.[65]

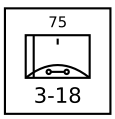


S-60 Battery: The S-60 is a Soviet single 57 mm gun on an open mount on a towed carriage. A battery has six gun mounts. The basic fire-control system is optical, but the gun is often used with the SON-9 (Fire Can) radar. The S-60 entered service in 1950 with

Soviet forces, replacing the 61-K 73 mm gun. It was exported to many Soviet allies and saw combat with Arab forces in the 1967 and 1973 Arab-Israeli War, with communist forces in the Vietnam War, and with Iraqi forces in the Iran-Iraq War and the Gulf War. However, it was not used by communist forces in the Korean War. When paired with the SON-9 (Fire Can) radar, it was considered one of the more dangerous AAA systems faced by US aircraft in Vietnam.[66]

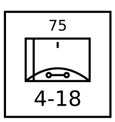


ZSU-57-2 Section: The ZSU-57-2 has two 57 mm cannons in a lightly armored but open-topped turret on an armored and tracked hull. A section has two vehicles. The cannons are similar to the S-60 cannons and the fire-control system is optical. The ZSU-57-2 entered service in the Soviet Army in 1955 and replaced the BTR-40A and BTR-152A self-propelled 14.5 mm AA guns in the AA batteries of tank regiments. It was in turn replaced by the ZSU-23-4 from 1965. The ZSU-57-2 also served widely with Soviet allies and saw combat with Egypt and Syria in the 1967 and 1973 Wars, with Syria in the 1982 Lebanon War, with North Vietnam in the 1972 Easter Offensive and the 1975 Spring Offensive, and with various factions in the Yugoslav Wars in the 1990s.[67]

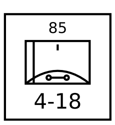


M51 Battery: The M51 Skysweeper has a single 75 mm gun in an open mount in a towed carriage. A battery has four mounts. The mount incorporates the T38 radar fire-control system and the gun has a relatively high rate of fire of 45 rounds per minute. Proximity-fuzed shells are available. The M51 entered service with the US Army in 1951 and was used for defense of strategic locations in the USA and in Europe, but was retired in 1959. It also served with Greece, Japan, and Turkey, with the last examples being retired until the mid-1970s. [68]

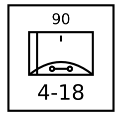
HEAVY AAA UNITS



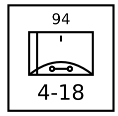
Flab Kan 38 Battery: The Swiss Flab Kan 38 (Fliegerabwehr Kanone 38) has a single 75 mm L/49 gun in an open mount on a towed carriage. It is a licensed copy of the French Canon de 75 contre aéronefs modèle 1939 Schneider. A battery has six gun mounts. Its fire control system is optical, but in Swiss service from 1958 it was paired with the Mk.7 Blue Cedar radar. The Flab Kan 38 entered service with the Swiss Air Force in 1939 and was retired in 1967.[69]



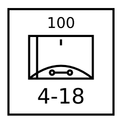
KS-12 Battery: The KS-12 (M1939 52-K) and the very similar KS-12A (M1944 KS-1) have a single 85 mm gun in an open mount on a towed carriage. A battery has six gun mounts. The rate of fire of the KS-12 is 10 rounds per minute. Its fire-control system is optical, but the guns are often paired with the SON-9 (Fire Can) radar. The KS-12 entered service with Soviet forces in 1939. After WW2, it began to be replaced in Soviet service by the KS-19 100 mm and KS-30 130 mm guns, but was widely used by Soviet allies and saw combat with communist forces in the Korean War and the Vietnam War.[70]



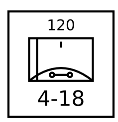
90 mm M2 Battery: The 90 mm M2 has a single 90 mm gun on a towed carriage. A battery has four guns. The fire control system of the M2 is optical, but an SCR-584 FCR-A normally controls each battery. The mechanical loading system permits a rate of fire of 24 rounds per minute. Proximity-fuzed shells are available. The M2 entered service with the US Army in 1943, following on from the similar M1 in 1940, and served as the main US heavy AA gun in WW2 and the Korean War. It was retired in the mid 1950s. It also served with Canadian (until 1960), French (until 1960), and Turkish forces.[71]



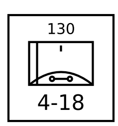
QF 3.7-inch Battery: The QF 3.7-inch has a single 94 mm gun on a towed carriage, but typically is used from a prepared firing site. The fire control system is optical, but an SCR-584 FCR-A normally controls each battery. In the final Mk VI version, the mechanical loading system permits a rate of fire of 20 rounds per minute. Proximity-fuzed shells are available. The QF 3.7-inch entered service with the British Army in 1937, served as its principal heavy AA gun during WW2 and beyond, and was withdrawn in 1957. It also served with Australia, Belgium, Canada, Sri Lanka, Cyprus, India, Ireland, Israel, Malta, Nepal, New Zealand, Pakistan, South Africa, Yugoslavia, and Portugal.[72]



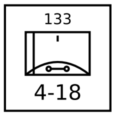
KS-19 Battery: The KS-19 has a single 100 mm gun in an open mount on a towed carriage. A battery has six gun mounts. The fire-control system of the KS-19 is optical, but like other contemporary Soviet guns, it was often paired with the SON-9 (Fire Can) radar. The KS-19 entered service in 1948 with Soviet forces as a replacement for the KS-12 85 mm gun. It also served with many Soviet allies and saw combat with communist forces in the Korean War and Vietnam War and Iraqi forces in the Iran-Iraq and Gulf Wars.[73]



120 mm M1 Battery: The M1 has a single 120 mm gun in an open mount on a towed carriage, but typically is used from a prepared firing site. A battery has four guns. The fire control system of the M1 is optical, but an SCR-584 FCR-A normally controls each battery. The rate of fire is 12 rounds per minute. Proximity-fuzed shells are available. The M1 entered service with the US Army in 1944, and though deployed to the Pacific Theater did not engage on enemy aircraft. After the war, it was widely deployed to defend of strategic targets against Soviet Tu-4 bombers, but was withdrawn by 1960. It also served with Canadian forces.[74]



KS-30 Battery: The KS-30 has a single 120 mm gun in an open mount on a towed carriage. A battery has six gun mounts. Development of the KS-30 started after WW2, and it has features adopted from captured German guns as well as British QF 3.7-inch guns. The fire-control system is optical, but like other contemporary Soviet guns, it was often paired with the SON-9 (Fire Can) radar. The rate of fire is 12 rounds per minute. The KS-30 entered service in 1955 with Soviet forces, was used to defend strategic sites, and served until 1962. It was not widely exported, but also served with the North Vietnamese forces in the Vietnam War and Iraqi forces in the Iran-Iraq and Gulf Wars.[75]



QF 5.25-inch Battery: The QF 5.25-inch has a single 133 mm gun in a fixed, fortified turret. It is a dual-purpose gun, used both against aircraft and for coastal defense, adapted from a naval gun. A battery has four guns. The fire control system is optical, but an SCR-584 or Blue Cedar FCR-A normally controls each battery. The hydraulic loading system permits a rate of fire of 10 rounds per minute. The QF 5.25-inch entered service with the British Army in 1942 and was installed in and around London. Late in WW2, seven were installed in Australia and three in New Guinea. Eleven were installed in Gibraltar starting in 1953 and were manned until the early 1980s.[76]

AAA UNITS TABLE

Type	Gun	Country	Year	Size[a]	Defense Strength	Sighting Range	Mobility	VPs	AAA Class	Altitude	Range			Hit			Damage Rating	FCR	LRF	Night IR Sights	SAM
								3D/2D/D			S	M	L	S	M	L					
M2	12.7 mm	US	1933	Platoon	3	12	U	3/2/1	L	5	2	3	4	3	2	1	1	—	—	—	—
DShK	12.7 mm	SU	1938	Platoon	3	12	U	3/2/1	L	5	2	3	4	3	2	1	1	—	—	—	—
M16	12.7 mm × 4	US	1944	Section	<u>3</u>!	12	U	6/3	L	5	2	3	4	4	3	2	2	—	—	—	—
M17	12.7 mm × 4	US	1944	Section	<u>3</u>!	12	U	6/3	L	5	2	3	4	4	3	2	2	—	—	—	—
M55	12.7 mm × 4	US	1945	Platoon	3	12	T	4/3/1	L	5	2	3	4	5	4	3	2	—	—	—	—
Vz.53	12.7 mm × 4	CS	1953	Platoon	3	12	T	4/3/1	L	5	2	3	4	5	4	3	2	—	—	—	—
BTR-152A Vz.53	12.7 mm × 4	CS	1955	Section	<u>3</u>!	12	G/P	6/3	L	5	2	3	4	4	3	2	2	—	—	—	—
Mobile M45	12.7 mm × 4	US	1967	Section	2	12	G/P	4/2	L	5	2	3	4	4	3	2	2	—	—	—	—
ZPU-1	14.5 mm	SU	1949	Battery	3	12	T	3/2/1	L	6	2	3	5	3	2	1	1	—	—	—	—
ZPU-2	14.5 mm × 2	SU	1949	Battery	3	12	T	4/3/1	L	6	2	3	5	4	3	2	1	—	—	—	—
ZPU-4	14.5 mm × 4	SU	1949	Battery	3	12	T	5/3/2	L	6	2	3	5	5	4	3	2	—	—	—	—
Mobile ZPU-1	14.5 mm	SU	1949	Section	2	12	G/P	2/1	L	6	2	3	5	2	1	0	1	—	—	—	—
Mobile ZPU-2	14.5 mm × 2	SU	1949	Section	2	12	G/P	3/2	L	6	2	3	5	3	2	1	1	—	—	—	—
Mobile ZPU-4	14.5 mm × 4	SU	1949	Section	2	12	G/P	4/2	L	6	2	3	5	4	3	2	2	—	—	—	—
BTR-40A	14.5 mm × 2	SU	1950	Section	<u>3</u>!	12	G/P	4/2	L	6	2	3	5	3	2	1	1	—	—	—	—
BTR-152A ZPU-2	14.5 mm × 2	SU	1951	Section	<u>3</u>!	12	G/P	4/2	L	6	2	3	5	3	2	1	1	—	—	—	—
BTR-152A ZPU-4	14.5 mm × 4	SU	1951	Section	<u>3</u>!	12	G/P	5/3	L	6	2	3	5	4	3	2	2	—	—	—	—
M113 ZPU-2	14.5 mm × 2	LB	1975	Section	<u>4</u>!	12	U	4/2	L	6	2	3	5	3	2	1	1	—	—	—	—
M113 ZPU-4	14.5 mm × 4	LB	1975	Section	<u>4</u>!	12	U	5/3	L	6	2	3	5	4	3	2	2	—	—	—	—
M167	20 mm × 6	US	1967	Platoon	3	12	T	7/5/2	L	8	2	4	6	6	5	4	3	R/HF	—	—	—
M163	20 mm × 6	US	1968	Section	<u>4</u>!	12	U	8/4	L	8	2	4	6	5	4	3	3	R/HF	—	—	—
TCM-20	20 mm × 2	IL	1970	Platoon	3	12	T	5/3/2	L	8	2	4	6	4	4	3	2	—	—	—	—
M3 TCM-20	20 mm × 2	IL	1970	Section	<u>3</u>!	12	U	6/3	L	8	2	4	6	3	3	2	2	—	—	—	—
Rh-202	20 mm × 2	DE	1972	Battery	3	12	T	5/3/2	L	8	2	4	6	5	4	3	3	—	—	—	—
Panhard M3 VDA	20 mm × 2	FR	1975	Section	<u>4</u>	12	U	7/4	L	8	2	4	6	4	4	3	2	R/UF	—	—	—
53 T1	20 mm	FR	1976	Platoon	3	12	T	4/3/1	L	8	2	4	6	4	3	2	2	—	—	—	—
Tarasque	20 mm	FR	1977	Platoon	3	12	T	4/3/1	L	8	2	4	6	4	3	2	2	—	—	—	—
ZU-23	23 mm × 2	SU	1960	Battery	3	12	T	6/4/2	L	9	2	5	8	5	4	3	3	—	—	—	—
Mobile ZU-23	23 mm × 2	SU	1960	Section	2	12	G/P	5/3	L	9	2	5	8	4	3	2	3	—	—	—	—
ZSU-23-4	23 mm × 4	SU	1965	Section	<u>4</u>	12	U	9/5	L	9	2	5	8	5	4	3	3	W/VF	—	—	—
BTR-152A ZU-23	23 mm × 2	PS	1975	Section	<u>3</u>!	12	G/P	4/2	L	9	2	5	8	4	3	2	3	—	—	—	—
Sinai 23 Ayn-al-Saqr	23 mm × 2	EG	1992	Section	<u>4</u>	12	U	10/5	L	9	2	5	8	5	4	3	3	R/UF	—	—	Ayn-al-Saqr
Sinai 23 Stinger	23 mm × 2	EG	1992	Section	<u>4</u>	12	U	12/6	L	9	2	5	8	5	4	3	3	R/UF	—	—	FIM-92
M2 Blazer	25 mm × 5	US	—	Section	<u>4</u>	12	U	14/7	L	10	3	6	9	5	4	3	4	S/VF	Y	Y	FIM-92D
LAV-AD	25 mm × 5	US	1997	Section	<u>4</u>	12	U	14/7	L	10	3	6	9	5	4	3	4	—	Y	Y	FIM-92D
M53/59	30 mm × 2	CS	1959	Section	<u>4</u>!	12	G/P	7/4	L	11	4	7	9	3	2	1	4	—	—	—	—
AMX-13 DCA	30 mm × 2	FR	1969	Section	<u>4</u>	12	U	10/5	L	11	4	7	9	4	3	2	3	R/VF	—	—	—
AMX-30 DCA	30 mm × 2	FR	—	Section	<u>4</u>	12	U	10/5	L	11	4	7	9	4	3	2	3	R/VF	—	—	—
AMX-30SA	30 mm × 2	FR	1979	Section	<u>4</u>	12	U	10/5	L	11	4	7	9	5	4	3	3	R/VF	—	—	—
Tunguska	30 mm × 2	SU	1984	Section	<u>4</u>	12	U	20/10	L	11	4	7	9	5	4	3	4	W/VF	—	—	SA-19
Tunguska-M	30 mm × 2	SU	1990	Section	<u>4</u>	12	U	22/11	L	11	4	7	9	5	4	3	4	W/VF	—	—	SA-19

AAA UNITS TABLE

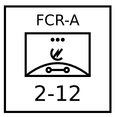
Type	Gun	Country	Year	Size[a]	Defense Strength	Sighting Range	Mobility	VPs 3D/2D/D	AAA Class	Altitude	Range S	Range M	Range L	Hit S	Hit M	Hit L	Damage Rating	FCR	LRF	Night IR Sights	SAM
Oerlikon GDF-001	35 mm × 2	CH	1963	Battery	3	12	T	7/5/2	M	15	4	8	12	4	3	2	4	—	—	—	—
Gepard	35 mm × 2	DE	1976	Section	4	12	U	11/6	M	15	4	8	12	5	4	3	4	W/VF	—	—	—
Oerlikon GDF-005	35 mm × 2	CH	1985	Battery	3	12	T	8/5/3	M	15	4	8	12	5	4	3	4	—	Y	—	—
61-K	37 mm	SU	1939	Battery	3	12	T	5/3/2	M	12	4	8	11	3	2	1	4	—	—	—	—
M15	37 mm + 12.7 mm × 2	US	1942	Section	3!	12	U	4/2	M	12	2	4	8	3	2	1	4	—	—	—	—
Bofors L/60	40 mm	SE	1935	Battery	3	12	T	5/3/2	M	15	4	8	12	2	2	1	4	—	—	—	—
M19	40 mm × 2	US	1944	Section	3!	12	U	6/3	M	15	4	8	12	2	2	1	4	—	—	—	—
M34	40 mm	US	1951	Section	2	12	U	3/2	M	15	2	4	8	1	1	1	4	—	—	—	—
Bofors L/70	40 mm	SE	1952	Battery	3	12	T	6/4/2	M	15	4	8	12	3	2	1	4	—	—	—	—
M42	40 mm × 2	US	1953	Section	3!	12	U	6/3	M	15	4	8	12	2	2	1	4	—	—	—	—
Bofors L/70 BOFI	40 mm	SE	—	Battery	3	12	T	8/5/3	M	15	4	8	12	5	4	3	4	—	—	—	—
Bofors L/70 BOFI-R	40 mm	SE	—	Battery	3	12	T	9/6/3	M	15	4	8	12	6	5	4	4	W/VF	—	—	—
S-60	57 mm	SU	1950	Battery	3	12	T	8/5/3	M	18	5	10	15	2	2	1	5	—	—	—	—
ZSU-57-2	57 mm × 2	SU	1955	Section	4!	12	U	6/3	M	18	5	10	15	2	1	1	5	—	—	—	—
Flab Kan 38	75 mm	CH	1938	Battery	4	18	T	9/6/3	H	24	6	12	18	1	1	0	5	—	—	—	—
M51	75 mm	US	1953	Battery	3	18	T	9/6/3	M	30	6	12	18	4	4	3	5	W/MF	—	—	—
KS-12	85 mm	SU	1939	Battery	4	18	T	9/6/3	H	27	6	12	18	1	1	0	6	—	—	—	—
90 mm M2	90 mm	US	1943	Battery	4	18	T	9/6/3	H	43	7	14	21	2	2	1	6	—	—	—	—
QF 3.7-Inch	94 mm	UK	1939	Battery	4	18	T	9/6/3	H	45	9	18	27	2	2	1	6	—	—	—	—
KS-19	100 mm	SU	1948	Battery	4	18	T	10/7/3	H	39	7	14	21	1	1	0	6	—	—	—	—
120 mm M1	120 mm	US	1944	Battery	4	18	T	9/6/3	H	57	11	22	33	2	2	1	7	—	—	—	—
KS-30	130 mm	SU	1955	Battery	4	18	T	11/7/4	H	64	12	24	36	1	1	0	7	—	—	—	—
QF 5.25-Inch	133 mm	UK	1942	Battery	4	18	T	11/7/4	H	46	12	24	36	2	2	1	7	—	—	—	—

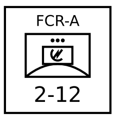
FCR UNITS

Most medium and heavy AAA batteries can be associated with an add-on fire-control radar (FCR), which improves their accuracy and allows them to engage unsighted targets at night and in adverse weather. These units are presented below with their counters.

The FCR Unit Table summarizes the properties of FCR units. As well as the basic properties common to all ground units, the table gives: the FCR class; the frequency band; and the modifier to the hit roll.[77]

FCR-A





Towed and Mobile FCR-A Platoon: FCR-As typically use 1940s technology. They can be towed or mobile.

SCR-584: The SCR-584 is an early US towed FCR-A based on the British 3 GHz magnetron. It replaced the much larger and less accurate SCR-268 and its basic technology was used in a series of successors. The SCR-584 entered service with US and British anti-aircraft artillery units in 1944 and was commonly used to control batteries of 90 mm M2, QF 3.7-inch, 120 mm M1, and QF 5.5-inch guns. It was also supplied to the Soviet Union and used to control batteries of 85 mm KS-12, and 100 mm KS-19 guns.[78]

SCR-784: The SCR-784 towed FCR-A is a more compact version of the SCR-584. It entered service with the US Army in 1945, replacing the SCR-584 and controlling 90 mm M2 and 120 mm M1 guns.[79]

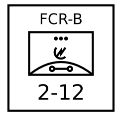
SON-4: The Soviet SON-4 (Whiff) towed FCR-A is based on the SCR-584 radars supplied by the US during WW2. It is commonly used with 85 mm KS-12 and 100 mm KS-19 batteries. It entered services in the Soviet Army in 1948 and started to be replaced by the SON-9 in 1955.[80]

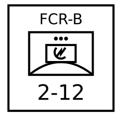
Blue Cedar: The Blue Cedar or "Radar, Anti-Aircraft No.3 Mk.7" towed FCR-A is a post-war development of the Mk.4 radar with post-war technology, resulting in a more compact and reliable system. It entered services with the British Army in 1952, controlling batteries of QF 3.7-inch and QF 5.25-inch guns, and served until these guns were retired in 1957. It also served with the Swiss Air Force from 1958, controlling 7.5 cm guns, until 1965, when the radars were replaced by Super Fledermaus systems and the guns by 35 mm Oerlikon GDFs. The radar was also used by the Netherlands, Sweden, Finland, and South Africa.[81]

SON-9: The Soviet SON-9 (Fire Can) towed FCR-A was derived from the earlier SON-4, which in turn was derived from the US-supplied SCR-584. It is commonly used with 57 mm S-60, 85 mm KS-12, 100 mm KS-19, and 130 mm KS-30 batteries. The SON-9 entered service in 1955 with Soviet forces and was widely exported to Soviet allies. It has seen combat, including with North Vietnam in the Vietnam War, Egypt and Syria in the 1967 War, the War of Attrition, and the 1973 War.[82]

SON-30: The Soviet SON-30 (Fire Wheel) towed FCR-B was derived from the earlier SON-9. It entered service in 1957 and was used to control batteries of 130 mm KS-30 guns.[83]

FCR-B

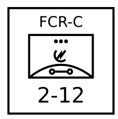


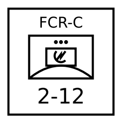


Towed and Mobile FCR-B Platoon: FCR-Bs typically use 1950s technology. They can be towed or mobile.

SON-50: The SON-50 (Flap Wheel) mobile FCR-B is new development and a contemporary of the RPK-2/1RL33 (Gun Dish) radar used in the ZSU-23-4. It has an integrated IFF system. It is used to control batteries of S-60 57 mm and KS-30 130 mm guns and entered service in the 1960s.[84]

FCR-C



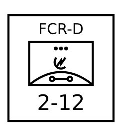


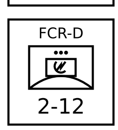
Towed and Mobile FCR-C Platoon: FCR-Cs typically use 1960s technology. They can be towed or mobile.

Super Fledermaus: The Contraves Super Fledermaus FCR-C can control a section of two Oerlikon GDF 35 mm guns; it cannot control a whole battery. It entered service in 1965 with the Swiss Air Force and also was used by the Argentine Air Force, Brazil, Iran, Israel, Japan, and South Africa. It was used by Israel in the 1967 Six Day War and afterward, controlling batteries of Bofors 40 mm L/70 guns in the defense of strategic sites. The Super Fledermaus also saw combat with the Argentine Air Force in the 1982 South Atlantic War controlling Oerlikon GDF guns in the defense of BAM Malvinas (Port Stanley Airport).[85]

Flycatcher: The Signaal Flycatcher FCR-C can control up to six Bofors L/60 or L/70 guns or SAM launchers. Flycatcher entered service with the Royal Netherlands Air Force in 1979, serving for airfield defense and each controlling three modified Bofors 40 mm L/70 gun mounts. It subsequently served with the Indian Armed Forces from 1985, the Irish Defense Forces from 2003, the Royal Netherlands Army from 1987, the Thai Armed Forces from 1981, Turkey, and Venezuela.[86]

FCR-D





Towed and Mobile FCR-D Platoon: FCR-Ds typically use 1970s technology. They can be towed or mobile.

Skyguard: The Contraves Skyguard FCR-D can control two Oerlikon GDF 35 mm guns and additionally a SAM launcher with AIM-9 Sparrow, RIM-9 Sea Sparrow, or Aspide missiles. Skyguard entered service in 1977 with the Swiss Air Force. It

FCR UNIT TABLE

Type	Size	Defense Strength	Sighting Range	Mobility	VPs	FCR		
					3D/2D/D	Class	Frequency	Modifier
Towed FCR-A	Platoon	2	12	T	6/4/2	A	LF	−1
Mobile FCR-A	Platoon	2	12	G/P	6/4/2	A	LF	−1
Towed FCR-B	Platoon	2	12	T	6/4/2	B	MF	−2
Mobile FCR-B	Platoon	2	12	G/P	6/4/2	B	MF	−2
Towed FCR-C	Platoon	2	12	T	7/5/2	C	VF	−2
Mobile FCR-C	Platoon	2	12	G/P	7/5/2	C	VF	−2
Towed FCR-D	Platoon	2	12	T	8/5/3	D	MW	−3
Mobile FCR-D	Platoon	2	12	G/P	8/5/3	D	MW	−3

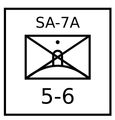
was subsequently very widely exported and used by the Argentine Army, Austria, Bahrain, Bangladesh, Brazil, Canada, Chile, China, Cyprus, Egypt, Greece, Indonesia, Iran, Italy, South Korea, Kuwait, Malaysia, Oman, Pakistan, Saudi Arabia, Spain, Taiwan, and the UK. It saw combat with the Argentine Army in the 1982 South Atlantic War in the defense of BAM Malvinas (Port Stanley Airport) and BAM Condor (Goose Green Airfield) controlling Oerlikon GDF 35 mm guns.[87]

INFANTRY SAM UNITS

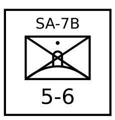
Infantry SAM units are squads of infantry specialized in the destruction of attacking aircraft with portable SAMs. These units are described below with their counters.

The Infantry SAM Unit Table summarizes the properties of infantry SAM units. In addition to the basic properties shared by all ground units, it gives: the number of ready, volley, and reload missiles; the multi-target capability; whether the unit has quick-reaction capability; whether the unit has IFF capability; whether the unit has night IR sights; and the optical lock-on number and range. Each squad has one launcher and, unless otherwise specified, one ready missile and two reload missiles. Infantry SAM squads may be attached to infantry, infantry weapons, infantry mortar, or infantry HQ platoons.

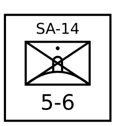
The Infantry SAM Table summarizes the properties of the actual missiles: guidance mode; launch roll; turn rate; flight time; visibility; ECCM, chaff, and flare ratings; whether the missile is active homing (AH) or has a home-on-jam (HOJ) capability; boost phase in FPs; base speed and sustainer duration in game turns; minimum altitude (and modifier for targets at T level); hit rolls for direct and proximity hits; and damage ratings for direct and proximity hits. Most infantry SAMs do not have true proximity fuzes, and so proximity hits in the game correspond to glancing direct hits in reality.



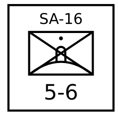
SA-7A Squad: The 9K32 Strela-2 (SA-7A Grail) was the first Soviet portable SAM. It has a rear-aspect, uncooled IR seeker. It entered service with Soviet forces in 1968, but quickly was replaced by the improved Strela-2M (SA-7B).[88]



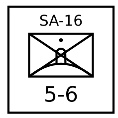
SA-7B Squad: The 9K32M Strela-2M (SA-7B Grail) is similar to the Strela-2 (SA-7A Grail), but uses a rear-aspect, electrically-cooled IR seeker. In Warsaw Pact armies, the Strela-2M was sometimes used with the PRP portable RWR receiver that was capable of passively identifying enemy aircraft by their radar emissions. The Strela-2M entered service with Soviet forces in 1970, replacing the Strela-2. It has seen combat in almost every conflict since 1970 in which the Soviet Union or its successor states have been participants or supporters. In particular, the Strela-2M was used heavily by North Vietnamese forces in the 1973 Easter Offensive, by Arab forces in the 1973 Yom Kippur War, by Angolan forces in the South African Border War, by the Afghan Mujahedeen in the Soviet-Afghan War (having been supplied by the CIA), by Iran in the Iran-Iraq War in the 1980s, and by Serbian and Serbian-supported forces in the Yugoslav Wars in the 1990s.[89]



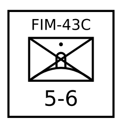
SA-14 Squad: The 9K34 Strela-3 (SA-14 Gremlin) is a development of the earlier Strela-2M. It has a rear-aspect, cryogen-cooled IR seeker. In Warsaw Pact armies, the Strela-3 was sometimes used with the PRP portable RWR receiver that was capable of passively identifying enemy aircraft by their radar emissions. The Strela-3 entered service with the Soviet Army in 1974. It has seen combat in many conflicts since then in which the Soviet Union or its successor states have been participants or supporters. In particular, it was used by Iraqi forces in the 1991 Gulf War.[90]



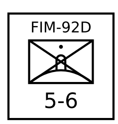
SA-16 Squad: The 9K310 Igla-1 (SA-16 Gimlet) uses as single-channel IR seeker similar to that in the SA-14 in a new missile body. The warhead is similar to the SA-7/14, but also exploded any remaining rocket fuel. In the game, this is simulated by increasing the damage rating by one if the missile attacks before expending more than half of its FPs (round down). It incorporates an IFF interrogator, but this is only used in Warsaw Pact armies. The Igla-1 entered service with the Soviet Army in 1981. It has been widely exported and have seen combat in many conflicts since then in which the Soviet Union or its successor states have been participants or supporters. [91]



SA-18 Squad: The 9K38 Igla (SA-18 Grouse) uses a similar missile and launcher similar to the 9K310 Igla-1 (SA-16), but with a new a dual-channel IR seeker to reduce its susceptibility to flares. Again, the warhead is similar to the SA-7/14, but also exploded any remaining rocket fuel. In the game, this is simulated by increasing the damage rating by one if the missile attacks before expending more than half of its FPs (round down). It incorporates an IFF interrogator, but this is only used in Warsaw Pact armies. The Igla entered service with the Soviet Army in 1983. It has been widely exported and have seen combat in many conflicts since then in which the Soviet Union or its successor states have been participants or supporters. It was used by Iraqi forces in the 1991 Gulf War, by the Peruvian Army in the 1995 Cenepa War, and by Serbian forces in the 1999 Kosovo War.[92]



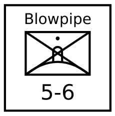
FIM-43C Squad: The FIM-43C Redeye Block III was the first US infantry SAM to reach production and enter service. It has a rear-aspect, cryogen-cooled IR seeker. The FIM-43C entered service with the US Army in 1968 and subsequently was used by Australia, Chad, Denmark, West Germany, Greece, Israel, Jordan, Saudi Arabia, Somalia, Sudan, Sweden, and Turkey. It was also supplied by the CIA to the Afghan Mujahedeen and the Nicaraguan Contras.[93]



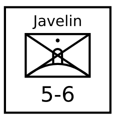
FIM-92A/B/C/D Squad: FIM-92A squadFIM-92B squadFIM-92C squad

The FIM-92A Stinger has an all-angle, second-generation, cryogen-cooled IR seeker. Its warhead weight 3 kg and was significantly more lethal than in earlier missiles. The FIM-92B Stinger POST adds a UV seeker to distinguish flares from aircraft. The FIM-92C/D Stinger RMP have software updates to further improve the performance against flares. All have an IFF interrogator to help avoid fratricide. The FIM-92A entered service with the US Army in 1981, the FIM-92B in 1986, the FIM-92C in 1989, and the FIM-92D in 1992. They were exported to dozens of US allies including the Afghan Mujahedeen and the Angolan UNITA. They saw service in the 1982 South Atlantic War, the Soviet-Afghan War, the Angolan Civil

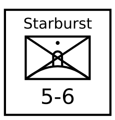
War, the Libyan Invasion of Chad, the Tajik Civil War, the Chechen War, the Sri Lankan Civil War, the Syrian Civil War, and the Russian Invasion of Ukraine.[94]



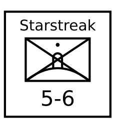
Blowpipe Squad: Blowpipe is an MCLOS optically-guided missile. While in theory this gave it an all-aspect capability, in practice guiding the missile to the target in combat proved to be very difficult. Blowpipe entered service with the British Army and Royal Marines in 1975. It was also used by the Argentine Army, Canadian Army, Chilean Army and Navy, Ecuadorean Army, Guatemalan Army, Portuguese Army, Malawian Army, Malaysian Army, Nigerian Army, Omani Army, Royal Thai Air Force and Army, and the UAE Army. It saw combat in the 1982 South Atlantic War with both sides and had a low success rate. In 1986, it was trialed by the Afghan Mujahedeen, but again was not considered to be effective. Finally, it was also used in the 1995 Cenepa War by Ecuador.[95]



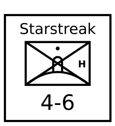
Javelin Squad: Given the poor performance of Blowpipe in the 1982 South Atlantic War, Javelin was quickly developed as an upgraded version of Blowpipe with SACLOS guidance in place of MCLOS guidance. Javelin replaced Blowpipe in the British Army and Royal Marines in 1984. It was also used by the Botswana Army, the Canadian Army, the Malaysian Army, the Peruvian Army, and the South Korean Army.[96]



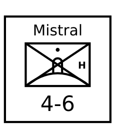
Starburst Squad: Starburst is Javelin with the radio-link replaced by a with laser link. It replaced Javelin in the British Army and Royal Marines in 1989 and was also used by the Canadian Army, the Malaysian Army, the Qatari Army, and the Royal Thai Army. Starburst served with the British Army in the 1991 Gulf War, but was not used in combat.[97]



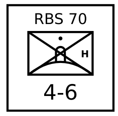
Starstreak Squad: Starstreak is a high-speed laser-guided missile. As the missile approaches the target, it launches three independent explosive darts. The darts have contact fuzes, but not proximity fuzes. Starstreak replaced Javelin in the British Army and Royal Marines in 2000 and is also used by the South African Army and the Armed Forces of Ukraine. Starstreak served with the British Army in the 2003 Invasion of Iraq, but was not used in combat. It has seen combat in the Russian Invasion of Ukraine.[98]



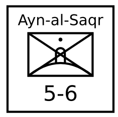
Starstreak LML Squad: The Starstreak Light Multiple Launcher (LML) has three ready-to-fire Starstreak missiles on a pedestal mount with a single sighting unit. The launcher and its associated equipment are heavy and will normally be transported by a vehicle unit. Starstreak LML entered service with the British Army and Royal Marines in 2000. It is also used by Indonesia, Malaysia, and South Africa.[99]



Mistral Squad: The Mistral is a French IR-homing missile on a pedestal mount. The missile has a considerably larger warhead than either the Stinger or Igla missiles, but this comes at a cost in portability. The launcher and its associated equipment are heavy and will normally be transported by a vehicle unit. The Mistral entered service with the French Army in 1990. It was subsequently exported to many other countries.[100]



RBS 70 Squad: The RBS 70 is a Swedish laser-guided missile on a pedestal mount. The launcher and its associated equipment are heavy and will normally be transported by a vehicle unit. The RBS 70 entered service with the Swedish Army in 1977. It was subsequently used by Argentina, Australia, Bahrain, Brazil, the Czech Republic, Iran, Ireland, Lithuania, Norway, Pakistan, Singapore, Thailand, Tunisia, the UAE, Ukraine, and Venezuela.[101]



Ayn-al-Saqr Squad: The Ayn-al-Saqr ("Eye of the Hawk") missile is a licensed version of the 9M32M Strela-2m (SA-7B Grail) manufactured in Egypt. It equips one version of the Sinai 23 air-defense vehicle. It entered service with the Egyptian Army in 1984.[102]

INFANTRY SAM UNIT TABLE

						VPs		Missiles							Lock-On		
Name	Size	Year	Defense Strength	Sighting Range	Mobility	3D/2D/D	SAM	Ready	Volley	Reload	Multi-Target	Quick Reaction	IFF	Night IR Sights	Optical	Range	Other Names
SA-7A	Squad	1968	5	6	U	–/–/3	SA-7A	1	1	2	—	Y	—	—	7	9	9K32 Strela-2 (Grail)
SA-7B	Squad	1972	5	6	U	–/–/4	SA-7B	1	1	2	—	Y	—	—	7	9	9K32M Strela-2M (Grail)
SA-14	Squad	1974	5	6	U	–/–/6	SA-14	1	1	2	—	Y	—	—	7	12	9K34 Strela-3 (Gremlin)
SA-16	Squad	1981	5	6	U	–/–/8	SA-16	1	1	2	—	Y	Y	—	7	12	9K310 Igla-1 (Gimlet)
SA-18	Squad	1983	5	6	U	–/–/10	SA-18	1	1	2	—	Y	Y	—	8	16	9K38 Igla (Grouse)
FIM-43C	Squad	1968	5	6	U	–/–/4	FIM-43C	1	1	2	—	Y	—	—	7	9	Redeye Block III
FIM-92A	Squad	1981	5	6	U	–/–/9	FIM-92A	1	1	2	—	Y	Y	—	8	12	Stinger
FIM-92B	Squad	1986	5	6	U	–/–/10	FIM-92B	1	1	2	—	Y	Y	—	8	12	Stinger POST
FIM-92C	Squad	1989	5	6	U	–/–/10	FIM-92C	1	1	2	—	Y	Y	—	8	12	Stinger RMP
FIM-92D	Squad	1992	5	6	U	–/–/10	FIM-92D	1	1	2	—	Y	Y	—	8	12	Stinger RMP
Blowpipe	Squad	1975	5	6	U	–/–/6	Blowpipe	1	1	2	—	Y	Y	—	7	9	
Javelin	Squad	1984	5	6	U	–/–/8	Javelin	1	1	2	—	Y	Y	—	7	9	Javelin GL
Starburst	Squad	1989	5	6	U	–/–/8	Starburst	1	1	2	—	Y	Y	—	7	9	Javelin S15
Starstreak	Squad	2000	5	6	U	–/–/12	Starstreak	1	1	2	—	Y	Y	—	7	12	
Starstreak LML	Squad	2000	4	6	U	–/–/14	Starstreak	3	1	0	—	Y	Y	—	7	12	
Mistral	Squad	1990	4	6	U	–/–/11	Mistral	1	1	2	—	Y	—	—	7	12	
RBS 70	Squad	1977	4	6	U	–/–/8	RBS 70	1	1	2	—	Y	Y	—	7	10	
Ayn-al-Saqr	Squad	1984	5	6	U	–/–/4	Ayn-al-Saqr	1	1	2	—	Y	—	—	7	12	

INFANTRY SAM TABLE

																	Hit		Damage		
Name	Year	Guidance Mode	Launch Roll	Turn Rate	Flight Time	Visibility	ECCM	Chaff	Flare	AH	HOJ	Instant Arming	Boost Phase	Base Speed	Sustainer	Minimum Altitude	Direct	Proximity	Direct	Proximity	Other Names
SA-7A	1968	I	6	BT	1	6	—	—	5	—	—	Y	—	10	0	T	5	6	4	2	9M32 Strela-2 (Grail)
SA-7B	1972	I	7	BT	1	6	—	—	5	—	—	Y	—	10	0	T	5	6	4	2	9M32M Strela-2M (Grail)
SA-14	1974	M	7	BT	1	6	—	—	4	—	—	Y	—	9	0	T	6	7	4	2	9M34 Strela-3 (Gremlin)
SA-16	1981	M	8	BT/2	1	6	—	—	4	—	—	Y	—	12	0	T	6	7	4*	2*	9K310 Igla-1 (Gimlet)
SA-18	1983	A	8	BT/2	1	6	—	—	2	—	—	Y	—	12	0	T	7	8	4*	2*	9K38 Igla (Grouse)
FIM-43C	1968	M	7	BT	1	6	—	—	5	—	—	Y	—	10	0	T	6	7	4	2	Redeye Block III
FIM-92A	1981	A	8	BT/2	1	2	—	—	3	—	—	Y	—	14	0	T	7	8	5	3	Stinger
FIM-92B	1986	A	8	BT/2	1	2	—	—	1	—	—	Y	—	14	0	T	7	8	5	3	Stinger POST
FIM-92C	1989	A	8	BT/2	1	2	—	—	1	—	—	Y	—	14	0	T	7	8	5	3	Stinger RMP
FIM-92D	1992	A	8	BT/2	1	2	—	—	1	—	—	Y	—	14	0	T	7	8	5	3	Stinger RMP
Blowpipe	1975	OG	8	HT	1	6	—	3	0	—	—	—	—	8	0	T	3	7	6	5	
Javelin	1984	OG	8	ET	1	5	—	2	0	—	—	Y	—	10	0	T	5	8	6	5	Javelin GL
Starburst	1989	LG	8	ET	1	5	—	1	0	—	—	Y	—	10	0	T	5	8	6	5	Javelin S15
Starstreak	2000	LG	8	ET	1	5	—	1	0	—	—	Y	—	24	0	T	7	9	6	4	
Mistral	1990	A	8	ET	1	5	—	—	2	—	—	Y	—	18	0	T	7	9	6	3	
RBS 70	1977	LG	8	ET	1	6	—	2	0	—	—	—	—	12	0	T	5	8	6	3	
Ayn-al-Saqr	1984	I	7	BT	1	6	—	—	5	—	—	Y	—	10	0	T	5	7	4	3	

EXAMPLE ORDERS OF BATTLE

This appendix contains a number of example orders of battle and their correspondence in game units.

The orders of battle for battalions are presented with the headquarters unit in the top row, next the organic companies, one per row, then any battalion-level assets on the next row or rows, and finally any attached units on the last line. Within each row, infantry or towed units appear on the left and their transport units on the right.

Unit Identifiers: The unit identifiers for line platoons are the platoon number, followed by the company letter or number, followed by the battalion number. If only one battalion is used, the battalion number may be omitted. Transport platoons will have a "T" prefix. For example, "1B3" means the first platoon of B company of the third battalion, and "T1B3" is its transport platoon. Companies normally have letters in Western armies and numbers in Warsaw Pact armies. Additionally, AD is used for air defense, AR for artillery, AT for anti-tank, R for reconnaissance, and S for scout, support, or security. The unit identifiers for battalion HQ units are HQ followed by the battalion number.

Unit Terminology: Within the game, units are referred to as brigades, battalions, companies/batteries, and platoons according to their size. However, some forces use other terms, but not always with the same correspondence. The Unit Terminology Table gives examples of some of these uses.

UNIT TERMINOLOGY TABLE

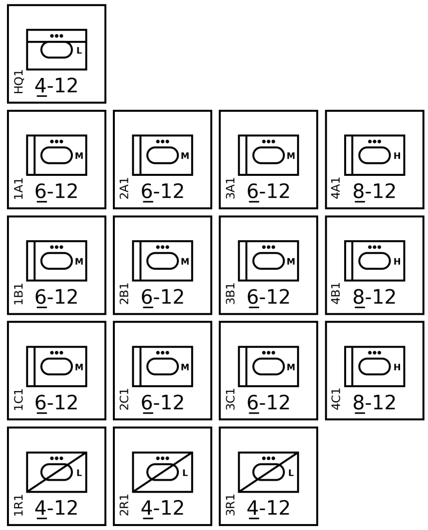
Force	Type	Brigade	Battalion	Company or Battery	Platoon
British Army	Armor	Brigade	*Regiment*	*Squadron*	*Troop*
	Artillery	—	*Regiment*	Battery	—
Soviet Army	Other	*Regiment*	Battalion	Company	Platoon
	Artillery	*Regiment*	Battalion	Battery	—
US Army	Cavalry	*Regiment*	*Squadron*	*Troop*	Platoon
	Pre-1965 Infantry	*Regiment*	Battalion	Company	Platoon
	Pre-1965 Armored	*Combat Command*	Battalion	Company	Platoon
USMC	Infantry	*Regiment*	Battalion	Company	Platoon

BRITISH ARMY (EARLY 1960S)

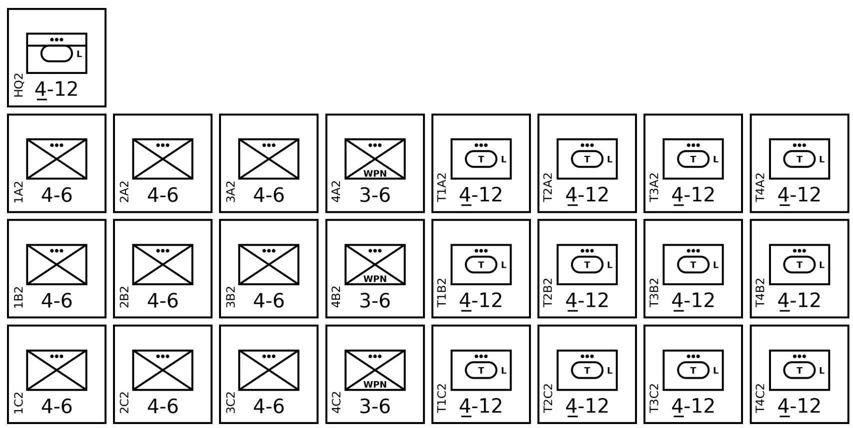
The British Army in the early 1960s was in transition. It operated both medium Centurion tanks (upgraded with the L7 105-mm gun) and heavy Conqueror tanks, and both mechanized and motorized infantry.[103]

Armoured Regiment: This regiment has three squadrons A/B/C each with three tank troops with Centurion medium tanks with 105-mm guns and one tank troop with Conqueror heavy tanks with 120-mm guns. The Conqueror was originally intended to counter the Soviet IS-3 heavy tank, but after the Centurion was upgraded from the 84-mm 20-pdr gun to the 105-mm gun this need diminished. The reconnaissance squadron R has three light reconnaissance troops with Ferret armored cars.

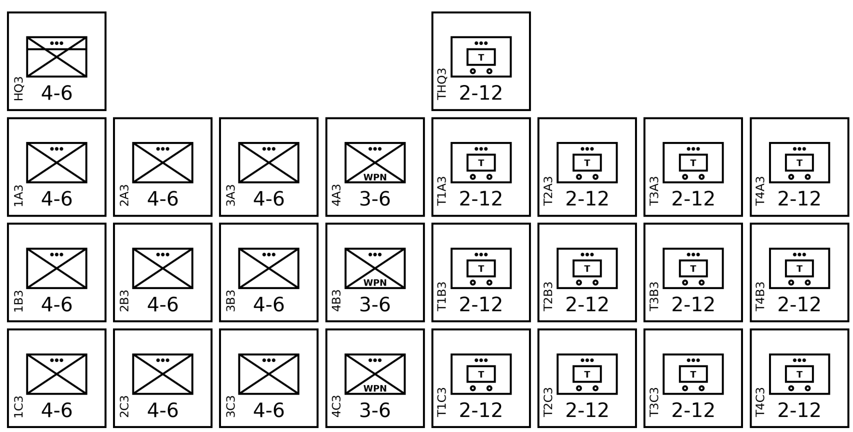
Note that in armored formations in the British Army, the terms troop, squadron, and regiment are used in place of platoon, company, and battalion.



APC Infantry Battalion: This battalion has three companies A/B/C each with three infantry platoons, one infantry weapons platoon, and corresponding light APC platoons with Saracen light APCs to provide mobility and protection. The organization parallels that of an ordinary infantry battalion, but with Saracens instead of trucks.

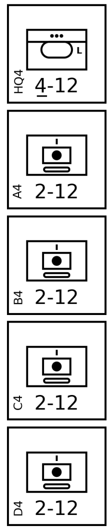


Infantry Battalion: This battalion has three companies A/B/C each with of three infantry platoons, one infantry weapons platoon, with corresponding truck platoons to provide mobility.



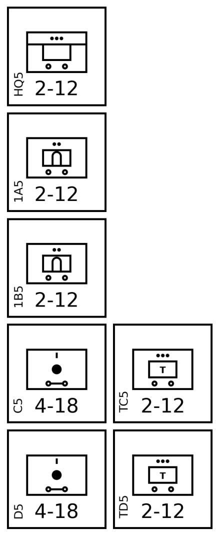
Field Artillery Regiment: This regiment has four batteries A/B/C/D each with six Cardinal 155-mm self-propelled guns.

Note that in artillery formations in the British Army, the term regiment is used in place of battalion.



Heavy Artillery Regiment: This regiment has four batteries A/B/C/D, of which A/B batteries each have two Honest John mobile missile launchers (and are each represented by section) and C/D batteries each have four towed 8-inch M1 guns with trucks.

Note that in artillery formations in the British Army, the term regiment is used in place of battalion.

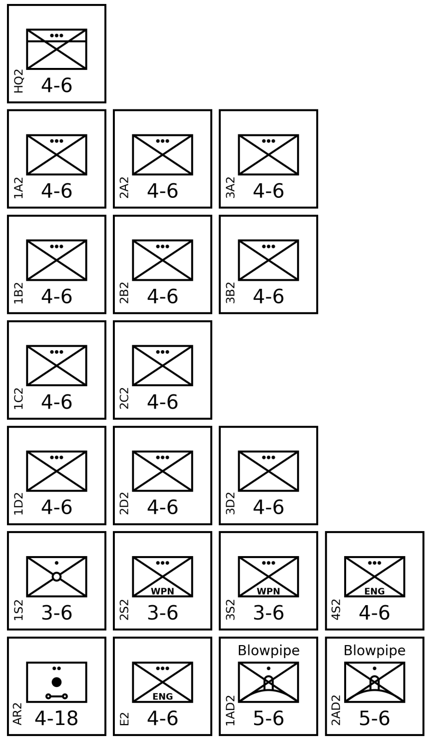


BATTLE OF GOOSE GREEN

The 1982 Battle of Goose Green in the South Atlantic War was fought between 2 Para Battalion of the British Army and 12 Infantry Regiment of the Argentine Army, both with attached elements.[104]

2 Para Battalion: 2 Para consisted of: three line companies A/B/D each with three platoons of infantry; one patrol company C with two infantry platoons; and a support company S with one infantry mortar squad (as only two of the eight mortars in the mortar platoon were available), one infantry weapons platoon of sustained-fire GPMGs, one infantry weapons platoon with MILAN portable ATGMs, and one infantry engineers platoon.

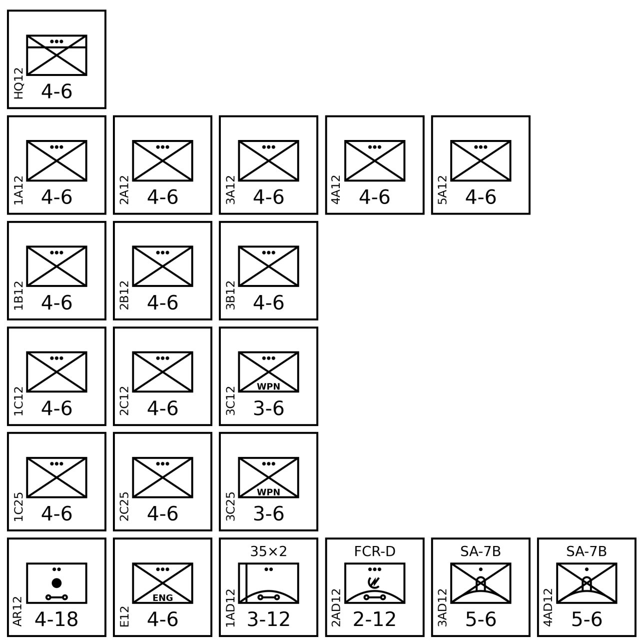
The attached units were: a towed artillery section of three 105-mm Light Guns from Alma Battery of 29 Commando Regiment, Royal Artillery; an infantry engineers platoon of 59 Independent Commando Squadron, Royal Engineers; and two squads of Blowpipes from 43 Air Defence Battery, Royal Artillery.



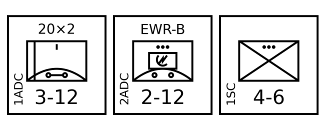
Task Force Mercedes: The Argentine Army elements at Goose Green were called *Fuerza de Tareas Mercédes* (Task Force Mercedes).

The main body of the force was *Regimiento de Infantería 12* (12th Infantry Regiment), with three line companies A/B/C, and reinforced with C company of 25th Infantry Regiment.

The attached units were: a towed artillery section with three 105-mm pack howitzers from A Battery of 4th Airborne Artillery Regiment; an infantry engineers platoon from the 9th Engineering Company; a section with two Oerlikon GDF 35-mm AA guns with a Skyguard FCR from B Battery of 601st Anti-Aircraft Artillery Group; and a pair of SA-7B squads.



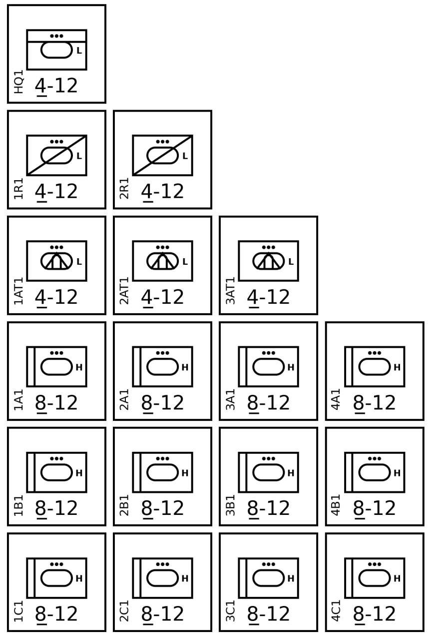
Military Airbase Condor: The Argentine Air Force renamed the airstrip at Goose Green as *Base Aérea Militar Cóndor* (Condor Military Air Base). The Air Force elements at the base included a battery of six 20-mm Rh-202 AA guns with an Elta EWR and a security platoon.



BRITISH ARMY (MID 1980S)

The British Army in the mid 1980s.[105]

Armoured Regiment: This regiment [battalion] has three squadrons [companies] A/B/C each consisting of four tank troops [platoons] each with fourteen Chieftain heavy tanks. The reconnaissance (R) squadron [company] has two troops [platoons] each equipped with four Scorpion light tanks. The anti-tank squadron [company] has tree troops [platoons] each equipped with three FV438s with the Swingfire missile.



ADDITIONAL RULES

RULE 1: GROUND UNITS

A ground *unit* is a combat or logistics formation of infantry, armor, transports, AAA, SAM launchers, or radars. This section discusses them in general.[106]

A. Type and Size: A ground unit will have a type, for example, “heavy tank” or “ZSU-23-4”, and a size, which will be one of squad, section, platoon, or battery. Typically, a squad has five to ten soldiers and/or one vehicle, gun, or launcher; a section has ten to twenty soldiers and/or two vehicles, guns, or launchers; and a platoon or battery has thirty to fifty soldiers and/or three to six vehicles, guns, or launchers. The term “platoon” is more commonly used for infantry and tank units and “battery” for artillery and air-defense units, but within the game they are equivalent.

Companies, battalions, and other larger units above platoons or batteries are represented by multiple units. For example, an infantry company might be represented by three infantry platoon units.

All ground-combat and transport platoons are also available as sections and squads and all AAA platoons and batteries are also available as sections. A section or squad is identical in every way to a platoon or battery that has taken 1D or 2D damage, respectively, except that no VPs are awarded for this notional damage and non-AAA squads and sections may not engage in barrage fire.[107]

In the game, ground units are uniformly referred to as squads, sections, platoons, and batteries. However, other terms are used for these units even within English-speaking armies. For example, in many Commonwealth armies a section is equivalent to a squad, a troop is equivalent to a platoon, and a squadron is equivalent to a company. In the US Armored Cavalry, a troop is equivalent to a company and a squadron is equivalent to a battalion.

B. Basic Properties: The type and size of a ground unit determines its basic properties: damage resilience, defense strength, target class, mobility, and sighting range.

The damage resilience is the amount of damage a ground unit can suffer before being killed. It is determined by the size of the unit: a squad has a damage resilience of 1D; a section has a damage resilience of 2D; and a platoon or battery has a damage resilience of 3D.[108]

The numerical defense strength is used to calculate the odds for air-to-ground attacks and for ground-to-ground combat.

The target class is used to determine the appropriate attack strength of any weapons used in air-to-ground attacks. The class can be a hard, open, or soft according to the degree to which the unit enjoys armored protection. Many weapons have different attack strengths against targets of different classes. A hard target has armored protection from both direct (front, side, and rear) and top attacks. Hard targets are further described as light, medium, or heavy according to their degree of protection, and have corresponding defense strengths.[109] An open target has armored protection from direct attacks but not from top attacks. An open target is normally treated as a hard target, but against napalm, fire, FAE, and air-burst HE bombs, it is considered to be a soft target.[110] A soft target does not have armored protection their personnel and/or weapons.[111]

C. Representation: The briefing documents show the graphical representation of ground units as counters.[112] The size of a unit is indicated by one, two, or three dots (for squads, sections, or platoons) or one vertical bar (for batteries) in an upper position. Hard targets are indicated by an underlined defense strength and open targets by an underline defense strength followed by an exclamation point.

Advanced Rule

D. Mobility: The mobility of a ground unit restricts the hexes it can occupy according to the terrain in and around those hexes. Infantry units, tracked armored units, wheeled armored units other than armored trucks, and tracked unarmored units have unrestricted (U) mobility. Trucks, armored trucks, and other wheeled unarmored units have good (G) or poor (P) mobility. All other units have towed (T) mobility.[113]

A unit with unrestricted mobility may occupy: any urban hex; any clear hex; or any forest hex. A unit with good mobility may occupy: any urban hex; any clear hex; or any forest hex with a road or trail. A unit with poor mobility may occupy: any urban hex; any clear or forest hex with a road, trail, or runway; or any clear hex adjacent to an urban hex or another clear hex with a road, trail, or runway. A unit with towed mobility may occupy hexes according to the mobility of its transport.

When necessary, a scenario must specify whether trucks, armored trucks, and other wheeled unarmored units have good or poor mobility, according to the terrain and the capabilities of the vehicles represented, and assign transport units to towed units.[114] A scenario may modify the mobility classes of ground units and the associated restrictions. [115]

E. Composite Platoons: At the start of the game, an infantry, infantry weapons, infantry mortar, or infantry HQ platoon may attach one or two infantry squads to form a composite platoon. The attached squads are commonly infantry SAM or FAC squads. Once formed, a composite platoon may not be split into its constituent units.[116]

An attached squad is not represented by a separate counter, does not count for stacking, and may not be sighted, identified, or attacked directly. Instead, it is considered to form part of the composite platoon. An attached squad retains its normal capabilities (for example, an attached SAM squad may engage aircraft and an attached FAC squad may sight and mark targets).

A composite platoon has the same damage resilience, defense strength, target class, mobility, and sighting range as its constituent platoon. If a composite platoon is sighted, no information is given on any attached squads. If a composite platoon is identified, any attached squads are also identified (for example, “infantry platoon with an attached FAC squad”). If a composite platoon is suppressed, each attached squad is suppressed too. If a composite platoon suffers damage, then for each D of damage, roll one die for each attached squad and on a 3− the attached squad is killed. If a composite platoon is killed, each attached squad is killed too. VPs are awarded for damage to each of the constituent units.[117]

F. Transport: An infantry or towed unit may be transported by a specific associated transporting IFV, APC, armored truck, truck, or tracked-carrier unit of at least the same size. When necessary, a scenario will establish the association between units and their specific transports. [118]

A transported unit is not represented by a separate counter, does not count for stacking, and may not be sighted, identified, or attacked directly. It may not attack other ground units or aircraft, may not launch SAMs, may not use radar, and may not act as a FAC.

If a transporting unit is sighted, no information is given on the transported unit. If a transporting unit is identified, no information is given on a transported infantry unit, but any transported towed unit is also identified (for example, “truck platoon towing an artillery battery” or “truck platoon towing a ZPU-4 battery”). If a transporting unit suffers damage, then its transported unit suffers the same damage. VPs are awarded for damage to both the transporting and transported unit.

A unit may start the game either transported or not. If not transported, it may be placed according to either its mobility or that of its associated transport.[119]

An infantry unit in the same hex as its associated transport may declare that it is mounting the transport at the start of a Ground Unit Interaction Phase. Both units must not be suppressed, and the transport must not have suffered more damage than the infantry unit. At the end of the Ground Unit Interaction Phase five game turns later, the unit is considered to be transported, and its counter is removed from the map.

An infantry unit being transported may declare that it is dismounting at the start of a Ground Unit Interaction Phase. The transport unit must not be suppressed. At the end of the Ground Unit Interaction Phase two game turns later, the unit is considered to be no longer transported, and its counter is placed on the map in the same hex as its transport. If placing the counter would violate stacking requirements, the unit must abort the dismounting process.

During these processes, the infantry unit may not perform any other action and the transport may not move and may not carry out ground attacks, but may perform barrage fire if it has this capability. The infantry unit may abort either of these process at the start of any Ground Unit Interaction Phase. If either are suppressed during the process, the unit must abort the process in the next Ground Unit Interaction Phase.[120]

A towed unit may not mount or dismount during the game.

RULE 2: ANTI-AIRCRAFT ARTILLERY

A. Laser Rangefinders (LRFs): If an AAA unit has an FCR-R and an LRF, then it does not suffer a +1 modifier if the FCR is jammed or shut down. If an AAA unit has an FCR-W and an LRF, then it only suffers a +1 modifier (not a +2 modifier) if the FCR is jammed or shut down.

NOTES

1. **Robin Olds**

The full quotation contains a second insight: "You're not going to win a war by shooting down a damn MiG, you're going to win a war by bombing the bejesus out of whatever you were sent to bomb. Of course, we weren't there to win it anyway, so it was all a waste of time." (Sherwood, John. Fast Movers: Jet Pilots and the Vietnam Experience, chapter 1)

2. **Ground-Combat and Transport Units**

Air Strike and *Eagles of the Gulf* provide eleven types of ground-combat and transport unit: light, medium, and heavy armor, infantry, infantry FAC, infantry SAM, towed artillery, infantry HQ, armored HQ, truck, and light truck. We provide many more. Does this change outcomes? No. Effectively, all ground units are basically a protection class, defense strength, and sighting range, and the new units map to the existing ones. Does this change the experience? I think it does; the wider range of units enhances the gaming experience.

3. **Infantry Weapons Platoon**

An infantry weapons platoon is not strictly necessary, but it adds to the look-and-feel, and we might consider giving them longer range in ground-to-ground combat. Having a single type is probably more appropriate than having separate types for grenade launchers, recoilless rifles, ATGMs, and machine guns. One might think that the .50 cal or DShK infantry AAA unit could serve as a machine-gun weapons platoon, but machine-gun weapons platoons have low tripods for ground combat rather than high ones for AAA.

4. **Infantry HQs**

In the *Air Strike* scenarios HQ platoons are included at a ratio of about one to every nine normal platoons, so they seem to be battalion HQs.

5. **Tanks**

Protection Levels:

Light, medium, and heavy armor correspond to defense strengths of 4, 6, and 8. Considering the vehicles depicted in scenarios in *Air Strike* and *Eagles of the Gulf*, it seems that typically:

– Light = protection from 14.5 mm and smaller rounds.
– Medium = protection from autocannons and guns up to about 90 mm.
– Heavy = protection against guns of 100 mm or more.

Obviously, this is a gross simplification as each vehicle has its own strengths and weaknesses, and its armor varies between its front, side, rear, and top.

To extend this to other tanks and AFVs, we use Wargames Research Group (WRG) 1950–1985 miniatures rules, which give armor classes of I to X for a range of AFVs:

CLASS I
Sexton
Archer
VP.90
AMX.105A
AMX.155F3
BTR.40
BTR.152V
BTR.50P
BTR.60P
ASU.57
ZSU.57-2
SU.76
AT-P
M.7
M.42
M.44
M.56
M.107
M.110
Ontos
XR.311
½ tracks
RBY.1

CLASS II
Ferret
Fox
Saracen
FV.432
Abbot
SP.70
FV.180
AMX.105B
AMX.155GCT
AMX.13DCA
AML
VAB
YP.408
Spz.kurz.
PVB.302
VK.155
BRDM
BMD
BMP.1
BTR.152K
BTR.50PK
BTR.60PB
PT.76
ZSU.23-4
SP.73
SP.74
Commando
M.8
M.18
M.52
M.55
M.59
M.108
M.109
M.113
M.114
LVTP.7
Lynx
Cougar
Grizzly
L.33

CLASS II "s"
Dutch AIFV

CLASS III
Scorpion
Scimitar
Striker
Spartan
Saladin
EBR
AMX.10P
AMX.10RC
AMX.13
AMX VC1
Luchs.
HS.30
IKV.91
Staghound
M.5
M.10
M.24
M.41
Sheridan

CLASS III "s"
FVS
XM.723
UK.MICV

CLASS IV
Cromwell
AMX.30
Marder
JPz.Rak.
Gepard
TAM
Pz.4H
ASU.85
Sherman

CLASS V
Vijayanta
Comet
JPz.Kan
Panzer 68
Type 61
T.34/85
SU.100

CLASS VI
Leopard 1
Type 74
ISU.122
ISU.152
M.46
M.47

CLASS VI "s"
Leopard.1A4

CLASS VI"S"
AMX.32

CLASS VII
Centurion 3
T.54
T.55
M.48
M.103

CLASS VIII
Centurion 13
T.62
M.60

CLASS IX
T.72
T.64
T.10
IS.3

CLASS IX "S"
Leopard 2
EPC
T.80

CLASS X
Chieftain
Conqueror
Shir.1
Strv.103

CLASS X "S"
UK.MBT.80
Shir.2
XM.1
Merkava
Leopard 3

The correspondence seems to be:

– Open = 4! = class I.
– Light = 4 = classes II to III.
– Medium = 6 = classes IV to VIII.
– Heavy = 8 = classes IX and X.

Since these rules are from 1979 and from unclassified sources, their reliability for the more modern vehicles is somewhat in doubt. Nevertheless, they are probably fine for older vehicles.

Heavy Tanks: The WRG 1950–1985 rules give class IX or X (i.e., heavy) protection to the Chieftain, Conqueror, EPC (Leclerc), IS-3, Leopard 2, Merkava 1, Shir 2 (Challenger 1), Strv 103, T-10, T-64, T-72, and T-80. Beyond these, we have:

– Al-Kalid/VT-1A and Type 99: These are based on the Type 90-II prototype, which in turn is based on the T-72 (heavy).
– Ariete: This weighs 54 tonnes, only slightly less than the M1A1 (heavy) which also has a crew of four, and therefore presumably has similar protection.
– Arjun: The weights 59–68 tonnes and has crew of four, similar to the M1A1 (heavy), and so presumably has similar protection.
– Challenger 2: This is derived from the Challenger 1 (heavy).
– IS-2: This has a similar weight to the IS-3 (heavy) and a similar armament (122 mm D-25), so presumably similar protection.
– K1: This is derived from the M1 (heavy).
– K2: This presumably has at least the same protection as the K1 (heavy).
– M-84, PT-91, and T-90: These are derived from the T-72 (heavy).
– Merkava 2/3/4: These are derived from the Merkava 1 (heavy).
– T-14: This presumably has at least the same protection as the T-90 (heavy).
– T-84: This is derived from the T-80 (heavy).
– Type 10: This weighs 44 tonnes in its standard configuration and has a crew of three, and so presumably has protection at least equal to earlier tanks with similar weights such as the T-72 (heavy).
– Type 90: The has a crew of three and weights 50 tonnes, and so presumably has protection at least the equal of the lighter T-72 (heavy).
– Type 99: The has a crew of three and weights 51–55 tonnes, and so presumably has protection at least the equal of the lighter T-72 (heavy).

Medium Tanks: The WRG 1950–1985 rules give class IV to IIX (i.e., medium) protection to the AMX-30, ASU-85, Centurion, Comet, Cromwell, ISU-122, ISU-152, Kanonen JPz, Leopard 1, M103, M4 Sherman, M46, M47, M48, M48, M60, Panzer 68, SU-100, T-34, T-54, T-55, T-62, TAM, Type 61, Type 74, and Vijayanta. Beyond these, we have:

– M26: The M46 (medium) is largely an M26 with an improved engine.
– M-50 and M-51 Sherman: These are based on the M4 (medium) and have improved armament and engines.
– Panzer 61: This is essentially an early version of the Panzer 68 (medium).
– Type 59/69/79: The Type 59 was Chinese version of the T-54A (medium). The Type 69 and 79 had improved equipment, some of which was derived from captured T-62s, but the same level of protection.
– Type 85/88/96: These have new hulls and turrets. It's not clear whether their level of protection is significantly improved over the Type 59/69/79 (medium).
– Type 15: Wikipedia gives little information on the armor, but the weight is 36 tonnes (with the armor package), and that seems to correspond more to a medium tank.
– Vickers MBT: This is similar to the Vijayanta (medium).

Light Tanks: The WRG 1950–1985 rules give class II and III (i.e., light) protection to the AMX 10RC, AMX 13, Commando, Cougar, Fox, SpPz Luchs, M8, M24, M41, M114A2, M551, Panhard AML, Panhard EBR, PT-76, Saracen, Scimitar, and Scorpion. Beyond these, we have:

– Stryker MGS: With additional ceramic armor, this has protection against 14.5 mm but not heavier rounds, which I believe counts as III.
– Strv 74: This has a similar weight to the M41 (light).
– Type 62: Wikipedia give the hull armor as being similar to the M41 (light).
– Type 63: Wikipedia give the hull armor as being only 10 mm, which corresponds to light.
– Type 05: Based on Type 05 IFV (light).
– Type 08: Based on Type 08 IFV (light).

Open Tanks: These are uncommon, but we need an open target class for M3 half-track APCs, so it is easy to apply it here too. The WRG 1950–1985 rules give class I (i.e., open) protection to the ASU-57 and SU-76.

6. **IFV Units**

We might distinguish APCs and IFVs in ground-to-ground combat. Bradley IFVs made mincemeat of Iraqi tank regiments; M113 APCs would not have had the same impact.

Medium IFVs: The WRG 1950–1985 rules give class IV (i.e., medium) protection to the Marder. Beyond these, we have:

- BMP-3: Wikipedia says the BMP-3 has frontal protection from 30 mm rounds at 200 m.
- Boxer: Wikipedia states that with the baseline armor it is protected against medium-caliber rounds. Additional armor is also available.
- Strf 9040/CV90: Wikipedia states that the CV90 has frontal protection against 30 mm APFSDS rounds. This is considerably more than typical light AFVs.
- LAV III (add-on-armor): Wikipedia says the basic version only has protection against small-arms rounds, but an additional armor kit developed in 2009 gives it protection against 30 mm rounds.
- M2A2/M2A3/M2A4: Wikipedia states their these M2A2 also has protection against 30 mm APDS rounds and RPGs. The M2A3 and M2A4 have visually similar armor.
- Puma: Wikipedia says the frontal arc has protection against medium-caliber rounds and shaped-charge projectiles. With additional armor, the sides gain similar protection.

Light IFVs: The WRG 1950–1985 rules give class II or III (i.e., light) protection to the AMX-10P, BMD, BMP-1, Commando, Dutch AIFV (YPR-765), FV432 (FV432 RARDEN), HS.30 (SPz Lang), M113 (M113 FSC), SPz Kurz, UK MICV (Warrior), and XM723 (M2/M3 Bradley). Beyond these, we have:

- BMP-2: Wikipedia says the BMP-2 has similar protection to the BMP-1.
- LAV-25: Wikipedia says this only has protection against small-arms rounds.
- LAV III: Wikipedia says the basic version only has protection against small-arms rounds, but an additional armor kit developed in 2009 gives it protection against 30 mm rounds.
- Pbv 301: Wikipedia states the armor ranges from 8 to 50 mm, with upper front armor beinng 34 mm (20 mm at 55 degree). This seems to be light.
- Piranha IFV: Presumably similar protection to LAVs.
- Stryker ICVD: This is derived from the Stryker (light).
- Type 86: Wikipedia states this is reverse-engineered from the BMP-1.
- Type 92: Wikipedia states that it is protected against small-arms rounds and the frontal armor can stop 12.7 mm rounds at 100 meters and 25 mm rounds at 1000 meters. This seems to correspond to light armor. The weight is also 21 tonnes, which is considerable less than most medium IFVs.
- Type 05: Wikipedia states that it has protection against small-arms rounds and some protection against 12.7 mm rounds. This corresponds to light armor.
- Type 08: Wikipedia states that it is protected against small-arms rounds and the frontal armor can stop 12.7 mm rounds at 200 meter. This corresponds to light armor.
- VCBI: Wikipedia says this only has protection against 14.5 mm rounds.

There are many variants of the Piranha, and the in-service date is a guess.

7. **APC Units**

Heavy APCs: These are converted tanks.

- Achzarit: Wikipedia states that this based on the T-54/T-55 (medium), but has additional composite armor. Its weight at 44 tonnes is much heaver than the T-55 at 36 tonnes, despite the lack of turret, which suggests the additional armor is quite significant. Therefore, we consider it to be heavy.
- Namer: This is based on the Merkava IV (heavy).
- Nakpadon: This is an improved version of the Nagmashot and Nagmachon. Wikipedia states that its weight is around 55 tonnes, which strongly suggests its armor qualifies as heavy.
- Puma: The photos in Wikipedia show additional armor. Its weight at 50 tonnes is almost identical to the Centurion, despite the lack of turret, which suggests the additional armor is quite significant.

The in-service dates for the Nakpadon (1998) and Puma (1993) are guesses.

Medium APCs:

- Boxer: Wikipedia states that with the baseline armor it is protected against medium-caliber rounds. Additional armor is also available.
- Nagmashot: Based on the Centurion (medium).
- Nagmachon: Based on the Centurion (medium).

The in-service dates for the Nagmashot (1990) and Nagmachon (2000) are guesses.

Light APCs: The WRG 1950–1985 rules give class II or III (i.e., light) protection to the BTR-50PK, BTR-60PB, Commando, Grizzly, FV432, LVTP-7, M113, M59, Saracen, Spartan, VAB, and YPR-408. Beyond these, we have:

- Bronco/Warthog: Wikipedia says it has protection against 7.62 mm rounds.
- BVS10/Viking: Wikipedia says it has protection against small-arms fire and can have additional armor to protect against 14.5 mm rounds.
- BTR-70: Wikipedia states the frontal armor is only 9 mm.
- BTR-80: Wikipedia states that this (and previous models) only have protection from small-arms fire.
- LAV II Bison: Wikipedia states that the Coyote is protected against small-arms rounds, and the Bison seems to be similar.
- LVTP-5: Wikipedia states that the maximum armor is 16 mm.
- M75: Wikipedia states the maximum armor is 38 mm. This is more than the thickness of some light tanks, but it is almost vertical.
- OT-64 SKOT: Wikipedia states the maximum armor is 13 mm.
- OT-810: Wikipedia states the maximum armor is 14.5 mm.
- Piranha APC: Presumably similar protection to LAVs.
- M1126 Stryker: Wikipedia states it has some protection against 14.5 mm rounds from the front and all-round protection against small-arms fire.
- TPz Fuchs: Wikipedia states that it is protected against small-arms rounds.
- Type 63: Wikipedia states that it is protected against small-arms rounds.
- Type 77: Wikipedia states that it is based on the PT-76 (light), and its weight of 15–5 tonnes matches other light APCs.
- Type 89: Wikipedia states that it is protected against small-arms rounds.
- Type 08: Based on Type 08 IFV (light).

There are many variants of the Piranha, and the in-service date is a guess.

Open APCs: The WRG 1950–1985 rules give class I (i.e., open) protection to the AT-P, BTR-50P, BTR-60P, and Half-tracks (M3).

8. **Armored Truck Platoon**

Mobility:

- The BTR-40 is based on the GAZ-63 truck.
- The BTR-152 is based on the ZIS-151 or ZIL-157 truck.
- The SPK/VPK are based on SAAB/Volvo flat-bed trucks.
- The Saxon has only four wheels, which makes me suspect it has limited mobility.

Light Armored Trucks: The WRG 1950–1985 rules give class II or III (i.e., light) protection to the BTR-152K. Beyond this, we have:

- Saxon: Wikipedia states that it is protected against small-arms rounds.

Open APCs: The WRG 1950–1985 rules give class I (i.e., open) protection to the BTR-40 and BTR-152. Beyond these, we have:

- SKP/VKP/SKPF/VKPF: Wikipedia states the armor is 8 to 20 mm, which would be light if the vehicle were not open.

Sources:

- BTR-40: Wikipedia.
- BTR-152: Wikipedia.
- Saxon: Wikipedia.
- SKP/VKP/SKPF/VKPF: Wikipedia

9. **Armored Anti-Tank Missile Vehicles**

Anti-tank units are mainly chrome. However, in ground combat, we might give them more effectiveness against vehicular or towed targets and less against infantry targets.

Medium Anti-Tank Missile Vehicles: The WRG 1950–1985 rules give class IV (i.e., medium) protection to the Raketen JPz. Beyond this, we have:

- Jaguar: This is Raketen JPz (medium) upgraded with HOT missiles.

Light Anti-Tank Missile Vehicles: The WRG 1950–1985 rules give class II or III (i.e., light) protection to the BRDM and Striker. Beyond these, we have:

- Fennek MRAT: Based on the Fennek (light).
- FV438: This is derived from the FV432 APC (light).

- M150 and M901: These are derived from the M113 APC (light).
- M1134 Stryker ATGM: These are derived from the Stryker APC (light).
- VAB HOT: This is derived from the VAB APC (light).
- YP-408 PRAT: This is derived from the YPR-408 APC (light).
- YPR-765 PRAT: This is derived from the YPR-765 APC (light).

10. **Armored Reconnaissance Vehicles**

Again, reconnaissance vehicles are mainly chrome. However, in ground combat we might give them reduced effectiveness.

Light Reconnaissance Vehicles: The WRG 1950–1985 rules give class II or III (i.e., light) protection to the BRDM-2, Ferret, Lynx (M113½ Lynx), and M114. Beyond these, we have:

- LAV II Coyote: Wikipedia states that it is protected against small-arms rounds, which corresponds to light.
- Fennek: Wikipedia states that it is protected against small-arms rounds, which corresponds to light.

Light Reconnaissance Vehicles: The WRG 1950–1985 rules do not give any reconnaissance vehicles with class I (i.e., open) protection. However, we have:

- M20: Open version of the M8 (light) with no turret and armed only with a .50-cal machine gun.

11. **Armored Mortars**

For the protection class, we have:

- FV432: based on FV432 (light).
- M106/M125/M1064: based on M113 (light).
- M1129 Stryker MCV: based on Stryker (light).

We might consider these as open.

12. **Armored Artillery**

Light Armored Artillery: The WRG 1950–1985 rules give class II (i.e., light) protection to the Abbot, AMC-155GCT (AMX-30 AuF1), L33, M52, M108, M109, SP.73 (2S3), SP.74 (2S1), and VK.155 (BKan 1). Beyond these, we have:

- 2S19 Msta-S: Wikipedia states this all-round protection from 15 mm of steel armor. This corresponds to light armor.
- AS-90: Wikipedia states this has up to 17 mm of steel armor. This corresponds to light armor.
- Panzerhaubitze 2000: Wikipedia states this all-round protection from 14.5 mm of steel armor. This corresponds to light armor.

Open Armored Artillery: The WRG 1950–1985 rules give class I (i.e., open) protection to the AMX-105A, M7, and M44. Beyond these, we have:

- The M-50 has frontal and side armor, like the M7 and unlike for example the M107 and M108.
- The M40 also has frontal and side armor. The rear door is probably armored too, but is open when the gun is used.

13. **Armored Rocket Artillery**

Light Armored Rocket Artillery: For the protection class, we have:

- M270: I can find no definite information on the armor protection.
- BM-1: According to Wikipedia, the T-72 hull was chosen to give adequate protection close to the front lines. However, it is not clear whether the rocket launcher has adequate protection.

14. **Tracked Artillery**

To count as a hard target, both the crew and weapons must be protected by armor.

- The WRG 1950–1985 rules give class I (i.e., open) protection to the AMX.155F3 (AMX-13 F3), M107, and M110. However, the only crew member protected seems to be the driver. The other crew members travel and operate the gun without protection.
- Similar considerations apply to the 2S4, 2S7 (although some of the crew travel under armor), M40, M41, and Mk F3. In all of these, the gun mount is open.

15. **Mobile Artillery**

To count as a hard target, both the crew and weapons must be protected by armor.

- CAESAR: Some models have lightly-armored cabins, but the truck as a whole does not seem to be protected.
- Archer: The crew and engine compartments is protected against small-arms rounds, but the gun does not seem to be protected. Unlike the CAESAR, the gun is operated by the crew from the protected cabin. This is close to reaching the level of protection of light armored artillery.

16. **Mobile Rocket Artillery**

To count as a hard target, both the crew and weapons must be protected by armor.

- M142 HIMARS: The cabin has some protection, but the rest of the truck seems to be unprotected.
- BM-30: The cabin appears to have some protection, but the rest of the truck seems to be unprotected.
- Earlier Soviet vehicles appear to be completely unarmored.

17. **Mobile Rocket Artillery**

To count as a hard target, both the crew and weapons must be protected by armor, which is not the case here.

- M752 Lance: This seems to be based on the unarmored M548.
- AMX-30 Pluton: The vehicle is based on an AMX-30 hull and seems to be armored, but the missile is exposed.

Tracked missile artillery is mainly included to serve as a target.

18. **Mobile Missile Artillery Batteries**

Missile artillery is mainly included to serve as a target.

Regarding the protection class, none of these have complete protection against small arms.

19. **AAA Values**

This long note summarizes how I have determined the AAA values for AAA units. VP values are discussed separately.

I compare AAA values for AAA units from *Air Strike*, *Eagles of the Gulf*, *The Speed of Heat*, and Malcolm Pipe's 2026 compilation to the actual characteristics of the guns they represent. My aim is to empirically understand the underlying model sufficiently to identify anomalous values and to apply it to other guns.

AAA Values for Batteries: I start by attempting to understand the AAA values for batteries.

The upper part of the following table shows the original data for gun batteries from *Air Strike* (AS), *Eagles of the Gulf* (EOTG), *The Speed of Heat* (TSOH). I also include data for these guns from Malcolm Pipes' 2026 compilation (MP). For the ZPU-2, ZPU-4, Twin Rh-202, and KS-19, these sources are not in agreement among themselves and so there are multiple entries. I have added columns with the caliber, burst rate of fire per mount (i.e., the rate of fire per barrel multiplied by the number of barrels per mount), and muzzle velocity. Values in red will be subsequently modified.

AAA VALUES FOR BATTERIES

Type	Maximum Altitude	Ranges	Hit Rolls	Damage	FCR/LRF	Caliber (mm)	Rate of Fire (RPM)	Muzzle Velocity (m/s)	Sources
				Original Values					
M2	5	2/3/4	3/2/1	1	—	12.7	500	890	AS/MP
DShK	5	2/3/4	3/2/1	1	—	12.7	500	850	AS/MP
M55	5	2/3/4	5/4/3	2	—	12.7	2000	890	EOTG/MP
ZPU-1	6	2/3/5	3/2/1	1	—	14.5	600	1000	EOTG/TSOH/MP
ZPU-2	6	2/3/5	4/3/2	1	—	14.5	1200	1000	AS
ZPU-2	6	2/3/5	4/3/1	3	—	14.5	1200	1000	MP
ZPU-4	6	2/3/5	5/4/3	2	—	14.5	2400	1000	AS/MP
ZPU-4	6	2/3/5	4/3/2	2	—	14.5	2400	1000	TSOH
TCM-20	8	2/4/6	4/4/3	2	—	20	1450	860	EOTG
Single Rh-202	8	2/4/6	3/2/1	2	—	20	1030	1044	AS
Twin Rh-202	8	2/4/6	5/4/3	2	—	20	2060	1044	AS
Twin Rh-202	8	2/4/6	3/2/1	3	—	20	2060	1044	MP
M167	8	2/4/6	6/5/4	3	R	20	3000	1030	AS/MP
ZU-23	9	2/5/8	4/3/2	2	—	23	2000	980	AS/TSOH/MP
Gemag	10	3/6/9	6/5/4	4	R	25	3600	1040	AS
Oerlikon GDF-001	12	4/8/10	5/4/3	4	—	35	1100	1175	AS/MP
61-K (M-38)	12	4/8/11	3/2/1	4	—	37	165	880	AS/TSOH/MP
Bofors L/60	15	4/8/12	2/2/1	4	—	40	135	865	EOTG/MP
Bofors L/70	15	4/8/12	3/3/2	4	—	40	330	1000	AS/MP
BOFI-R	15	4/8/12	4/4/3	4	W	40	330	1000	AS
S-60	18	5/10/15	2/2/1	5	—	57	110	1000	AS/TSOH/MP
KS-12	27	6/12/18	2/1/0	6	—	85	10	790	AS/TSOH
KS-19	39	7/14/21	1/1/0	6	—	100	15	900	AS
KS-19	39	7/14/21	2/1/0	6	—	100	15	900	MP
				Empirical Model Values					
Base hit rolls depend on rate of fire			1/1/0				15		
			2/2/1				120		
			3/2/1				500		
			4/3/2				1000		
			4/4/3				1500		
			5/4/3				2000		
Hit roll modifiers for FCR/LRF			+1		R				
			+1		L				
			+2		W				
Hit roll modifiers for proximity fuzes			+1						
Hit roll modifiers for section			−1						
Base damage depends on caliber				1		≤14.5			
				2		20–23			
				3		25–30			
				4		35–40			
				5		57			
				6		75–100			
				7		≥120			
Damage modifier for rate of fire				+1			≥2000		
				Modified or Adopted Values					
ZPU-2			4/3/2	1		— adopted from AS			
ZPU-4			5/4/3			— for rate of fire			
Twin Rh-202			5/4/3	3		— for rate of fire			
ZU-23			5/4/3	3		— for rate of fire			
Oerlikon GDF-001			4/3/2			— for rate of fire			
Oerlikon GDF-001	15	4/8/12				— for muzzle velocity			
Bofors L/70			3/2/1			— rate of fire			
BOFI-R			6/5/4			— for FCR-W and proximity fuzes			
KS-12			1/1/0			— for rate of fire			
KS-19			1/1/0			— for rate of fire			
				New Values					
53 T1 Platoon	8	2/4/6	4/3/2	2	—	20	1000	1050	
Tarasque Platoon	8	2/4/6	4/3/2	2	—	20	1000	1050	
Oerlikon GDF-005	15	4/8/12	5/4/3	4	L	35	1100	1175	
BOFI	15	4/8/12	5/4/3	4	L	40	330	1000	
M51	30	6/12/18	4/4/3	5	W	75	45	854	
Flab Kan 38	24	6/12/18	1/1/0	5	—	75	25	700	
90 mm M2	32	7/14/21	2/2/1	6	—	90	24	820	
QF 3.7-inch Mk VI	45	9/18/27	2/2/1	6	–	94	20	1044	
120 mm M1	57	11/22/33	2/2/1	7	–	120	12	945	
KS-30	64	12/24/36	1/1/0	7	—	130	12	970	
QF 5.25-inch	46	12/24/36	2/2/1	7	—	133	10	850	

Qualitatively, altitude and range increase with caliber, hit rolls increase with rate of fire, and damage rating increases with both caliber and rate of fire. This is as expected. Quantitatively:

- Approximately doubling the rate of fire increases the hit rolls by one, with typical values of 1/1/0 at about 15 rounds per minutes, 2/2/1 at about 150 rounds per minute, 3/2/1 at about 500 rounds per minute, 4/3/2 at about 1000, 4/4/3 at about 1500, and 5/4/2 at about 2000 and above. Intermediate rates of fire can have one of these hit rolls increased or decreased by one.
- FCR-R effectively modifies the hit rolls by +1 (e.g., the M167 and Gemag compared to the twin Rh-202).
- FCR-W effectively modifies the hit rolls by +2.
- A laser rangefinder (LRF) effectively modifiers the hit rolls by +1.
- Proximity fuzes increase the hit rolls by +1 (e.g., BOFI and BOFI-R).
- As we will see below, sections have their hit rolls modified by −1.
- As we shall see below, sections have their hit rolls limited to 5/4/3.
- The base damage rating is 1 for ≤14.5 mm, 2 for 20–23 mm, 3 for 25–30 mm, 4 for 35–40 mm, 5 for 57 mm, and 6 for 85–100 mm. I have extended this to 7 for ≥120 mm.
- The damage rating is increased by one if the rate of fire is 2000 rounds per minute or more (e.g., M55, ZPU-4, Gemag, and M167).

These model values are shown in the second section of the table.

Most of the original values can be used as they are, but a few of the hit rolls and damage ratings do not fit this general pattern and need adjusting. Also, for guns for which the sources give different values, a single set needs to be adopted. These are shown in the third section of the table and are discussed in the corresponding notes below.

Values for new guns are shown in the fourth section of the table. They are based on the empirical model discussed here and are explained in more detail in the corresponding notes.

AAA Values for Sections: I now turn to sections. Again, my aim is to empirically understand the underlying model sufficiently to identify anomalous values and to apply it to other guns.

The upper part of the following table shows the original data for gun sections from *Air Strike* (AS), *Eagles of the Gulf* (EOTG), and *The Speed of Heat* (TSOH), and Malcolm Pipes' 2026 compilation (MP). Again, I have added columns with the caliber, burst rate of fire per mount, and muzzle velocity. Values in red will be subsequently modified.

AAA VALUES FOR SECTIONS

Type	Maximum Altitude	Ranges	Hit Rolls	Damage	FCR/LRF	Caliber (mm)	Rate of fire (RPM)	Muzzle Velocity (m/s)	Source
Original Values									
M16	5	2/3/4	4/3/2	2	—	12.7	2000	890	EOTG/MP
Mobile ZPU-4	6	2/3/5	3/3/2	2	—	14.5	2400	1000	AS
Mobile ZPU-4	6	2/3/5	5/4/3	2	—	14.5	2400	1000	MP
M163	8	2/4/6	5/4/3	3	R	20	3000	1030	AS
M163	8	2/4/6	6/5/4	3	R	20	3000	1030	MP
M3 TCM-20	8	2/4/6	3/3/2	2	—	20	1450	860	EOTG
Panhard	8	2/4/6	2/2/1	2	—	20	1450	860	AS
Panhard	8	2/4/6	3/3/2	2	—	20	1450	860	MP
Mobile ZU-23	9	2/5/8	2/2/1	2	—	23	2000	980	AS/MP
Nile 23	9	2/5/8	3/3/2	2	—	23	2000	980	EOTG/MP
ZSU-23-4	9	2/5/8	5/4/3	3	W	23	4000	980	AS/MP
Blazer	10	3/6/9	5/4/3	4	W	25	3600	1040	AS
Blazer	9	3/6/9	5/4/3	4	W	25	3600	1040	MP
M53/59	11	4/7/9	4/3/2	3	—	30	1000	1000	AS/MP
AMX-30	11	4/7/9	5/4/3	3	W	30	1200	970	AS
ZSU-30-2	11	4/7/9	5/4/3	4	W	30	5000	960	AS/MP
Gepard	12	4/8/10	5/4/3	4	W	35	1100	1175	AS
Gepard	12	4/8/12	5/4/3	5	W	35	1100	1175	MP
ZSU-57-2	18	5/10/15	2/1/1	5	—	57	240	1000	EOTG/MP
Modified or Adopted Values									
Mobile ZPU-4			4/3/2			— to match the M16 and ZPU-4			
M163			5/4/3			— to match the M167			
Panhard			4/4/		R	— to match the M3 TCM-20			
Mobile ZU-23			4/3/2	3		— to match the ZU-23			
Nile 23			4/3/2	3		— to match the ZU-23			
Blazer	10					— adopt *Air Strike* values			
M53/59			3/2/1			— to match the AMX-30 DCA			
AMX-30 DCA			4/3/2		R	— does not have FCR-W			
Gepard	15	4/8/12				— for muzzle velocity			
Gepard				4		— adopt *Air Strike* values			
New Values									
M17	5	2/3/4	4/3/2	2	—	12.7	2000	890	
BTR-152A Vz.53	5	2/3/4	4/3/2	2	—	12.7	2000	850	
Mobile M45	5	2/3/4	4/3/2	2	—	12.7	2000	890	
Mobile ZPU-2	6	2/3/5	3/2/1	1	—	14.5	1200	1000	
BTR-40A	6	2/3/5	3/2/1	1	—	14.5	1200	1000	
BTR-152A ZPU-2	6	2/3/5	3/2/1	1	—	14.5	1200	1000	
BTR-152A ZPU-4	6	2/3/5	4/3/2	2	—	14.5	2400	1000	
M113 ZPU-2	6	2/3/5	3/2/1	1	—	14.5	1200	1000	
M113 ZPU-4	6	2/3/5	4/3/2	2	—	14.5	2400	1000	
BTR-152A ZU-23	9	2/5/8	4/3/2	3	—	23	2000	980	
Sinai 23	9	2/5/8	4/3/2	3	—	23	2000	980	
AMX-13 DCA	11	4/7/9	4/3/2	3	R	30	1200	970	
AMX-30SA	11	4/7/9	4/3/2	3	R	30	1200	970	
M15	12	2/4/8	3/2/1	4	—	37	120	792	
M19	15	4/8/12	2/2/1	4	—	40	270	865	
M34	15	2/4/12	1/1/1	4	—	40	135	865	
M42	15	4/8/12	2/2/1	4	—	40	270	865	

Comparing similar weapons, batteries and sections have the same maximum altitudes, ranges, and damage ratings. However, sections (with two mounts) typically have a −1 modification to their hit rolls compared to batteries (with four to six mounts). In a very few cases, the difference is 0, and in a very few others, it is −2. A difference of 1 in the hit rolls is commensurate with the approximate doubling of the rate of fire from a section to a battery.

To determine the appropriate AAA for sections, the following recipe seems to work in most cases:

- Take the value for an equivalent battery apply an adjustment of −1 to the hit rolls to account for the reduced rate of fire in the section. (Only do this if the rate of fire is reduced; see the comment on the ZSU-23-4 and ZSU-57-2 below.)
- Limit the hit rolls to 5/4/3. The preceding guideline would predict hit rolls of 6/5/4 for the Blazer and 7/6/5 for the ZSU-23-4 and ZSU-30-2, so this limit is to conform to the original values. I suggest that in the underlying model, beyond a certain rate of fire, other factors take over to limit the probability of a hit.

Again, most of the original data can be used as they are, but a few of the values are need adjusting. Also, for guns for which the sources give different values, a single set needs to be adopted. These are shown in the second section of the table and are discussed in the corresponding notes below.

Values for new guns are shown in the third section of the table. They are based on the empirical model discussed here and are explained in more detail in the corresponding notes.

20. **M2 Platoon**

Caliber, rate of fire, and muzzle velocity: 12.7 mm, 500 RPM, and 890 m/s.

AAA values: 5-2/3/4-3/2/1-1

I adopt the AAA values, defense strength, and sighting range from *Air Strike* and Malcolm Pipes. The adopted AAA values are in agreement with the empirical model.

Sources:

- Wikipedia on the M2.
- WeaponSystems on the M2.

21. **DShK Platoon**

Caliber, rate of fire, and muzzle velocity: 12.7 mm, 500 RPM, and 850 m/s.

AAA values: 5-2/3/4-3/2/1-1

I adopt the AAA values, defense strength, and sighting range from *Air Strike* and Malcolm Pipes. The adopted AAA values are in agreement with the empirical model.

Service in the Korean War is from Singh, ch. 1.

Sources:

- Wikipedia on the DShK.
- WeaponSystems on the DShK.
- Singh, M. (Pen & Sword, 2020): “Anti-Aircraft Artillery in Combat 1950–1972: Air Defence in the Jet Age”.

22. **M16 Section**

Caliber, rate of fire, and muzzle velocity: 12.7 mm, 2000 RPM, and 890 m/s.

AAA values: 5-2/3/4-4/3/2-2

I adopt the AAA values, defense strength, and sighting range from *Eagles of the Gulf* and Malcolm Pipes. The adopted AAA values are in agreement with the empirical model.

Sources:

- Wikipedia on the M16, M45, and M2.
- WeaponSystems on the M16 and M2.

23. **M17 Section**

Caliber, rate of fire, and muzzle velocity: 12.7 mm, 2000 RPM, and 890 m/s.

AAA values: 5-2/3/4-4/3/2-2

I give this same AAA values, defense strength, and sighting range as the M16. The adopted AAA values are in agreement with the empirical model.

Sources:

- Wikipedia on the M17, M45, and M2.
- Tank Encyclopedia on the M17.

24. **M55 Battery**

Caliber, rate of fire, and muzzle velocity: 12.7 mm, 2000 RPM, and 890 m/s.

AAA values: 5-2/3/4-5/4/3-2

I adopt the AAA values, defense strength, and sighting range from *Eagles of the Gulf* and Malcolm Pipes. The adopted AAA values are in agreement with the empirical model.

Sources:

- Wikipedia on the M45 and M2.
- WeaponSystems on the M2.

25. **Vz.53 Platoon**

Caliber, rate of fire, and muzzle velocity: 12.7 mm, 2000 RPM, and 850 m/s.

AAA values: 5-2/3/4-5/4/3-2

I give this same AAA values, defense strength, and sighting range as the M55. The adopted AAA values are in agreement with the empirical model.

Malcolm Pipes gives data for the M53, but it appears to be a truck-mounted section (the “type” is given as “truck” and the hit rolls are the same as an M16 section), but one inconsistent value is its defense strength of 3, which corresponds to other towed batteries, rather than 2, which corresponds to other truck-mounted sections.

Sources:

– Wikipedia on the Vs.53 and DShK.
– WeapoNEWS on the Vz.53.
– WeaponSystems on the DShK.

26. **BTR-152A Vz.53 Section**

Caliber, rate of fire, and muzzle velocity: 12.7 mm, 2000 RPM, and 850 m/s.

AAA values: 5-2/3/4-4/3/2-2

I give this same AAA values, defense strength, and sighting range as the M16. The adopted AAA values are in agreement with the empirical model.

The in-service date assumes they were supplied or converted shortly after the 1955 arms deal. Some *Eagles of the Gulf* scenarios for 1956 War give the Egyptian Army M16s, but I do not think these served with the Egyptian Army; I think these might well be BTR.152A Vz.53s.

The BTR-152 is based on a truck, and so I give it "G/P" mobility rather than "U".

Sources:

– Wikipedia on the BTR-152A, Vs.53, and DShK.
– WeapoNEWS on the Vz.53.
– Army Guide on the BTR-152.
– Miranda, M., 2015: "Gun Truck"

27. **Mobile M45 Battery**

Caliber, rate of fire, and muzzle velocity: 12.7 mm, 2000 RPM, and 890 m/s.

AAA values: 5-2/3/4-4/3/2-2

I give this same AAA values as the M16. The adopted AAA values are in agreement with the empirical model. I give this the same defense strength and sighting range as the mobile ZPU-4 in *Air Strike*.

Sources:

– Wikipedia on the M45, M2, and gun trucks.
– WeaponSystems on the M2.

28. **ZPU-1/2/4 Battery**

Caliber, rate of fire, and muzzle velocity: 14.5 mm, 600/1200/2400 RPM, and 1000 m/s.

ZPU-1 *AAA values:* 6-2/3/5-3/2/1-1

ZPU-2 *AAA values:* 6-2/3/5-4/3/2-1

ZPU-4 *AAA values:* 6-2/3/5-5/4/3-2

For the ZPU-1, I adopt the AAA values, defense strength, and sighting range from *Eagles of the Gulf*, *The Speed of Heat*, and Malcolm Pipes. The adopted AAA values are in agreement with the empirical model.

For the ZPU-2, *Air Strike* gives hit rolls of 4/3/2 whereas Malcolm Pipes gives it 4/3/1 and *Air Strike* gives a damage rating of 1 (equal to the ZPU-1) whereas Malcolm Pipes gives it a rating of 3 (more than the ZPU-4). The ZPU-2 should logically have both hit rolls and damage ratings intermediate to those of the ZPU-1 and ZPU-4, but this is impossible as these only differ by one. The solution in *Air Strike* seems to be a good compromise: give it the hit rolls of the ZPU-4 but the damage rating of the ZPU-1. I adopt the AAA values, defense strength, and sighting range from *Air Strike*. The adopted AAA values are in agreement with the empirical model.

For the ZPU-4, *Air Strike* and Malcolm Pipes give hit rolls of 5/4/3 whereas *The Speed of Heat* gives 4/3/2. All give damage ratings of 2. I see no reason to give this different hit rolls to those of the M55, given the similarities between the two in caliber, rate of fire, and muzzle velocity. Therefore, I give it hit rolls of 5/4/3, by analogy with the M55 and in agreement with the empirical model. I therefore adopt the AAA values, defense strength, and sighting range from *Air Strike* and Malcolm Pipes.

Sources:

– Wikipedia on the ZPU.
– WeaponSystems on the ZPU-1, ZPU-2, and ZPU-4, and KPV.

29. **Mobile ZPU-1/2/4 Section**

Caliber, rate of fire, and muzzle velocity: 14.5 mm, 600/1200/2400 RPM, and 1000 m/s.

Mobile ZPU-1 *AAA values:* 6-2/3/5-2/1/0-1

Mobile ZPU-2 *AAA values:* 6-2/3/5-3/2/1-1

Mobile ZPU-4 *AAA values:* 6-2/3/5-4/3/2-2

For the mobile ZPU-1 and ZPU-2 sections, I adopt the AAA values of the ZPU-1 and ZPU-2 batteries with a −1 modifier to the hit rolls for being sections. I adopt same the defense strength and sighting range as the mobile ZPU-4 section in *Air Strike* and Malcolm Pipes. The adopted AAA values are in agreement with the empirical model.

For the mobile ZPU-4, *Air Strike* gives hit rolls of 3/3/2 and Malcolm Pipes gives hit rolls of 5/4/3. The values from *Air Strike* seem to be too low and those of Malcolm Pipes seem to be too high. I see no reason to give this different hit rolls to those of the M16, given the similarities between the two in caliber, rate of fire, and muzzle velocity. Therefore, I give it hit rolls of 4/3/2, by analogy with the M16, in agreement with those of the ZPU-4 battery with a −1 modifier, and in agreement with the empirical model. I adopt the defense strength and sighting range from *Air Strike* and Malcolm Pipes.

Sources:

– Wikipedia on the ZPU and Technicals.
– WeaponSystems on the ZPU-1, ZPU-2, and ZPU-4, and KPV.

30. **BTR-40A Section**

Caliber, rate of fire, and muzzle velocity: 14.5 mm, 1200 RPM, and 1000 m/s.

AAA values: 6-2/3/5-3/2/1-1

I have adopted the AAA values of the mobile ZPU-2 section and the defense strength and sighting range of the M16. The adopted AAA values are in agreement with the empirical model.

The BTR-40 is based on a truck, and so I give it "G/P" mobility rather than "U".

Sources:

– Wikipedia on the BTR-40A and ZPU-2.

31. **BTR-152A Section**

Caliber, rate of fire, and muzzle velocity: 14.5 mm, 1200/2400 RPM, and 1000 m/s.

BTR-152A ZPU-2 *AAA values:* 6-2/3/5-3/2/1-1

BTR-152A ZPU-4 *AAA values:* 6-2/3/5-4/3/2-2

I have adopted the AAA values of the mobile ZPU-2/4 section and the defense strength and sighting range of the M16. The adopted AAA values are in agreement with the empirical model.

The BTR-152 is based on a truck, and so I give it "G/P" mobility rather than "U".

Army Guide states that the A/E mounted a ZPU-2 and the D a ZPU-4. Others (Wikipedia) that the A/D/E referred to the chassis (with the D using the BTR-152V and the E the BTR-152V1 chassis) rather than the gun mount. I am not sure how to resolve this, but I don't think it is that important. Instead, I will use "BTR-152A ZPU-2" and "BTR-152A ZPU-4," which extends obviously to other guns mounted on the BTR-152A and similar.

Sources:

– Wikipedia on the BTR-152A and ZPU-2.
– Army Guide on the BTR-152.
– Miranda, M., 2015: "Gun Truck"

32. **M113 ZPU-2/4 Section**

Caliber, rate of fire, and muzzle velocity: 14.5 mm, 1200/2400 RPM, and 1000 m/s.

M113 ZPU-2 *AAA values:* 6-2/3/5-3/2/1-1

M113 ZPU-4 *AAA values:* 6-2/3/5-4/3/2-2

I have adopted the AAA values of the mobile ZPU-2/4 section and the defense strength, sighting range, and mobility of the M163. The adopted AAA values are in agreement with the empirical model.

Sources:

– Wikipedia on the M113 and ZPU-2/4.

33. **M167 Platoon**

Caliber, rate of fire, and muzzle velocity: 20 mm, 3000 RPM, and 1030 m/s.

AAA values: 8-2/4/6-6/5/4-3 R/HF

I have adopted the AAA values, defense strength, and sighting range from *Air Strike* and Malcolm Pipes. The adopted AAA values are in agreement with the empirical model.

The 1976 Operator's Manual describes the use of the AN/TVS-2B or AN/TVS-5 night sight. These are first- and second-generation image intensifiers. They are mounted next to the tracking sight and appear to be useful only for ground targets. I do not count it as giving "night IR sights" capability.

MobileRadar states the AN/VPS-2 operates at 9.2–9.5 GHz.

Sources:

– Wikipedia on the M167 and M61 Vulcan.
– WeaponSystems on the M167 and M61.
– Department of the Army, TM 9-1005-286-10, 1976: "Operator's Manual (Crew) for Gun, Air Defense Artillery, Towed".
– MobileRadar on the AN/VPS-2.

34. **M163 Section**

Caliber, rate of fire, and muzzle velocity: 20 mm, 3000 RPM, and 1030 m/s.

AAA values: 8-2/4/6-5/4/3-3 R/HF

I have adopted the AAA values, defense strength (4), and sighting range (12) from *Air Strike*, but open rather than hard target class. The adopted AAA values are in agreement with the empirical model. Malcolm Pipes gives the section the same hit rolls as the M167 battery, despite the fewer mounts, which does not seem to be correct. The turret is open, so I give this a target class open rather than hard.

The 1969 report and 1976 manual mention the AN/VPS-2 range-only radar. MobileRadar states it operates at 9.2–9.5 GHz.

The 1976 Operator's Manual describes the use of the AN/TVS-2B night sight. This is a first-generation image intensifier. It is mounted next to the tracking sight and appears to be useful only for ground targets. I do not count it as giving "night IR sights" capability.

The PIVADS version introduced in 1984 had an improved sight, but I assume this makes no difference in the AAA values.

Sources:

– Wikipedia on the M163, M167, and M61 Vulcan.
– WeaponSystems on the M163, M167 and M61.
– Department of the Army, TM 9-2350-300-10, 1976: "Operator's Manual For Gun, Air Defense Artillery, Self-Propelled 20-mm, M163".
– Department of the Army, ACGC-64F, 1969: "Final Report: XM163 Vulcan Air Defense System".
– MobileRadar on the AN/VPS-2.

35. **TCM-20 Platoon**

Caliber, rate of fire, and muzzle velocity: 20 mm, 1450 RPM, and 860 m/s.

AAA values: 8-2/4/6-4/4/3-2

I have adopted the AAA values, defense strength, and sighting range from *Eagles of the Gulf*. The adopted AAA values are in agreement with the empirical model.

Sources:

– Wikipedia on the TCM-20 and HS.404.
– WeaponSystems on the TCM-20.

36. **M3 TCM-20 Section**

Caliber, rate of fire, and muzzle velocity: 20 mm, 1450 RPM, and 860 m/s.

AAA values: 8-2/4/6-3/3/2-2

I have adopted the AAA values, defense strength, and sighting range from *Eagles of the Gulf*. The adopted AAA values are in agreement with the empirical model.

Sources:

– Wikipedia on the M3 TCM-20, TCM-20, HS.404.
– WeaponSystems on the M3 TCM-20 and TCM-20.

37. **Rh-202 Battery**

Caliber, rate of fire, and muzzle velocity: 20 mm, 2060 RPM, and 1044 m/s.

Rh-202 *AAA values:* 8-2/4/6-5/4/3-3

I adopt the AAA values, defense strength, and sighting range from *Air Strike*, except I increase the damage rating from 2 to 3. *Air Strike* gives an unexpectedly low original damage rating of 2 for its rate of fire. A value of 3, increased by one from the baseline for the caliber, is probably more appropriate. The adopted AAA values are in agreement with the empirical model.

Malcolm Pipes gives values for the Rh-202, but it is not clear whether they are for the single or twin mount. He gives hit rolls of 3/2/1 (equal to the *Air Strike* values for the single mount) but a damage rating of 3 (equal to the value I have adopted for the twin mount). I do not adopt this combination of values either for the single or twin Rh-202.

Air Strike has single and twin Rh-202 batteries. Single mounts seem to be used for AFVs (e.g., Marder, Luchs, and Wiesel) and naval applications, but all AAA mounts seem to be twin. I therefore drop the single mounts.

Sources:

– Wikipedia on Rh-202.
– WeaponSystems on the Rh-202 twin mount and Rh-202 gun.

38. **Panhard M3 VDA Section**

Caliber, rate of fire, and muzzle velocity: 20 mm, 1450 RPM, and 860 m/s.

AAA values: 8-2/4/6-4/4/3-2 R/UF

I have adopted the AAA values, defense strength, and sighting range from *Air Strike*, except that I have adopted hit rolls of 4/4/3 and FCR-R.

The Panhard M3 VDA has the same twin guns as the M3 TCM-20. *Air Strike* has unexpectedly low original hit rolls of 2/2/1 compared to those of 3/3/2 for the M3 TCM-20 section. Malcolm Pipes gives it the same hit rolls, but omits the FCR-R. I adopt the hit rolls of the M3 TCM-20 (3/3/2) with a +1 modifier for FCR-R. The adopted AAA values are in agreement with the empirical model.

The Panhard M3 VDA in *Air Strike* seems to correspond to Wikipedia M3 VDA or WeaponSystems M3 VDAA.

However, the Panhard M3 VDA radar-equipped vehicle seems to be able to search and share speed and range information with the other vehicle in the section, which seems equivalent to FCR-R.

The radar is the EMD Rodeo according to Lesavre & de Launet and ESD RA-20 according to Wikipedia.

Sources:

– "Les armements de défense anti-antiaérienne par canons et armes automatiques," René Lesavre & Michel de Launet (Centres de hautes études de l'armement, 2007).
– Wikipedia on the Panhard M3 VDA and HS.820.
– WeaponSystems on the Panhard M3 VDAA.
– Jane's Land-Based Air Defence 1992-93 (extract)

39. **53 T1 Platoon**

Caliber, rate of fire, and muzzle velocity: 20 mm, 1000 RPM, and 860 m/s.

AAA values: 8-2/4/6-4/3/2-2

Lesavre & de Launet describe the original 1950's 53 T1, with an MG 151, and the update in the 1970's with the M693 (p. 37) F1 cannon.

Lesavre & de Launet (p. 50) state that the F1 is derived from the 20.693 = M693, which fired (p. 45) the 20 × 139 at 600–1000 RPM and 1050 m/s. Wikipedia on the F2 state that the M693 was derived from the HS 820.

Sources:

– Lesavre, R., & de Launet, M. (Centres de hautes études de l'armement, 2007): "Les armements de défense anti-antiaérienne par canons et armes automatiques".
– WeaponSystems on the M693.
– Wikipedia on the F2.

40. **Tarasque Platoon**

Caliber, rate of fire, and muzzle velocity: 20 mm, 1000 RPM, and 860 m/s.

AAA values: 8-2/4/6-4/3/2-2

Lesavre & de Launet (p. 49) describe the 53 T2 as having the same gun (M693) as the 1970's 53 T1, but with a more ergonomic mount.

Sources:

– Hogg, I. (Barnes & Noble Books, 2000): "Twentieth-Century Artillery".
– Lesavre, R., & de Launet, M. (Centres de hautes études de l'armement, 2007): "Les armements de défense anti-antiaérienne par canons et armes automatiques".
– WeaponSystems on the M693.
– Wikipedia on the F2, 20 mm Tarasque, and Tarasque mythological beast.

41. **ZU-23 Battery**

Caliber, rate of fire, and muzzle velocity: 23 mm, 2000 RPM, and 980 m/s.

AAA values: 9-2/5/8-5/4/3-3

I adopt the AAA values, defense strength, and sighting range are from *Air Strike*, *The Speed of Heat*, and Malcolm Pipes, but the hit rolls are modified as from 4/3/2 to 5/3/2 and the damage rating from 2 to 3.

Air Strike, *The Speed of Heat*, and Malcolm Pipes all give the ZU-23 hit rolls of 4/3/2, but give 5/4/3 to other AAA batteries with similar rates of fire (M55, twin Rh-202, M167, and Gemag) after accounting for modifiers for FCR. Furthermore, all of these other batteries have damage ratings increased by one above their baseline values. There does not seem to be a reason to penalize the ZU-23. It is undeniably basic, and one might think that would be reason to reduce its effectiveness. However, the TCM-20 is basic too and has both a lower rate of fire than the ZU-23 (1450 compared to 2000) and a smaller caliber (20 mm compared to 23 mm), yet its hit rolls are higher than the ZU-23 in *Air Strike*, *The Speed of Heat*, and Malcolm Pipes and its damage rating is the same.

Therefore, by analogy with these batteries and in agreement with the empirical model, I give it hit rolls of 5/4/3 and a damage rating of 3.

Sources:

– Wikipedia on the ZU-23.
– WeaponSystems on the ZU-23 and 2A14.

42. **Mobile ZU-23 Section**

Caliber, rate of fire, and muzzle velocity: 23 mm, 2000 RPM, and 980 m/s.

AAA values: 9-2/5/8-4/3/2-3

I adopt the AAA values, defense strength, and sighting range from *Air Strike* and Malcolm Pipes' compilation, but the hit rolls and damage are taken from the ZU-23 battery with a −1 modifier to the hit rolls for being a section.

The hit rolls and damage in *Air Strike* and Malcolm Pipes' compilation are 2/2/1 and 2. As with the ZU-23 battery, these seem low. The M16 has a similar rate of fire and smaller caliber, but much better hit rolls and the same damage rating, and this does not seem appropriate. Therefore, by analogy with the M16 and in agreement with the empirical model, I give it hit rolls of 4/3/2 and a damage rating of 3.

Sources:

– Wikipedia on the ZU-23 and Technicals.
– WeaponSystems on the ZU-23 and 2A14.

43. **BTR-152A ZU-23 Section**

Caliber, rate of fire, and muzzle velocity: 23 mm, 2000 RPM, and 980 m/s.

AAA values: 9-2/5/8-4/3/2-3

I adopt the AAA properties of the mobile ZPU-2/4 section and the defense strength and sighting range of the M16. The adopted AAA values are in agreement with the empirical model.

The BTR-152 is based on a truck, and so I give it "G/P" mobility rather than "U".

Army Guide states that the A/E mounted a ZPU-2 and the D a ZPU-4. Wikipedia states that the A/D/E referred to the chassis (with the D using the BTR-152V and the E the BTR-152V1 chassis) rather than the gun mount. I am not sure how to resolve this, but I don't think it is that important. Instead, I will use "BTR-152A ZPU-2" and "BTR-152A ZPU-4," which extends straightforwardly to other guns mounted on the BTR-152.

Sources:

– Wikipedia on the BTR-152A and ZU-23.
– Army Guide on the BTR-152.
– Miranda, M., 2015: "Gun Truck"

44. **ZSU-23-4 Section**

Caliber, rate of fire, and muzzle velocity: 23 mm, 4000 RPM, and 980 m/s.

AAA values: 9-2/5/8-5/4/3-3 W/VF

I adopt the AAA values, defense strength, and sighting range from *Air Strike* and Malcolm Pipes' compilation. The ZSU-23-4 has four barrels whereas the ZU-23-2 has two, so the normal −1 modifier for sections does not apply. Naively following the model and with the +2 modifier for FCR-W, the hit rolls would be 7/6/5, but they are limited to 5/4/3. The adopted AAA values are in agreement with the empirical model.

RadarTutorial states that the radar operates at 14.6–15.6 GHz (NATO J-band and IEEE Ku-band), and does not mention a difference between search and tracking. This makes sense, as there seems to be only one radar dish.

Wikipedia states that the ZSU-23-4M3 "Biryusa" upgrade from 1977 adds IFF.

Sources:

– Wikipedia on the ZSU-23-4.
– WeaponSystems on the ZSU-23-4 and 2A7.
– RadarTutorial on the RPK-2/1RL33 radar.

45. **Sinai 23 Section**

Caliber, rate of fire, and muzzle velocity: 23 mm, 2000 RPM, and 980 m/s.

Sinai 23 Ayn-al-Saqr *AAA values:* 9-2/5/8-5/4/3-3 R/UF

Sinai 23 Stinger *AAA values:* 9-2/5/8-5/4/3-3 R/UF

I adopt the AAA properties of mobile ZU-23, but with +1 to the hit rolls for the FCR-R giving 5/4/3. This means that the Sinai 23 (with two barrels) has the same hit rolls and damage rating as the ZSU-23-4 (with four barrels). Perhaps this is acceptable, since there are 30 years between these systems. Malcolm Pipes gives this hit rolls of 3/3/2 and no FCR-R. With FCR-R, these hit rolls would be 4/4/3, which is close to those adopted. The adopted AAA values are in agreement with the empirical model. I adopt the defense strength and sighting range from *Eagles of the Gulf* and Malcolm Pipes' compilation.

The turret is the Dassault TA-23E and seems to be similar to the LAV-AD turret, but with a pair of 23 mm guns instead of a GAU-12 25 mm gun.

I don't know which model of Stinger was used; I would guess the FIM-92A.

Malcolm Pipes has the Sinai 23 with SA7B from 1990 and the Nile 23 with Stingers from 1998. Wikipedia states that they were similar but competing vehicles, and the Sinai 23 was selected in preference to the Nile 23.

The brochure explicitly mentions a "day sight"; there is no sign of an IR sight.

A platoon seems to consist of three "satellite" fire units and one "leader" unit with a RA 20S radar. The radar has only 1 degree precision, so it is not a traditional FCR, but seems to give warning, cueing, and range for optical tracking. I will treat it as an FCR-R and give a +1 modifier.

The brochure states radar works in the E-band and is pulse-Doppler. There is no mention of IFF.

Perhaps the most accurate way to handle the radar would be to have a platoon of 3 fire units and 1 radar unit, and for each D determine if the radar unit is killed on a 3−.

Sources:

– Wikipedia on Sinai 23 and ZU-23.
– Brochure describing the Sinai 23.
– Description of RA 20S FCR.

46. **M2 Blazer**

Caliber, rate of fire, and muzzle velocity: 25 mm, 3600 RPM, and 1040 m/s.

AAA values: 10-3/6/9-5/4/3-4 S/VF LRF

The AAA values are adopted from the LAV-AD, as the two systems are so similar. The adopted AAA values are in agreement with the empirical model.

Jane's describe several Blazer turret options:

– Blazer 30: One-person turret with 30 mm GAU-13 and four Blowpipe/Javelins installer on a MOWAG Piranha.
– Blazer 25: Two-person turret with 25 mm GAU-12 and four Stingers installed on an M2.
– Blazer 25: One- or two-person turret with 25 mm GAU-12 and four Stingers installed on a MOWAG Piranha.
– Blazer 25: One- or two-person turret with 25 mm GAU-12 and two Stingers installed on Stormer.
– Blazer 25: Two-person turret with 25 mm GAU-12, four Stingers, two RBS-70s, and radar.

Jane's state that all have a digital fire-control system, laser rangefinder, FLIR, and ESD RA-20 S search radar. The Ericsson HARD radar is also mentioned. Since the M2 system did not proceed beyond a proposal, it is difficult to be sure what version or versions to include. I have elected to include a version with four Stingers and search radar. I do not see the US Army adopting any missile other than the Stinger, and the radar distinguishes this from the LAV-AD.

The Reddit post state that turret is similar to that of the LAV-AD.

The Reddit post states that the radar was a search radar. This would presumably give cueing and range information for the night IR sight.

Sources:

– Reddit post by unknown person.
– Jane's, extract posted on Secret Projects.
– Wikipedia on M2, GAU-12, and FAADS.

47. **LAV-AD Section**

Caliber, rate of fire, and muzzle velocity: 25 mm, 3600 RPM, and 1040 m/s.

AAA values: 10-3/6/9-5/4/3-4 LRF

I have adopted the AAA value, defense strength, and sighting range of the Blazer from *Air Strike*, but without FCR-W (−2 to the hit rolls), with a laser range finder (+1 to the hit rolls), and with night IR sights. Malcolm Pipes gives the same values, but reduces the maximum altitude from 10 to 9; I have not adopted this change. The adopted AAA values are in agreement with the empirical model.

Sources:

- Wikipedia on the LAV-AD, GAU-12, and FIM-92.
- Dickerson, C.R., 1991: "LAV-AD: Mobile Air Defense for the Ground Maneuver Force".

48. **M53/59 Section**

Caliber, rate of fire, and muzzle velocity: 30 mm, 1000 RPM, and 1000 m/s.

AAA values: 11-4/7/9-3/2/1-4

I adopt the AAA values from *Air Strike* and Malcolm Pipes' compilation, but reduce the hit rolls from 4/3/2 to 3/2/1. I adopt the defense strength (4), target class (open), and sighting range (12) from the M163.

The hit rolls for the M53/59 section from *Air Strike* and Malcolm Pipes' compilation are 4/3/2. The empirical model suggests 3/2/1, derived from the modified hit rolls of 4/3/2 for the Oerlikon GDF-001 battery with a −1 modifier for being a section and also from the hit rolls of 5/3/2 for the AMX-30SA with a −2 modifier to discounting the FCR-W. Therefore, by analogy with these units and in agreement with the empirical model, I adopt hit rolls of 3/2/1.

The M53/59 is based on a truck, and so I give it "G/P" mobility rather than "U".

Sources:

- Wikipedia on the M53/59.
- WeaponSystems on the M53/59 and M53.

49. **AMX-13 DCA Section**

Caliber, rate of fire, and muzzle velocity: 30 mm, 1200 RPM, and 970 m/s.

AAA values: 11-4/7/9-4/3/2-3 R/VF

I adopt the *Air Strike* values given for the AMX-30 DCA, which has identical guns, but with the hit rolls reduced by one as this has only FCR-R and not FCR-W. The adopted AAA values are in agreement with the empirical model.

Malcolm Pipes' compilation gives slightly different values for different 30 mm systems. For the Tunguska and M53/59, it gives a maximum altitude of 11 and ranges of 4/7/9. For the AMX-13, BOV 30, and K-30 it gives 10 and 3/6/9. However, there seems to be very little difference between the muzzle velocities and masses of the shells fired by these guns. I prefer to keep the *Air Strike* values for the AMX-13 DCA, which match those he gives for the Tunguska and M53/59.

Malcolm Pipes' compilation also gives a damage rating of 4. This is the same as the Tunguska, despite the latter having a much higher rate of fire. I prefer to use the *Air Strike* value for the AMX-30 DCA of 3.

I believe the Thomson-CSF Œil Noir DRVC 1A radar provides search and ranging, but not tracking. BAS'ART states:

> Distance fournie par le radar en mode télémétrie. Vitesse de rotation de la droite pièce-objectif fournie par le pointage sur l'objectif.

I translate this as:

> Distance furnished by the radar in telemetry mode. Tracking speed of the target furnished by the pointing of the sight.

Lesavre & de Launet state:

> Le pointage était réalisé optiquement à l'aide d'un viseur correcteur à partir des éléments fournis par le radar. Le chef de char et le viseur disposaient chacun d'un système de visée
>
> - Le chef de char utilisait un viseur collimateur anti-aérien SAGEM 1B, dit aussi GS1C, de grossissement 1, porté par un pantographe assurant le parallélisme de l'axe de visée avec l'axe du canon quand la masse oscillante se déplaçait en site. Ce viseur permettait au chef de char de désigner l'objectif au tireur.
> - Le tireur de son côté disposait d'un viseur collimateur anti-aérien SAGEM 4A, dit aussi GS 6C, de grossissement 6, également supporté par un pantographe.
>
> La correction de visée était définie par un calculateur analogique (35) qui élaborait la position future de l'avion et introduisait la correction nécessaire dans la télécommande de pointage, en tenant compte
>
> - de la télémétrie radar
> - des positions et vitesses angulaires de la tourelle poursuivant l'avion
> - des éléments balistiques
>
> Cette correction-but était calculée pendant que le tireur effectuait la poursuite en manœuvrant son palonnier.

I translate this as:

> Pointing was performed optically using a corrective viewfinder from the elements provided by the radar. The vehicle commander and the gunner each had a sight system
>
> - The vehicle commander used a SAGEM 1B anti-aircraft collimator sight, also called GS1C, with a magnification of 1×, mounted in a pantograph ensuring the parallelism of the aiming axis with the axis of the barrel when the oscillating mass moved in place. This viewfinder allowed the vehicle commander to designate the target to the gunner.
> - The gunner on his side had a SAGEM 4A anti-aircraft collimator viewfinder, also called GS 6C, with a magnification of 6×, also mounted on a pantograph.
>
> The aiming correction was determined by an analog computer which worked out the future position of the aircraft and introduced the necessary correction into the sights, taking into account
>
> - the radar telemetry
> - the positions and angular velocities of the turret tracking the aircraft
> - the ballistic elements
>
> This target correction was calculated while the gunner tracked the target with his controls.

These descriptions are consistent with the radar of the Panhard M3 VDA, which is different but more or less contemporaneous.

BAS'ART states that the radar operates at 1.71–1.75 GHz (NATO D-band and IEEE S-band) and is a PD radar with MTI.

BAS'ART states that each divisional AA regiment had one battery of 12 units.

Sources:

- BAS'ART on the AMX-13 "Bitubes de 30 mm automoteur antiaérien" (Wayback).
- "Les armements de défense anti-antiaérienne par canons et armes automatiques," René Lesavre & Michel de Launet (Centres de hautes études de l'armement, 2007).
- Wikipedia on the AMX-13 DCA and HS.831A.
- WeaponSystems on the AMX-13 DCA (Wayback) and HS.831A (Wayback).

50. **AMX-30 DCA Section**

Caliber, rate of fire, and muzzle velocity: 30 mm, 1200 RPM, and 970 m/s.

AAA values: 11-4/7/9-4/3/2-3 R/VF

I adopt the *Air Strike* values given for the AMX-30 DCA, but with the hit rolls reduced by one as this has only FCR-R and not FCR-W. The adopted AAA values are in agreement with the empirical model.

See the notes on the AMX-13 DCA for a discussion of the AAA values and the fire-control system and radar.

Wikipedia and Lesavre & de Launet describe the development of the AMX-30 DCA from the AMX-13 DCA.

Sources:

- "Les armements de défense anti-antiaérienne par canons et armes automatiques," René Lesavre & Michel de Launet (Centres de hautes études de l'armement, 2007).
- Wikipedia on the AMX-30 DCA and HS.831A.
- WeaponSystems on the AMX-30 DCA (Wayback) and HS.831A (Wayback).

51. **AMX-30SA Section**

Caliber, rate of fire, and muzzle velocity: 30 mm, 1200 RPM, and 970 m/s.

AAA values: 11-4/7/9-5/4/3-3 R/VF

I adopt the *Air Strike* values given for the AMX-30 DCA, but with the hit rolls reduced by one as this has only FCR-R and not FCR-W. The adopted AAA values are in agreement with the empirical model.

See the notes on the AMX-13 DCA for a discussion of the AAA values and the fire-control system and radar.

Lesavre & de Launet mention the name of the upgraded radar, but do not give other details.

Sources:

- "Les armements de défense anti-antiaérienne par canons et armes automatiques," René Lesavre & Michel de Launet (Centres de hautes études de l'armement, 2007).
- Wikipedia on the AMX-30DCA/SA and HS.831A.
- WeaponSystems on the AMX-30DCA (Wayback) and HS.831A (Wayback).

52. **Tunguska Section**

Caliber, rate of fire, and muzzle velocity: 30 mm, 5000 RPM, and 960 m/s.

AAA values: 11-4/7/9-5/4/3-4 W/VF

I adopt the AAA values from *Air Strike* and Malcolm Pipes. The hit rolls would be 7/6/5 naively following the model, but they are limited to 5/4/3. The adopted AAA values are in agreement with the empirical model.

The Tunguska is called the ZSU-30-2 in *Air Strike*.

Wikipedia states that 1RL144 radar system uses the E-band for search, the J-band for tracking, and has IFF. RadarTutorial states it uses S and X band.

Sources:

- Wikipedia on the Tunguska.
- WeaponSystems on the Tunguska, 2A38, and 9M311.
- RadarTutorial on the 1RL144.

53. **Tunguska-M Section**

Caliber, rate of fire, and muzzle velocity: 30 mm, 5000 RPM, and 960 m/s.

AAA values: 11-4/7/9-5/4/3-4 W/VF

I adopt the AAA values from *Air Strike* and Malcolm Pipes. The Tunguska is called the ZSU-30-2 in *Air Strike*. The adopted AAA values are in agreement with the empirical model.

Sources:

- Wikipedia on the Tunguska.
- WeaponSystems on the Tunguska, 2A38, and 9M311.

54. **Oerlikon GDF-001 Battery**

Caliber, rate of fire, and muzzle velocity: 35 mm, 1100 RPM, and 1175 m/s.

AAA values: 15-4/8/12-4/3/2-4

The AAA values for the battery are from *Air Strike* with the hit rolls reduced from 5/4/3 to 4/3/2 and with the maximum altitude and maximum range increased to 15 and 12. The adopted AAA values are in agreement with the empirical model.

The Oerlikon GDF in *Air Strike* has unexpectedly high original hit rolls of 5/4/3 for its rate of fire. The values of 4/3/2 for the ZPU-2, which has a similar rate of fire, are probably more appropriate.

The increase in maximum altitude and maximum range give this a reach that matches the Bofors 40 mm L/70 and reflects the higher muzzle velocity of the 35 mm gun (1175 m/s compared to 1000 m/s). See also the GDF-005 and the Gepard.

Sources:

- Wikipedia on the GDF.
- WeaponSystems on the GDF and KDA.

55. **Gepard Section**

Caliber, rate of fire, and muzzle velocity: 35 mm, 1100 RPM, and 1175 m/s.

AAA values: 15-4/8/12-5/4/3-4 W/VF

The AAA values are from *Air Strike*, with the maximum altitude and maximum range increased to 15 and 12. The adopted AAA values are in agreement with the empirical model.

The increase in maximum altitude and maximum range give this a reach that matches the Bofors 40 mm L/70 and reflects the higher muzzle velocity of the 35 mm gun (1175 m/s compared to 1000 m/s). See also the GDF-001 and GDF-005.

In *Air Strike*, the damage rating is given as 4. In Malcolm Pipes' compilation, the damage rating is given as 5. However, the empirical model suggests 4 and this is the value given in Malcolm Pipes' compilation for the similar GDF-001 and GDF-005. I prefer the *Air Strike* value.

Wikipedia states that the Siemens MPRD-12 search radar operates in the S band (2.60–3.95 GHz) and the tracking radar in the Ku band (12.4–18.0 GHz). RadarTutorial states that it searches in the E band (2–3 GHz) and tracks in the J band (10–20 GHz) and has an integrated IFF.

Sources:

- Wikipedia on the Gepard.
- WeaponSystems on the Gepard and KDA.
- RadarTutorial on the MPRD-12.

56. **Oerlikon GDF-005 Battery**

Caliber, rate of fire, and muzzle velocity: 35 mm, 1100 RPM, and 1175 m/s.

AAA values: 15-4/8/12-5/4/3-4 LRF

The AA values are those of the GDF-001 with a +1 increase in the hit rolls for the laser rangefinder. The adopted AAA values are in agreement with the empirical model.

The maximum altitude and maximum range match those Malcolm Pipes' compilation and give this a reach that matches the Bofors 40 mm L/70 and reflects the higher muzzle velocity of the 35 mm gun (1175 m/s compared to 1000 m/s). See also the GDF-001 and the Gepard.

Sources:

- Wikipedia on the GDF.
- WeaponSystems on the GDF and KDA.

57. **61-K Battery**

Caliber, rate of fire, and muzzle velocity: 37 mm, 165 RPM, and 880 m/s.

AAA values: 12-4/8/11-3/2/1-4

The AAA values are from *Air Strike* and *The Speed of Heat*. The adopted AAA values are in agreement with the empirical model.

I can find no sources other than *Air Strike*, *Eagles of the Gulf*, and *The Speed of Heat* that refer to the 61-K 37 mm gun as the "M-38". Another designation for the gun is "M-1939", so I could understand "M-39".

Sources:

- Wikipedia on the 61-K.
- WeaponSystems on the 61-K.

58. **M15 Section**

Caliber, rate of fire, and muzzle velocity: 37 mm, 120 RPM, and 792 m/s.

AAA values: 12-2/4/8-3/2/1-4

The adopted AAA values are in agreement with the empirical model.

The M1 is somewhat inferior to the 61-K in rate of fire and muzzle velocity. There is a quote on Wikipedia about using the .50 cal machine guns to aim the 37 mm cannon.

Considering this as half an M16, the ranges would be 2/3/4 and the hit rolls 3/2/1. Considering this as half a 61-K battery, the ranges would be 4/8/11 and the hit rolls 2/1/0. We can combine these two by giving it ranges of 2/4/8 and hit rolls of 3/2/1. The damage rating is 4 to match the 37 mm.

Sources:

- Wikipedia on the M15, M1 37 mm, and M2 .50 cal.

59. **Bofors L/60 Battery**

Caliber, rate of fire, and muzzle velocity: 40 mm, 135 RPM, and 865 m/s.

AAA values: 15-4/8/12-2/2/1-4

The AAA values are from *Eagles of the Gulf*. The adopted AAA values are in agreement with the empirical model.

Sources:

- Wikipedia on the Bofors L/60.
- WeaponSystems on the Bofors L/60.

60. **M19 Section**

Caliber, rate of fire, and muzzle velocity: 40 mm, 270 RPM, and 865 m/s.

AAA values: 15-4/8/12-2/2/1-4

The AAA values are identical to those of the Bofors L/60 battery, since the number of guns is the same. The adopted AAA values are in agreement with the empirical model.

Sources:

- Wikipedia on the M19.

61. **M34 Section**

Caliber, rate of fire, and muzzle velocity: 40 mm, 135 RPM, and 865 m/s.

AAA values: 15-2/4/8-1/1/1-4

Considering this as half an M19, the ranges would be 4/8/12 and the hit rolls 1/1/0. Since no FCR can be added, the effective range is 8. We therefore give it ranges of 2/4/8 and hit rolls of 1/1/1. The damage rating is 4 to match the M19. The adopted AAA values are in agreement with the empirical model.

Sources:

– Wikipedia on the M34 and Bofors 40 mm L/60.
– AFV Data Base on the M34.

62. **Bofors L/70 Battery**

Caliber, rate of fire, and muzzle velocity: 40 mm, 330 RPM, and 1000 m/s.

AAA values: 15-4/8/12-3/2/1-4

The AAA values are from *Air Strike*, but with the hit rolls reduced from 3/3/2 to 3/2/1 to match the rate of fire. With this modification, the adopted AAA values are in agreement with the empirical model.

Singh (ch. 4) describes the Israeli use.

Sources:

– Wikipedia on the Bofors L/70.
– WeaponSystems on the Bofors L/70.
– Singh, M. (Pen & Sword, 2020): "Anti-Aircraft Artillery in Combat 1950–1972: Air Defence in the Jet Age".

63. **M42 Section**

Caliber, rate of fire, and muzzle velocity: 40 mm, 270 RPM, and 865 m/s.

AAA values: 15-4/8/12-2/2/1-4

The AAA values are identical to those of the Bofors L/60 battery, since the number of guns is the same. The adopted AAA values are in agreement with the empirical model.

Sources:

– Wikipedia on the M42.

64. **Bofors L/70 BOFI Battery**

Caliber, rate of fire, and muzzle velocity: 40 mm, 330 RPM, and 1000 m/s.

AAA values: 15-4/8/12-5/4/3-4

The BOFI in *Air Strike* corresponds to the BOFI-R described next.

The AAA values are adapted from *Air Strike* values for the Bofors L/70, but with the hit rolls modified as described in the note on the Bofors L/70 and then with modifiers of +1 for proximity fuzes and +1 for the laser rangefinder, giving hit rolls of 5/4/3. With this change, the adopted AAA values are in agreement with the empirical model.

I don't know when the BOFI became available or who else uses them.

Sources:

– Wikipedia on the BOFI and BOFI-R.
– WeaponSystems on the BOFI and BOFI-R.

65. **Bofors L/70 BOFI-R Battery**

Caliber, rate of fire, and muzzle velocity: 40 mm, 330 RPM, and 1000 m/s.

AAA values: 15-4/8/12-6/5/4-4 W/VF

The BOFI in *Air Strike* corresponds to the BOFI-R described here.

The AAA values are adapted from *Air Strike* values for the Bofors L/70, but with the hit rolls modified as described in the note on the Bofors L/70 and then modified by +1 for proximity fuzes and +2 for the BOFI-R radar, giving hit rolls of 6/5/4 for BOFI-R whereas *Air Strike* gives 5/5/4. With this change, the adopted AAA values are in agreement with the empirical model.

I don't know when the BOFI-R became available or who else uses them.

Wikipedia states the BOFI-R has: "Multisensor fire control system with a J band radar. Provides automatic acquisition and tracking with an effective range of 4 km without external radar input."

Sources:

– Wikipedia on the BOFI and BOFI-R.
– WeaponSystems on the BOFI and BOFI-R.

66. **S-60 Battery**

Caliber, rate of fire, and muzzle velocity: 57 mm, 110 RPM, and 1000 m/s.

AAA values: 18-5/10/15-2/2/1-5

The AAA values are from *Air Strike* and *The Speed of Heat*. The adopted AAA values are in agreement with the empirical model.

Sources:

– Wikipedia on the S-60.
– WeaponSystems on the S-60.

67. **ZSU-57-2 Section**

Caliber, rate of fire, and muzzle velocity: 57 mm, 240 RPM, and 1000 m/s.

AAA values: 18-5/10/15-2/1/1-5

For the ZSU-57-2 section, the AAA values are from *Eagles of the Gulf*. Note that the values are the same for the S-60 battery and the ZSU-57-2 section, as the ZSU-57-2 has two guns per mount. The adopted AAA values are in agreement with the empirical model.

Sources:

– Wikipedia on the ZSU-57-2.
– WeaponSystems on the ZSU-57-2.

68. **M51 Battery**

Caliber, rate of fire, and muzzle velocity: 75 mm, 45 RPM, and 854 m/s.

AAA values: 30-6/12/18-4/4/3-5 W/MF

The AAA hit rolls are based on the S-60, reduced by 1 for a lower rate of fire, increased by 1 for proximity fuzes, and increased by 2 for FCR-W, so 4/4/3. I give it the same range of 6/12/18 as the 85 mm KS-12, which fires a slightly heavier round at a slightly lower velocity. The adopted AAA values are in agreement with the empirical model.

I classify this as medium AAA, as it has powered traverse. However, I have not been able to confirm that this is fast enough for it to avoid the need to state pointing.

Wikipedia gives a rate of fire of 45 rounds per minute, a muzzle velocity of 854 m/s, an effective maximum altitude of 30k feet, and a horizontal range of 26 hexes (13 km).

The TM states that the search radar is X-band (6–11 GHz). RadarTutorial also states it is X-band or I/J-band. There is no mention of PD or MTI.

RadarTutorial notes that it is often used with the AN/MPQ-10 as a search radar, but this seems to be counter-battery radar.

Sources:

– Wikipedia on the M51 and T38 radar
– WeaponSystems on the M51.
– Werrell, K.P. (2005, Air University Press, 2nd edition): "Archie to SAM".
– Department of the Army (1956): TM 9-6092-1-1 "Antiaircraft Fire Control System M33C and M33D Operation, June 1956".
– RadarTutorial on the T-38 and AN/MPQ-10.

69. **Flab Kan 38 Battery**

Caliber, rate of fire, and muzzle velocity: 75 mm, 25 RPM, and 700 m/s.

AAA values: 24-6/12/18-1/1/0-5

Its rate of fire suggests hit rolls of 1/1/0. The adopted AAA values are in agreement with the empirical model.

Wikipedia gives the 75 mm Schneider a rate of fire of 20–30 RPM and a muzzle velocity of about 700 m/s.

Wikipedia gives an effective altitude of 27,000 ft. However, the KS-12 fires a heavier shell at a higher velocity and has a maximum altitude of 27,000 ft. To match this, I will give it 24,000 ft. The AAA ranges match those of the KS-12.

I classify the Flab Kan 38 as heavy but the M51 as medium, despite both having the same caliber. This is because the M51 has powered traverse whereas the Flab Kan 38 has manual traverse.

Sources:

– Wikipedia on the 75 mm Scheider and Swiss Fliegerabwehrtruppen.
– militaerfahrzeuge.ch on the Flab Kan 38

70. **KS-12 Battery**

Caliber, rate of fire, and muzzle velocity: 85 mm, 10 RPM, and 790 m/s.

AAA values: 27-6/12/18-1/1/0-6

The AAA values are from *Air Strike* and *The Speed of Heat*, with the hit roll modified to 1/1/0 for the rate of fire. In *Air Strike* and Malcolm Pipes' compilation, this has hit rolls of 2/1/0, but that seems too close to the rolls of 2/2/1 of the S-60 which has a much higher rate of fire. With this modification, the adopted AAA values are in agreement with the empirical model.

There are two KS-12 guns. The M-1939 is given a maximum altitude of 34k feet by Wikipedia and 25k feet by Singh. The M-1944 is given a maximum altitude of 38k ft by Wikipedia 29k ft by Singh. The game version have maximum altitudes of 27k feet, which is half-way between the figures given by Singh.

Wikipedia gives maximum horizontal ranges of 15.65 km and 18 km for the M-1939 and M-1944.

Wikipedia gives the rate of fire as 10–12 RPM.

It is not clear to me that the Soviets had proximity fuzes for this gun.

Sources:

- Wikipedia on the KS-12.
- WeaponSystems on the KS-12.
- Singh, M. (Pen & Sword, 2020): "Anti-Aircraft Artillery in Combat 1950–1972: Air Defence in the Jet Age".

71. **90 mm M2**

Caliber, rate of fire, and muzzle velocity: 90 mm, 24 RPM, and 820 m/s.

AAA values: 43-7/14/21-2/2/1-6

I give the M2 hit rolls of 1/1/0 for the rate of fire and then increased by 1 for proximity fuzes, a maximum altitude of 32 (following Werrell), and the ranges of the KS-19. The adopted AAA values are in agreement with the empirical model.

The M1 version entered service in 1940. The M2 was a development of the M1 and was the "standard weapon" from 1943. I presume all post-war examples are M2s.

Wikipedia gives the rate of fire as 24 RPM, about twice that of the KS-12.

Wikipedia gives a horizontal range of 19 km, and vertical range of 43.5k feet. Werrell states the effective ceiling is 32k feet. As the caliber and the muzzle velocity of the M2 are between the KS-12 (maximum altitude of 27) and the KS-19 (maximum altitude of 39), this seems to be about right.

Lesavre & de Launet describe the service with the French Army.

Sources:

- Wikipedia on the 90 mm M2.
- Werrell, K.P. (2005, Air University Press, 2nd edition): "Archie to SAM".
- Lesavre, R., & de Launet, M. (Centres de hautes études de l'armement, 2007): "Les armements de défense anti-antiaérienne par canons et armes automatiques".

72. **QF 3.7-inch Battery**

Caliber, rate of fire, and muzzle velocity: 94 mm, 20 RPM, and 1044 m/s.

AAA values: 45-9/18/27-2/2/1-6

I give the QF 3.7 hit rolls of 1/1/0 for the rate of fire and then increased by 1 for proximity fuzes, a maximum altitude of 45, and a maximum range of 27. The adopted AAA values are in agreement with the empirical model.

Wikipedia gives a maximum altitude of 45k feet, a muzzle velocity of 1044 m/s (significantly more than other similar guns), and a rate of fire of 20 rounds per minute.

Singh states that by 1956 the IDF had twelve 3.7-inch guns with FCR.

Singh states that Pakistan used the 3.7-inch gun in the 1971 War.

Sources:

- Wikipedia on the QF 3.7-inch.
- Singh, M. (Pen & Sword, 2020): "Anti-Aircraft Artillery in Combat 1950–1972: Air Defence in the Jet Age".

73. **KS-19 Battery**

Caliber, rate of fire, and muzzle velocity: 100 mm, 15 RPM, and 900 m/s.

AAA values: 39-7/14/21-1/1/0-6

The AAA values are from *Air Strike*. In Malcolm Pipes' compilation, this has hit rolls of 2/1/0, but that seems too close to the rolls of 2/2/1 of the S-60 which has a much higher rate of fire. The adopted AAA values are in agreement with the empirical model.

Wikipedia gives a rate of fire of 15 RPM, a vertical range of 48 altitude levels (15 km, with a proximity fuse), and a horizontal range of 39 hexes (21 km). The *Air Strike* values are 39 altitude levels and 21 hexes, so are cut back somewhat from the maximums.

It is not clear to me that the Soviets had proximity fuzes for this gun. The one suggestion that they do is that Wikipedia mentions different maximum altitudes for proximity and timed fuzes, but I can find no information on Soviet proximity fuzes.

Sources:

- Wikipedia on the KS-19.
- WeaponSystems on the KS-19.

74. **120 mm M1 Battery**

Caliber, rate of fire, and muzzle velocity: 120 mm, 12 RPM, and 945 m/s.

AAA values: 57-11/22/33-2/2/1-7

The hit rolls are adapted from the KS-30 (which has a similar rate of fire), with a +1 modifier for proximity shells. The adopted AAA values are in agreement with the empirical model.

Wikipedia gives rate of fire of 12 RPM, a maximum altitude of 57 altitude levels (17.5 km), and a maximum range of 50 hexes (25 km).

Sources:

- Wikipedia on the M1.

75. **KS-30 Battery**

Caliber, rate of fire, and muzzle velocity: 130 mm, 12 RPM, and 970 m/s.

AAA values: 64-12/24/36-1/1/0-7

The hit rolls are the same as the KS-19 (which has a similar rate of fire). In order to reach an altitude of 64, it needs a maximum range of at least 32. I will give it 36. The adopted AAA values are in agreement with the empirical model.

Wikipedia gives the rate of fire is 10–12 RPM, the vertical range as 72 altitude levels (22 km) and 54 hexes (29 km) horizontally. Wikipedia also says it provided a 25% increase over the effective ceiling of the KS-19, which would be 48 altitude levels. However, this has a heavier shell and slightly higher muzzle velocity than the 120 mm M1, so it should have a maximum altitude of more than 57. I will give it a maximum altitude of 64.

It is not clear to me that the Soviets had proximity fuzes for this gun.

Sources:

- Wikipedia on the KS-30.

76. **QF 5.25-inch Battery**

Caliber, rate of fire, and muzzle velocity: 133 mm, 10 RPM, and 850 m/s.

AAA values: 46-12/24/36-2/2/1-7

The AAA values are the same as for the KS-30, except the hit rolls are increased by one for proximity fuzes and the maximum altitude is reduced from 64,000 to 46,000 ft. This is the value given on Wikipedia, but it makes sense given the lower muzzle velocity of the 5.25-inch. The adopted AAA values are in agreement with the empirical model.

Sources:

- Wikipedia on the QF 5.25-inch and Princess Anne's Battery.

77. **FCR Units**

Bands: Radar frequencies are often referred to by the corresponding NATO (post 1978) and IEEE band designations.

Frequency (GHz)	NATO Band
1–2	D
2–3	E
3–4	F
4–6	G
6–8	H
8–10	I
10–20	J
20–40	K

Frequency (GHz)	IEEE Band
1.55–3.9	S
3.9–6.2	C
6.2–10.9	X
10.9–20.0	Ku
20.0–36.0	Ka

Frequencies: The table gives a summary of the frequencies of various FCRs along with the NATO and IEEE bands. Some radar systems use different individual radars for different roles. The roles are R for range, S for search, and T for tracking. The sources of the data are given in the entries for the individual radars.

Radar	Year	Role	Frequency (GHz)	NATO Band	IEEE Band	MTI	IFF	Notes
SCR-584	1944	S/T	2.7–2.8	E	S	N		
SCR-784	1945	S/T	2.7–2.8	E	S	N		
SON-4	1945	S/T	2.70-2.86	E	S	N		
Blue Cedar	1952	S/T	3.00–3.12	F	S	N		
SON-9	1955	S/T	2.70–2.86	E	S	N		
SON-30		S/T	—	E	S	N	Y	
SON-50		S/T	9.25–9.70	I	X	N	Y	
Super Fledermaus	1965	S/T	8.6-9.6	I	X	Y	N	MTI from 1970
Flycatcher	1979	S	9.2	I	X	Y	Y	MTI from 1992?
		T	9.2	I	X	—	—	
		T	26.5–30	K	Ka	—	—	From 1992?
Skyguard	1977	S/T	8.6–9.5	I	X		Y	
M163/M167	1967	R	9.2–9.5	I	X	—	N	
ZSU 23-4	1965	S/T	14.6–15.6	J	Ku	N	Y	IFF from 1977
AMX-13 DCA	1969	S/R	1.71–1.75	D	S	Y		
Panhard M3 DCA	1975	S/R	—	E	S	Y	N	
Sinai 23	1992	S/R	—	E	S	Y	N	
Tunguska	1984	S	—	E	S	Y	Y	
		T	—	J	X	—	—	
Gepard	1976	S	—	E	S	Y	Y	
		T	—	J	Ku	—	—	
M51	1953	T	—	I	X	N		
AMX-30SA BOFI-R	1979							

78. **SCR-584**

I classify this as FCR-A as the SON-9 is derived from it.

Wikipedia gives the frequency as 2.7–2.8 GHz (NATO E-band) and the search and tracking ranges for bomber-sized targets as 64 km and 29 km.

Werrell mentions its use with the 90 mm and 3.7-inch guns against V-1s in 1944.

Sources:

- Wikipedia on the SCR-584, 90 mm M1/M2 gun, 120 mm M1 gun, and QF 3.7-inch AA gun.
- Werrell, K.P. (2005, Air University Press, 2nd edition): “Archie to SAM”.
- RadarTutorial
- MobileRadar on the SCR-784

79. **SCR-784**

I classify this as FCR-A it is derived from the SCR-584.

The earliest manuals I can find are from 1945, so I suspect that is its in-service date.

Sources:

- Wikipedia on the SCR-784 and SCR-584.
- RadarTutorial on the SCR-784 and SCR-584
- MobileRadar on the SCR-584
- RadioNerds on the SCR-784.

80. **SON-4**

I classify this as FCR-A it is derived from the SCR-584.

RadarTutorial states that the frequency is 2.70–2.86 GHz.

RadarTutorial states that it is a tracking radar, and was typically used with a P-8 (Knife Rest A) or P-10 (Knife Rest B) search radar.

Sources:

- RadarTutorial on the SON-4.

81. **Blue Cedar**

I’m not sure whether this is an FCR-A or FCR-B.

The frequency is 3.00–3.12 GHz (NATO F-band and IEEE S-band).

The Swiss FCRs seem to have controlled 75 mm Flab Kan 38 L39 guns.

Sources:

- Wikipedia on the Blue Cedar.
- RadarTutorial on the Blue Cedar.
- Wikipedia on the Swiss Fliegerabwehrtruppen.

82. **SON-9**

Wikipedia gives the frequency as 2.7–2.8 GHz (NATO E-band). RadarTutorial gives it as 2.7–2.86 GHz.

Wikipedia states this was used with the 57, 85, 100, and 130 mm guns and was widely deployed in the Vietnam War. In *The Speed of Heat* the 57 mm S-60 and 85 mm KS-12 are often combined with an FCR-A, presumably the SON-9.

Sources:

- Wikipedia on the SON-9.
- RadarTutorial

83. **SON-30**

This looks very similar to the SON-9.

RadarTutorial gives the frequency as S band.

The photo on RadarTutorial shows an integrated IFF system.

Wikipedia and RadarTutorial both state it was used for 130 mm guns, but do not mention others.

RadarTutorial states that it had replaced the SON-4 and SON-9 by 1957, but the SON-9 only entered service in 1955. I suspect that it entered service in 1957.

Foss (p. 185) states that it was used for 130 mm guns.

Sources:

- Wikipedia on the SON-30.
- RadarTutorial on the SON-30.
- Foss, Christopher F., 1974 (London : Allan): “Artillery of the World”.

84. **SON-50**

This is truck-mounted.

RadarTutorial gives the frequency as 9.25–9.7 GHz (X band).

The photo on RadarTutorial shows an integrated IFF system.

RadarTutorial states it was used for 57 and 130 mm guns. Wikipedia states it was used for 57 mm guns.

RadarTutorial states that it is mounted on a URAL-375. Wikipedia then states that this was produced from 1961 to 1993. The in-service data can be no earlier than 1961. I guess it entered service in the 1960s; by the 1970s I think the focus had moved to SAMs.

Chant (p. 253) states that the SON-50 is believed to be derived from the Gun Dish radar on the ZSU-23-4 and has a TV back-up. I note, however, that the frequencies are very different: Flap Wheel is E band and Gun Dish is J band.

Sources:

- Wikipedia on the SON-50 and Ural-375
- RadarTutorial on the SON-50.
- Chant, Christopher, 1989 (Brassey’s Defence Publishers): “Air Defence Systems and Weapons World AAA and SAM Systems in the 1990s”.

85. **Super Fledermaus**

Wikipedia states there were two versions: the Feuerleitgerät (Flt Gt) 63 (from 1965) and the Feuerleitgerät 69 (from 1970). The latter had “background clutter suppression,” which probably is equivalent to MTI.

Wikipedia states that it can control *two* Oerlikon GDF mounts, hence the restriction to controlling sections and not batteries.

Wikipedia also states these the prototype of the Gepard used a Flt Gt 63. Since the FCR-W radar of the Gepard is equivalent to a +2 modifier and has VF frequency, it corresponds to an FCR-C. We assume the Super Fledermaus does too.

The Wikipedia page on the Super Fledermaus gives the frequency as 8.6–9.6 GHz (old NATO X-band), but also states that its search function operates in the E/F bands (2–4 GHz) and tracking function in the J band (10–20 GHz).

The service history is from Wikipedia and Singh (for Israeli use).

Sources:

- Wikipedia on the Super Fledermaus and its use with the Oerlikon GDF.
- Singh, M. (Pen & Sword, 2020): “Anti-Aircraft Artillery in Combat 1950–1972: Air Defence in the Jet Age”.

86. **Flycatcher**

Flycatcher is similar to the Cheetah radar, so presumably FCR-C.

According to Wikipedia, the search radar is NATO I/J band and the tracking radar is NATO K band. RadarTutorial states the search frequency is 9.2 GHz (NATO I band) and the tracking frequency is either I or K band. Van Elk states the tracking radar is an independent pulse-Doppler 26.5–31 GHz (old-NATO Ka band) system, but this seems to be referring to an upgrade in 1992. He also mentions that the radar is used in the Goalkeeper CIWS, which Wikipedia states searches in I band and tracks in both I and K bands. I conclude:

– Search is 9.2 GHz (I band)
– Tracking is 9.2 GHz (I band) or 26.5–31 GHz (K band)

Van Elk states that it has IFF and directed a maximum of 3 cannons or missile launchers.

Van Elk states that it succeeded the HSA L4/5 radar.

Sources:

– Wikipedia on the Flycatcher.
– van Elk, Bert (2020): Terug naar de toekomst: De Flycatcher.
– RadarTutorial

87. **Skyguard**

RadarTutorial states that the frequency is 8.6–9.5 GHz (NATO I-band or IEEE X-band). The description of the system suggests that both search and tracking radars use the same frequency.

Sources:

– Wikipedia on the Skyguard's use with the Oerlikon GDF.
– RadarTutorial

88. **SA-7A Squad**

The launcher and SAM values are from *Air Strike* with significant modifications:

– Zaloga gives an in-service date of 1968.
– *Air Strike* gives this a seeker type of I, but that does seem appropriate for an uncooled PbS seeker. Instead, I use a seeker type of E, by analogy with the uncooled seeker in the AIM-9B, and works at about 2 microns.
– *Air Strike* gives a turn rate of HT, but the control capability seems to be the same as in the SA-7B and SA-15. I will adopt BT.
– *Air Strike* gives this a speed of 8. Zaloga gives a lifetime of 8 seconds and an average speed of 1550 km/h. This corresponds to a range of 5.2 km or about 10 hexes.

89. **SA-7B Squad**

The launcher and SAM values are from *Air Strike* with significant modifications:

– *Air Strike* gives this a seeker type of I. That seems about right. The seeker was PbS and cooled electrically, like the AIM-9E which also has a type I seeker.
– *Air Strike* gives a turn rate of BT, which I adopt.
– *Air Strike* gives this a speed of 10. Zaloga states that it has a slightly more powerful rocket motor than the SA-7A, but the mean speed is the same at 1550 km/h. I adopt a speed of 10, like that of the SA-7A.

90. **SA-14 Squad**

The launcher and SAM values are from *Air Strike* with significant modifications:

– The seeker is cryogenically cooled at works at about 4 microns (Zaloga). *Air Strike* gives this a type A seeker, but that does seem appropriate; the first type A seekers in Soviet AAMs did not appear until 1985 in the AA-8C. Also, type A seekers have all-aspect capability Zaloga says that the seeker has similar performance to the RIM-43C, but only limited all-aspect capability. A type M seeker seems to be more appropriate.
– *Air Strike* gives a turn rate of ET, but the control capability seems to be the same as in the SA-7A and SA-7B. I will adopt BT.
– *Air Strike* gives this a speed of 12. Zaloga gives a lifetime of 8 seconds and an average speed of 1440 km/h (as it is slightly heavier than the SA-7). This corresponds to a range of 4.8 km or about 9 hexes.
– The warhead is the same as on the SA-7A/7B.

91. **SA-16 Squad**

The launcher and SAM values are from Malcolm Pipes with significant modifications:

– The warhead of the SA-16/18 is 1.17 kg, identical to the 1.17 kg of the SA-7 and similar to the 1.1 kg of the SA-14. Only later Iglas have heavier warheads. Therefore, I have reduced the damage ratings to match the SA-7.
– Carl Smeaton comments that the warhead explodes any remaining rocket fuel. I increase the damage rating by one if the target is reached in the first half of the FPs.
– I have reduced the lock-on range of the SA-16 and increased the flare rating to match those of the SA-14.
– Furthermore, I have given both the SA-14 and SA-16 the same speed and turn rate. Zaloga states that the engagement range increased by about 1 km with respect to the SA-7/14, so I am increasing the speed by 2 to 12. I will give both a turn rate of BT/2, matching Stinger.

92. **SA-18 Squad**

The launcher and SAM values are from Malcolm Pipes with some modifications:

– The warhead of the SA-16/18 is 1.17 kg, identical to the 1.17 kg of the SA-7 and similar to the 1.1 kg of the SA-14. Only later Iglas had heavier warheads. Therefore, I have reduced the damage ratings to match the SA-7.
– Carl Smeaton comments that the warhead explodes any remaining rocket fuel. I increase the damage rating by one if the target is reached in the first half of the FPs.
– I have given both the SA-14 and SA-16 the same speed and turn rate. Zaloga states that the engagement range increased by about 1 km with respect to the SA-7/14, so I am increasing the speed by 2 to 12. I will give both a turn rate of BT/2, matching Stinger.

93. **FIM-43C Squad**

The launcher and SAM values are from *Air Strike*.

94. **FIM-92A/B/C/D Squad**

The launcher and SAM values for the A model are from *Air Strike*. Those for the B/C/D models are from Malcolm Pipes compendium.

– I have increased the flare susceptibility of the FIM-92A from 2 to 3. It did not seem to have significant protection.
– Wikipedia gives an effective range of 5 miles, which is equivalent to a speed of 15. I have left the speed at 14, as that is close enough.
– Wikipedia gives a peak speed of 1930 mp and a self-destruct time of 17 seconds.
– *Air Strike* has the visibility as 2, but looking at photos I don't see much differ

95. **Blowpipe Squad**

The launcher and SAM values are from *Air Strike*.

96. **Javelin Squad**

The launcher and SAM values for Javelin are from *Air Strike*.

Javelin is also called "Javelin GL".

I have seen mention of a Javelin LML; did anyone use it or was it just a prototype?

97. **Starburst Squad**

The launcher and SAM values for Starburst are just those for Javelin but with the OG guidance replaced by LG guidance.

Starburst was originally called "Javelin S15".

98. **Starstreak Squad**

The launcher and SAM values are from Malcolm Pipes. The real darts do not have proximity fuzes, but the missile in the game can obtain a proximity hit. Perhaps a proximity hit here means a hit with only one dart?

99. **Starstreak LML Squad**

The SAM values are from Malcolm Pipes. I have guessed the launcher values are the same as for shoulder-launched version, just that the launcher has three ready missiles. I don't know if the launcher has reloads. Is the launcher really portable or does it have to be transported by a vehicle?

100. **Mistral Squad**

The launcher and SAM values are from *Air Strike*. Is the launcher really portable or does it have to be transported by a vehicle?

101. **RBS 70 Squad**

For the RBS 70: The launcher and SAM values are from *Air Strike*. Is the launcher really portable or does it have to be transported by a vehicle?

102. **Ayn-al-Saqr Squad**

The launcher and SAM values are taken to be equal to those of the SA-7B.

103. **British Army Units (Early 1960s)**

The British Army formations from early 1960s are from the document "Modern British TOEs" by Richard A. Rinaldi (2002).

104. **Goose Green**

The primary source for the orders of battle at Goose Green is Mark Adkin's *The Battle for Goose Green: A Battle is Fought to be Won* (Leo Cooper, 1992).

105. **British Army Units (Mid 1980s)**

The British Army formations from mid 1980s are from the document "Modern British TOEs" by Richard A. Rinaldi (2002).

106. **Rules**

Much of this section should be included in the 3A rules, but it's here for the moment.

107. **Squads and Sections as Damaged Platoons**

So for example sections/squads have 2D/1D damage resilience and +1/+2 modifiers to AAA attacks. Similar modifiers should be applied to ground-to-ground attacks.

108. **Damage Resilience in *Air Strike***

There is a form of damage resilience in *Air Strike*, in that units whose counters have white text are only able to take 1D of damage before being killed (see the last paragraph of rule 28). Such units include infantry SAMs, armored CCUs, and many vehicle SAMs. *The Speed of Heat* also mentions units being able to take only 2D damage, but it doesn't have any relevant units since it attaches SAM teams to infantry platoons rather than having them as separate (see below) and doesn't have a separate FAC unit. In both cases, the idea seems to have been incompletely implemented in the rules. I'm generalizing and homogenizing the idea. This is important as we need sections for mobile AAA units with two vehicles and for the Oerlikon GDF (as Skyguard can only control two guns and not a full battery). To a large degree, a section is equivalent to a platoon that has taken 1D and a squad is equivalent to a platoon that has taken 2D, so the mechanics of the game are really not impacted.

109. **Hard Targets**

Light, medium, and heavy armor correspond to defense strengths of 4, 6, and 8 in *Air Strike*.

110. **Open Targets**

The BTR-50P APC and the M3 half-tracked APC are examples of open targets. In the 1956 and 1967 Arab-Israeli Wars where both sides used M3 half-tracks. If an M3 is attacked by an air-burst or napalm bomb, the crew and passengers will suffer more or less as if they were in the open.

111. **Soft Targets**

Examples of soft targets include infantry, towed artillery, unarmored vehicles, and armored vehicles that do not fully protect their personnel and/or weapons such as the M-107 and M-110 tracked artillery vehicles.

Nothing like the M-107 or M-110 is included in *Air Strike*, but again I would assert that it is tracked mainly for mobility. Effectively, these are towed artillery installed in an exposed position on a tracked hull and have almost no protection for most of the crew.

Most tracked SAM vehicles are classified as hard targets in *Air Strike* and *Eagles of the Gulf*, despite many of them having completely exposed missiles. If an air-burst bomb explodes above a SAM-4 launcher, it is not going to shrug it off. I think many SAM vehicles are tracked mainly for *mobility* and perhaps for crew protection, but in terms of a mission-kill on their weapon systems, they are soft targets. I've not changed anything yet, but we need to think about this.

112. **Representation**

The representation is based on NATO symbols, including those for infantry, armor, anti-armor, air defense, guns, missiles, mortar, tube artillery, and rocket artillery. One addition is that a rectangle denotes an unarmored vehicle such as a truck or tracked carrier. A truck or mobile unit has two wheel symbols in the lower position, whereas a tracked carrier has a tracked symbol. The mortar symbol is modified to be simply an open circle, for compactness.

All vehicular units have either an armored or unarmored vehicle symbol, all infantry units have an infantry symbol, and towed units have none of these.

The letter T in the central position indicates a transport capacity (e.g., APCs, IFVs, cargo trucks, and tracked cargo carriers). Words and letters in the lower position indicate variants on a basic type: the letters H, M, L, and O on an armored unit indicate heavy, medium, light, and open armored protection; the word WPN or FAC on an infantry unit indicates a weapons or FAC unit; and the letter H on an infantry SAM unit indicates a heavy version of the launcher (e.g., the LML version of Starstreak).

113. **Mobility**

The inspiration for this rule was this photograph of the Mitla Pass from 1967, which shows trucks destroyed on and just off a road. Also, in the Falklands War roads could be traversed by Landrovers and AMLs, but off-road terrain was impassable to everything except the CVR(T)s and Bv 202s despite being open.

114. **Truck Mobility**

For example, a scenario set in the open desert of southern Iraq might specify that trucks have good mobility, but one set in the Falkland Islands might specify that they have poor mobility. Similarly, a scenario might specify that cargo trucks have poor mobility, but mobile artillery units have good mobility. Finally, a scenario might not assign an explicit transport unit to a towed artillery battery, but might state that it has been transported by helicopter and so can be placed as if it had a transport unit with good mobility.

115. **Modifying Mobility Classes**

One common modification is to explicitly indicate the hexes occupied by each ground unit; in this case this rule is largely superfluous (although if we allow ground units to move, then the restrictions can still be relevant).

Panhard AMLs are wheeled armored vehicles and so normally have unrestricted mobility. Nevertheless, in the Falklands War they were restricted to roads and trails, and this corresponds to poor mobility. A scenario could fix this problem by explicitly giving them poor mobility.

116. **Composite Platoons**

This rule comes from *The Speed of Heat*. It doesn't have counters for infantry SAMs, so in scenarios V-11 and V-22 it says treat infantry platoons as if they have an infantry SAM squad. This rule helps with stacking (since attached squads do not count for stacking) and with keeping your precious FACs and SAM teams anonymous. It also works nicely with the transport rule, since attached squads do not need their own transport but can ride with the platoon.

117. **Damage to Attached Squads**

The roll of 4– seems about right. The probability of killing an attached squad is 40% for 1D and 64% for 2D. Those numbers are fairly good approximations to 1/3 and 2/3. If the roll was 3–, they would be 30% and 51%, and that seems a little low.

Are the damage rolls secret?

118. **Transportation**

The idea of transporting units and cargo comes from the "Thunder and Fire" scenario from *Eagles of the Gulf*. The rules here simply generalize it slightly, incorporate the idea of mobility, and add rules for mounting and dismounting.

119. **Mobility**

Being placed according to the mobility of its transport allows a tracked transport to place a towed unit in a forest hex.

120. **Mounting and Dismounting Infantry**

The rules for mounting and dismounting are completely new. I'm very open to suggestions about this.